

Haller + Erne GmbH

heOGS - Tool configuration notes **Application Note**

HEI-71-206
Version R19

Version history

Version	History	Author	Date
1.0	First release	he	2016-12-05
1.1	Added step category information Added corded tool setup info Added Ccwlgnore OpenProtocol workaround	he	2016-12-15
R05	Added BOSCH EXAConnect II Added Crane TCI Multi	he	2017-03-15
R07	Added Nexo barcode scanner configuration	ic	2017-05-18
R08	Added Batch counter configuration	he	2017-05-30
R09	Added ID-Code ("Mode") configuration for Nexo	he	2017-07-05
R10	Fixed CS351 configuration (chapter 2.2) Added socket tray configuration notes Added KE350 multispindle application control Clarify usage of En signal for Nexo	he	2017-07-05
R11, R12	Added Manual Operations "GUI popup", "KEY input" & "TOOL acknowledge" Added MSTKN configuration sections Added User configuration, FTP output driver configuration, Barcode scanner & RFID scanner configuration	rb	2017-12-06
R13	Added Pick2Light setup Added Nexo Roaming warning, Firmware V1300SP1 info Added SCS Freedom ³ configuration	he	2018-05-28
R14	Updated OpenProtocol driver configuration	ic	2018-06-22
R15	Added info for Jäger positioning system (I/O over CS351/IM24V)	he	2018-11-09
R16	Added info for barcode tool Added info for modbus tool Added info for Euchner Key RFID reader Added more info for setting up Nexo scanner (from Murrhardt)	ic, he	2019-01-23
R17	Added Multi-Interface socket tray Added Sylvac BLE interface / tool (custom LUA tool) Added updated information for Nexo barcode scanning (Firmware V1300SP1)	he	2019-08-14
R18	Added Ergospin PLC settings for CS351	il	2020-08-11
R19	Added SCS Freedom ⁴ configuration info	he	2020-08-20

About this document

This document contains best practice information for setting up tools connected to the heOGS operator guidance software.

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1 Rexroth Nexo

The Nexo tool uses the open protocol interface to communicate with the heOGS software. In addition, data output can be enabled to send information out to the data collection package (Sys3xxGateway, QTrans or QualityR).

Nexo system requirements:

- Nexo Firmware >= V1300, recommended is >= V1300SP1

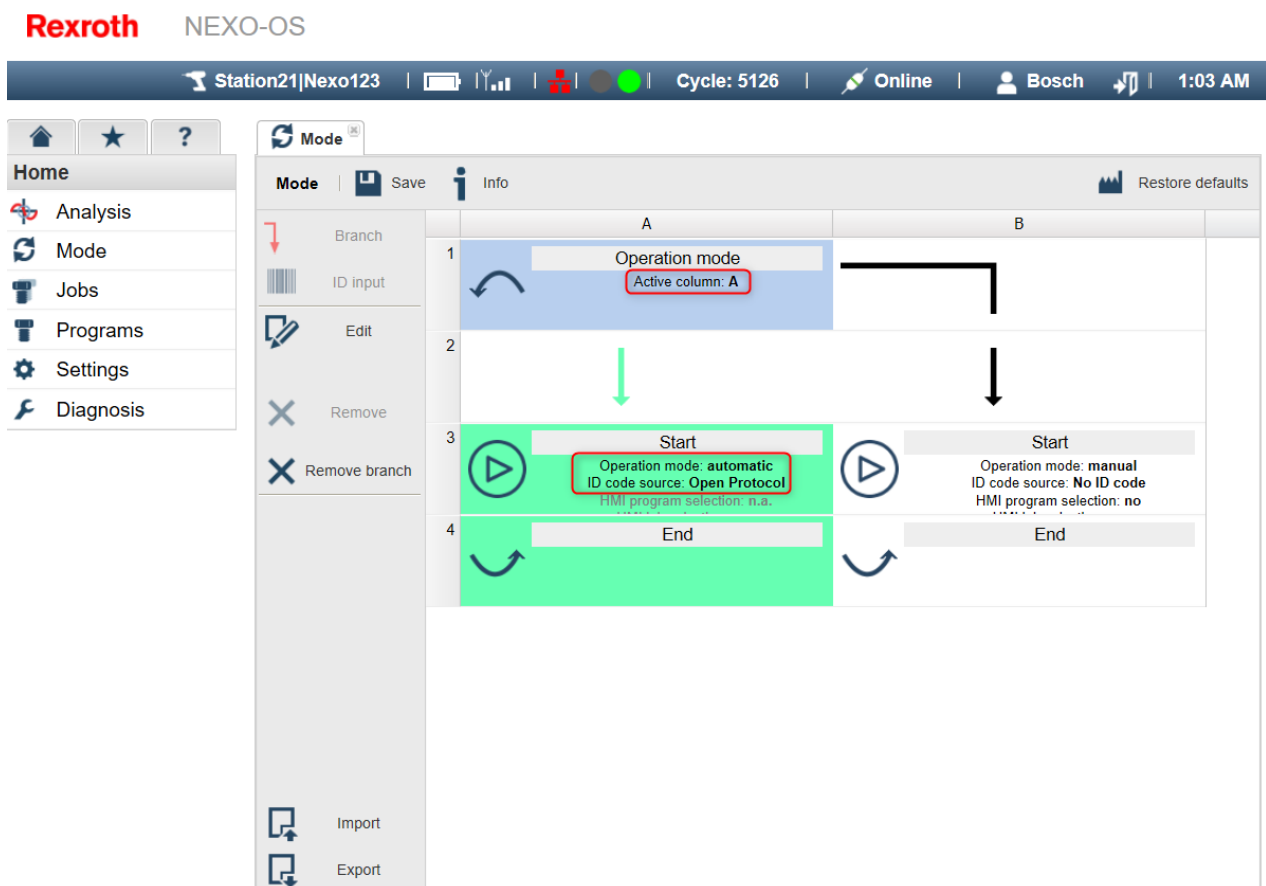


1.1 Tool configuration

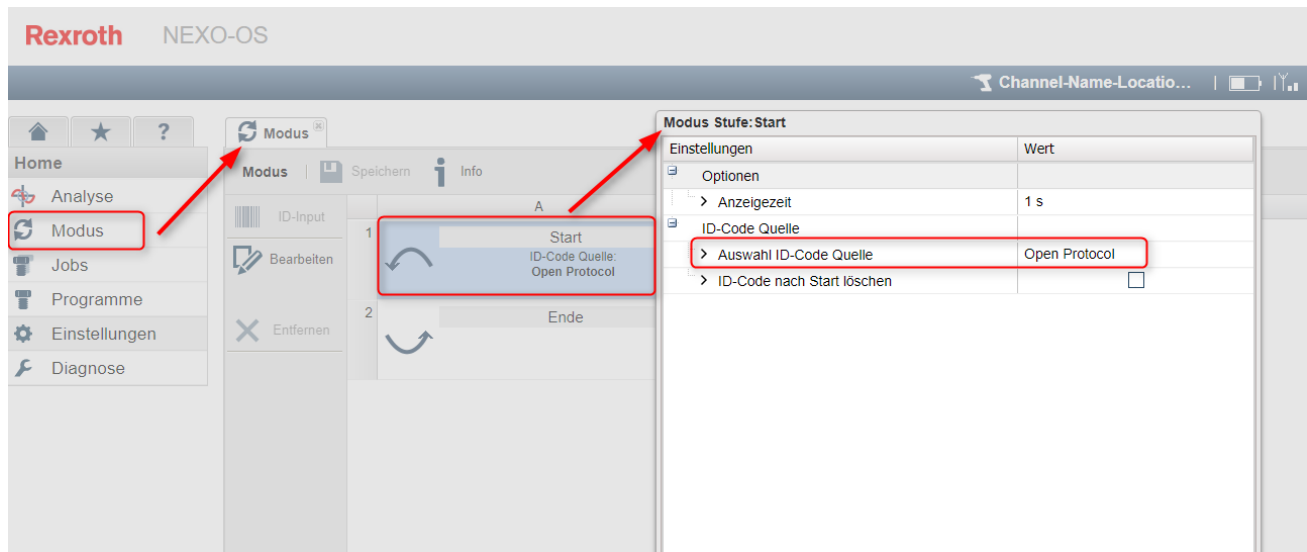
1.1.1 System configuration

1.1.1.1 Operation mode and ID-Code configuration

The Operation mode must be set to automatic (to allow controlling the tool through heOGS) and the ID code source must be set to "Open Protocol". In case multiple operation modes (columns) are defined (from firmware V1300 upwards), the correct column (with "automatic" operation mode) needs to be active.



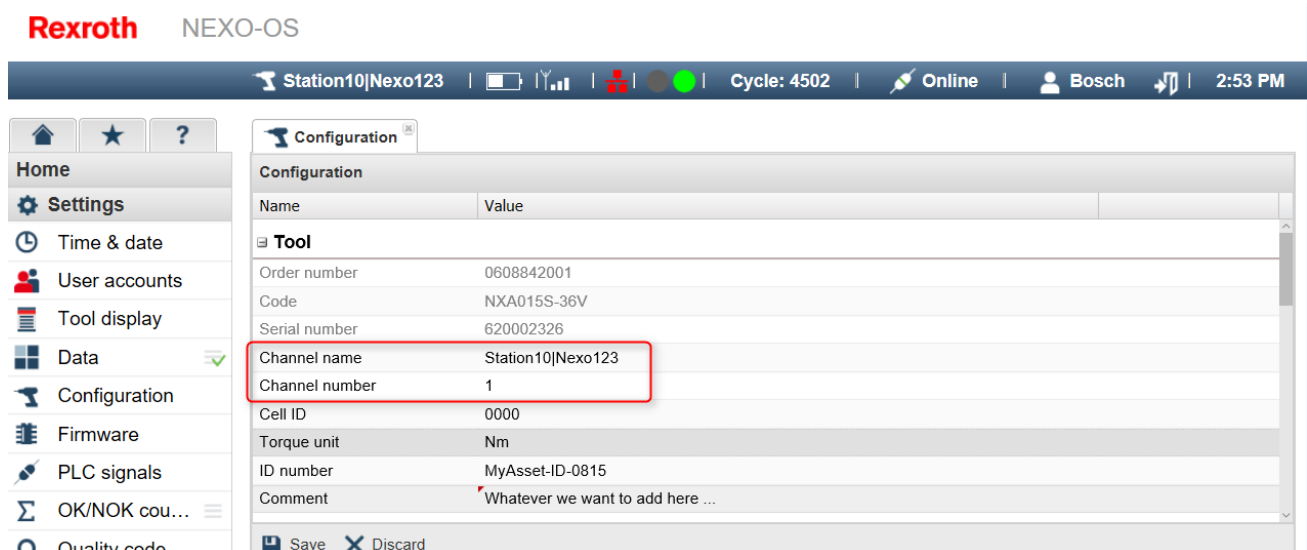
Typically, the ID code is sent to the tool through OpenProtocol. To make the tool accept these ID-codes and tag the results accordingly, please change the “Mode” settings as follows:



(double click the “Start”-step and change the ID-Code source to “OpenProtocol”).

1.1.1.2 Location information

The system information is primarily used to setup location information for the tool. Make sure to setup at least a meaningful channel name and a unique channel number per station in the “configuration” settings:



Channel name: Nexco channel name is used to send station and channel name to the data collecting software (see chapter 1.1.5). Station name and channel name are separated by | (pipe character):

Station_Name|Channel_Name - both Station and Channel name configured

|Channel_Name - no station name is reported (i.e. the IP address is used by default), the name given is used as channel name only.

Station_Name| - the name given is used as station name only, channel name is built as ChRack.Slot (Ch00.1)

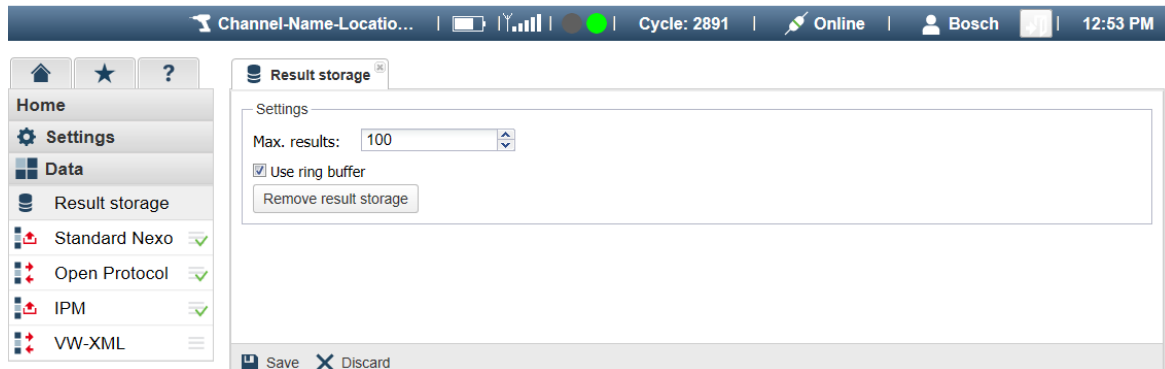
Name - if the pipe character is missing, then Station and Channel names are equal

Empty field - no station name is reported (i.e. the IP address is used by default), channel name is built as ChRack.Slot (Ch00.1)

1.1.1.3 Performance optimizations

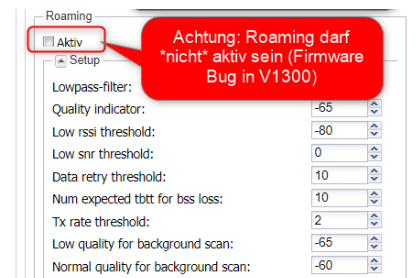
To optimize performance on the tools end, reduce the default number of tightening results stored in the “Result storage” settings:

Rexroth NEXO-OS



1.1.2 WLAN configuration

In Nexo Firmware < V1400 roaming does not work reliably. Therefore, it is required to disable roaming for Firmware versions < V1400 in Settings → WLAN:

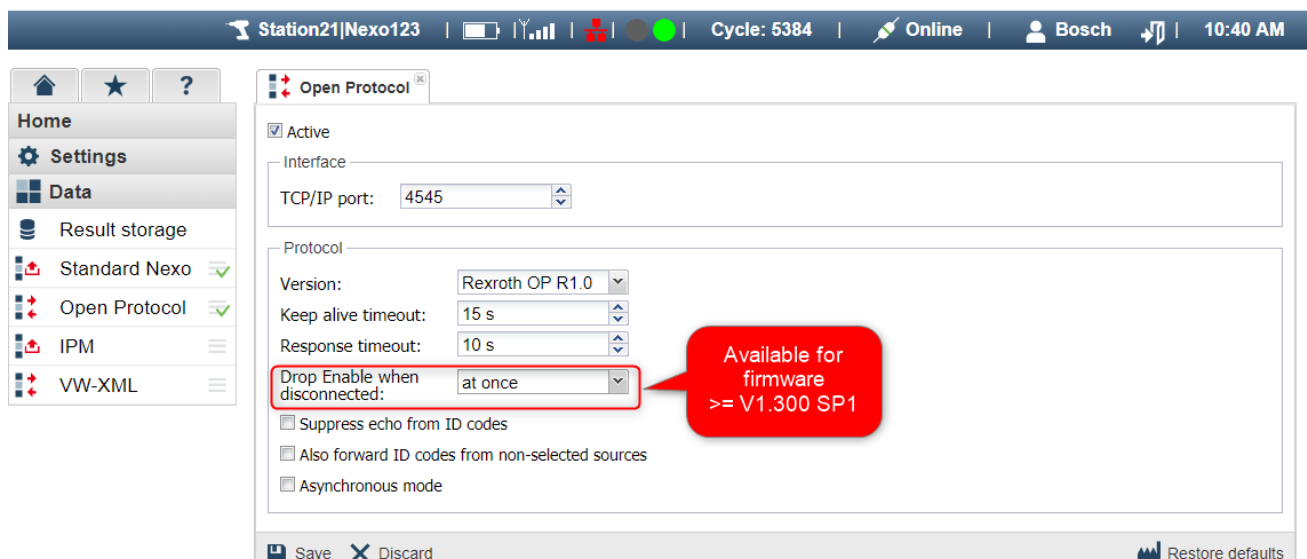


1.1.3 Open Protocol configuration

1.1.3.1 Setup and enable Open Protocol

The Open Protocol driver should be set up as follows:

Rexroth NEXO-OS



Important: For firmware >= V1300SP1 the setting “Drop Enable when disconnected” should be set to “at once” (which actually means “immediately”) to ensure guaranteed control over the tool from the heOGS software under all circumstances.

1.1.3.2 Setup PLC signals

To allow controlling the tool correctly, the PLC signals should be set up as follows:

Rexroth NEXO-OS

Channel-Name-Location... | Zyklus: 3071 | Online | Bosch | 8:58 AM

BMS-Signale

Eingänge

0.0	ActEn
0.1	CcwIgnore
0.2	
0.3	
0.4	
0.5	
0.6	
0.7	
1.0	
1.1	
1.2	
1.3	
1.4	
1.5	
1.6	
1.7	
2.0	Prog0
2.1	Prog1
2.2	Prog2
2.3	Prog3
2.4	Prog4
2.5	Prog5
2.6	Prog6
2.7	Prog7
3.0	En
3.1	<not available>

Schrauber

Steuersignale	Statussignale
Prog0	Ack0
Prog1	Ack1
Prog2	Ack2
Prog3	Ack3
Prog4	Ack4
Prog5	Ack5
Prog6	Ack6
Prog7	Ack7
En	EnAck
ActEn	ActEnAck (x1)
Cw	CwAck
Ccw	CcwAck
CwLock	CwLockAck
CcwLock	CcwLockAck
CwIgnore	StartBn (x1)
CcwIgnore	CCwSel (x1)
ResF	Rdy (x1)
ResRs	NF (x1)
NokAc	InCy (x1)
DisRp	InCyCcw
DisRpFtp	CyCmp
CntRes	Ok
CntSel0	Nok
CntSel1	TorqH

Ausgänge

0.0	CCwSel
0.1	InCy
0.2	StartBn
0.3	ActEnAck
0.4	Rdy
0.5	NF
0.6	
0.7	
1.0	
1.1	
1.2	
1.3	
1.4	
1.5	
1.6	
1.7	

Module

oprtcl tool vwXrr

Aktivieren Deaktivieren

BMS-Signale

Eingänge

0.0	Cw
0.1	Ccw
0.2	En
0.3	ResF
0.4	JobStart
0.5	ActScan
0.6	<=1>
0.7	<=1>

Schrauber

Steuersignale	Statussignale
Prog6	Ack0
Prog7	Ack1
En	Ack2
ActEn	Ack3
Cw	Ack4
Ccw	Ack5
	Ack6

Ausgänge

0.0	CCwSel
0.1	InCy
0.2	StartBn
0.3	ActEnAck
0.4	Rdy
0.5	NF
0.6	
0.7	

Important:

- **Never** assign signal "En" to opctrl input 3.0 – this may enable the tool without control of the heOGS software.
- **Never** assign signal "En" to tool input 0.2 – this will enable the tool without control of the heOGS software.
- **Always** assign signal "CcwIgnore" to opctrl input 0.1. This is used as a workaround to problems in the OpenProtocol network protocol – it prevents the operator from loosening if the network connection to the tool gets lost.

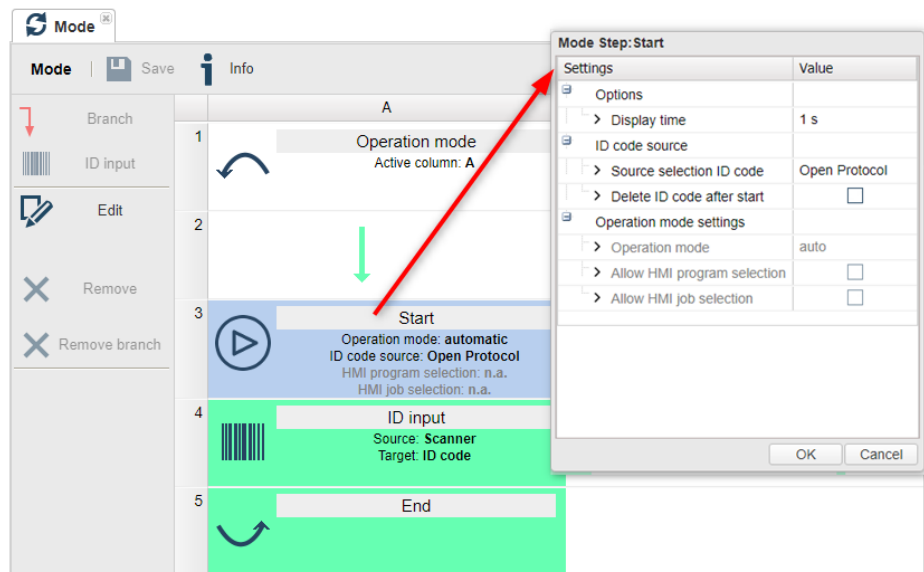
1.1.4 Barcode Scanner

Some Nexo models provide a built-in barcode scanner. This barcode scanner can be used instead of or in combination with any other ID-Code source. Using the Nexo scanner, it is therefore possible to start a workflow, select jobs or do other scan operations inside a workflow.

Please see the Nexo system manual for detailed information about how to use and configure the scanner. The following section shows a simple setup which allows to user to trigger the Nexo builtin scanner by pressing a button below the Nexo display.

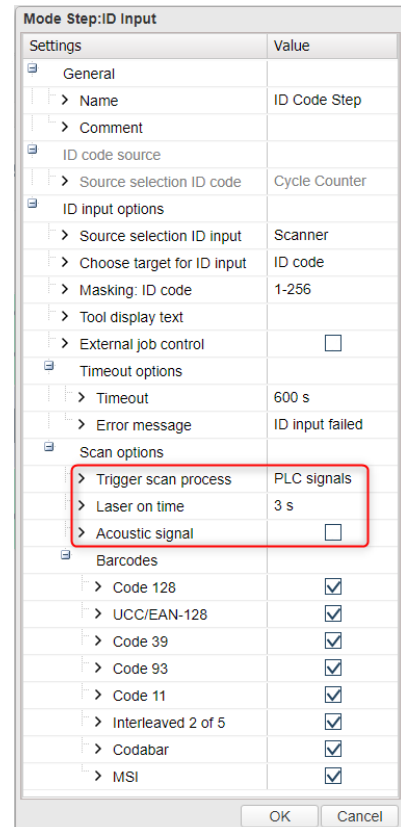
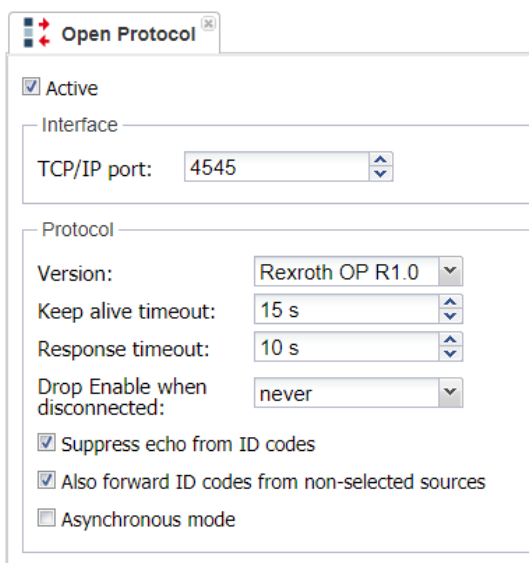
1.1.4.1 Configure Nexo to forward scanned barcodes to OGS

To use the Nexo built-in barcode scanner instead of an external scanner to start an OGS job, Nexo must be set to forward scanned barcodes to OGS. To make this work, the mode start steps ID-Code source must be set to “Open Protocol” and an additional ID-Input step is required with ID-Code source set to Nexo scanner as shown on the right.



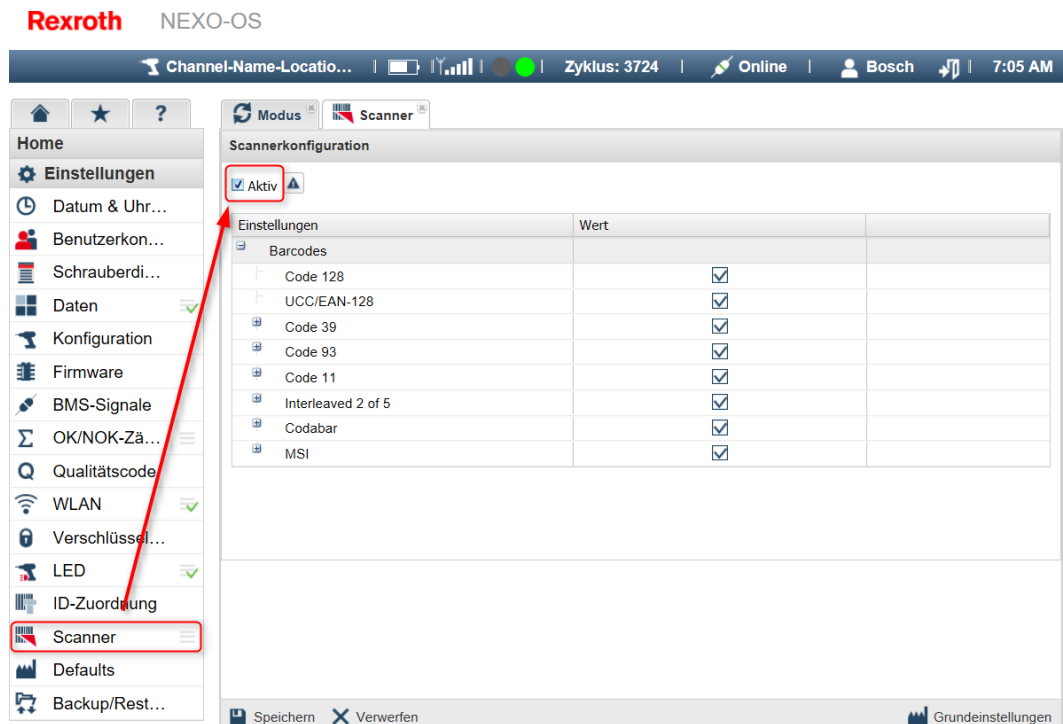
The Settings for the ID input step should be as shown on the right – this will enable scanning by setting the CW/CCW-Switch to the middle position (as long as the PLC assignment table is also setup correctly – see below, chapter 1.1.4.2).

To make the OpenProtocol driver correctly forward the ID-Code, check that the OpenProtocol settings (Settings → Data → OpenProtocol) are as follows (make sure to allow “also forward ID codes from non-selected sources”):

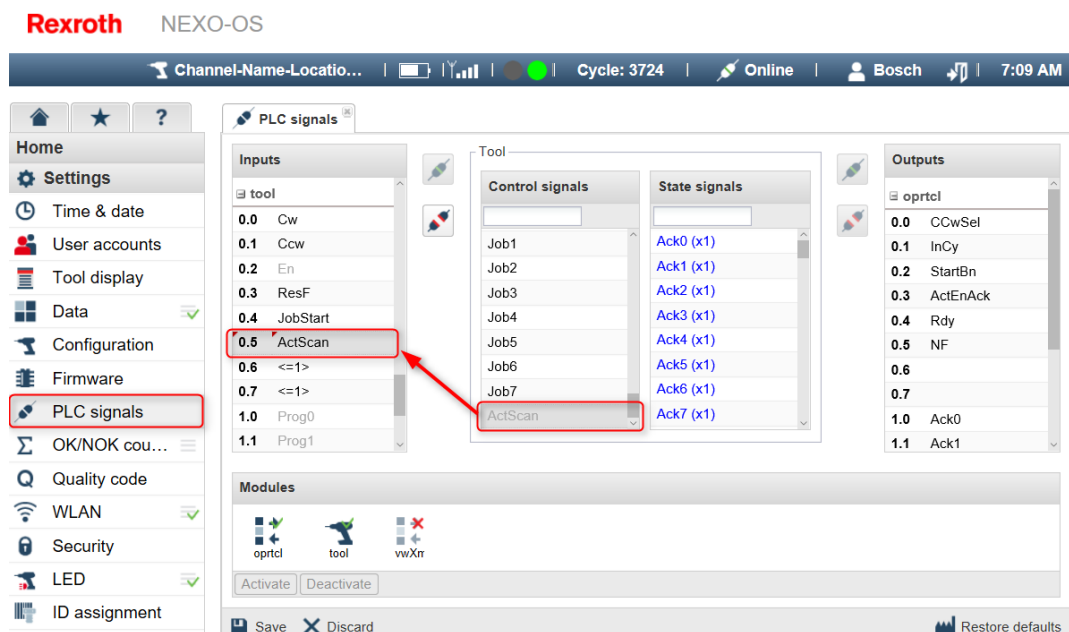


1.1.4.2 Configure Nexo to scan at a button press

Enable the barcode scanner here:



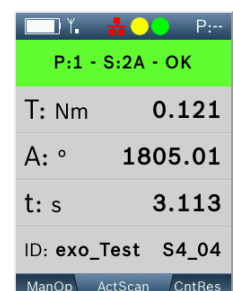
To enable scanning through the buttons below the Nexo display, assign the ActScan signal accordingly:



Note the following assignments:

- Tool 0.4: left button below display
- Tool 0.5: middle button below display
- Tool 0.6: right button below display

The tool will show the mapped signals, if the CW/CCW switch is set to the idle (middle) position (see screenshot on the right, note that here the signals ManOp, ActScan and CntRes are assigned to the 0.4, 0.5 and 0.6 tool inputs).



1.1.4.3 Configure heOGS

To enable barcodes from the tool, change the options in the heOGS tool driver (in station.ini) as follows:

```
[OPENPROTO]
CHANNEL_01=10.10.2.160
CHANNEL_01_MANUAL_OPPRG=98
; Barcode scanner (MID0051)
; - Set revision 1/2 to enable builtin barcode scanner.
; - Set to 0 (default) to disable usage of scanner
CHANNEL_01_BARCODE_MID0051_REV=2
; Channel parameter code
```

Please also see chapter 1.2 for more information on the tool driver settings.

1.1.5 Data output configuration

To send data out to a central QTrans/Sys3xxGateway/QualityR server, typically the following options are possible:

1. Use the "Standard Nexo" data output with the FTP transfer option (preferred). By default transmits all step data and tightening curves, but sometimes causes troubles with the network infrastructure (firewall transversal).
2. Use the "IPM" data output. By default reports "last step" data and curves.

The "Standard Nexo" data output with FTP transfer is the preferred option, if FTP is not allowed/supported (due to firewalling/natting), then use the "IPM" data instead.

Data reported to Sys3xxGateway will use the following mapping by default:

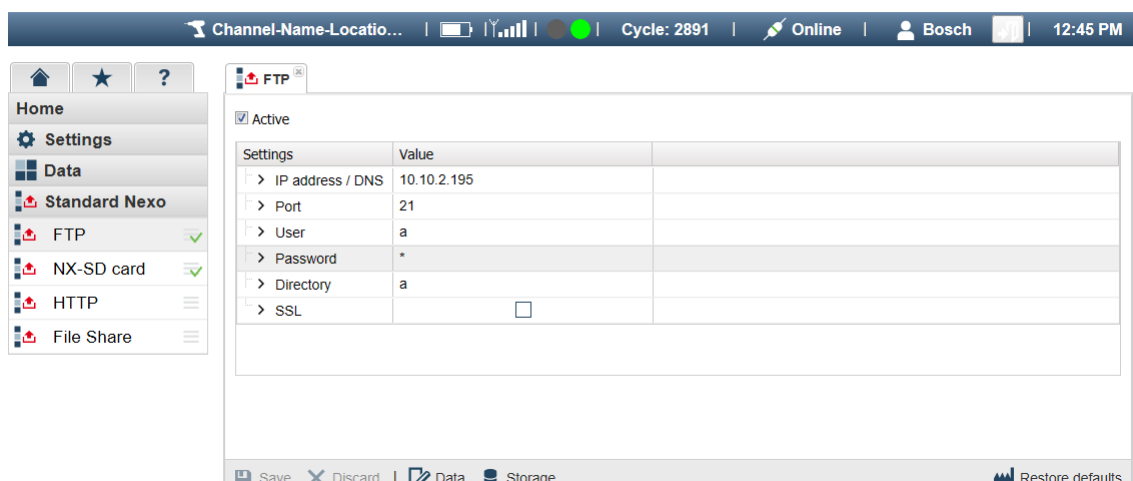
- Nexo IP address-->Default Sys3xxGateway station name
- Nexo channel name --> if non-empty is used as Sys3xxGateway station name
- Nexo channel number --> Sys3xxGateway channel number
- Tightening program name --> Used as operation name (QWX)

Warning: only enable a single data output, else you will see double-data set on the servers end!

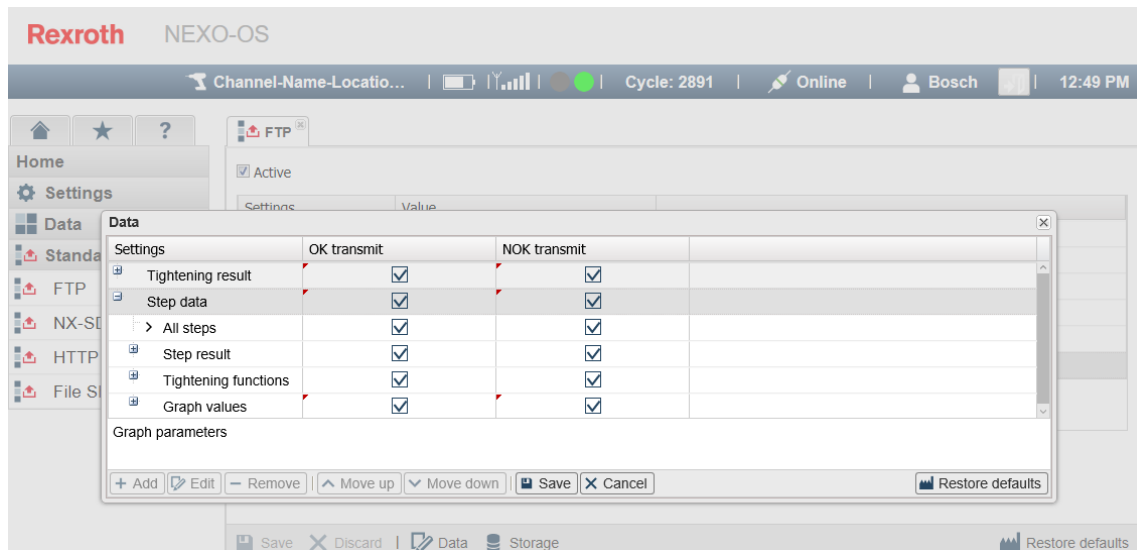
1.1.5.1 Configure Standard Nexo FTP

Configure the FTP data output as follows (User, Password and Directory are ignored for QTrans, Sys3xxGateway and QualityR, but might be needed for 3rd party servers):

Rexroth NEXO-OS



To make sure all data is correctly sent out to the server, check the “Data” (button on the bottom toolbar) contains the following (typically everything should be checked):



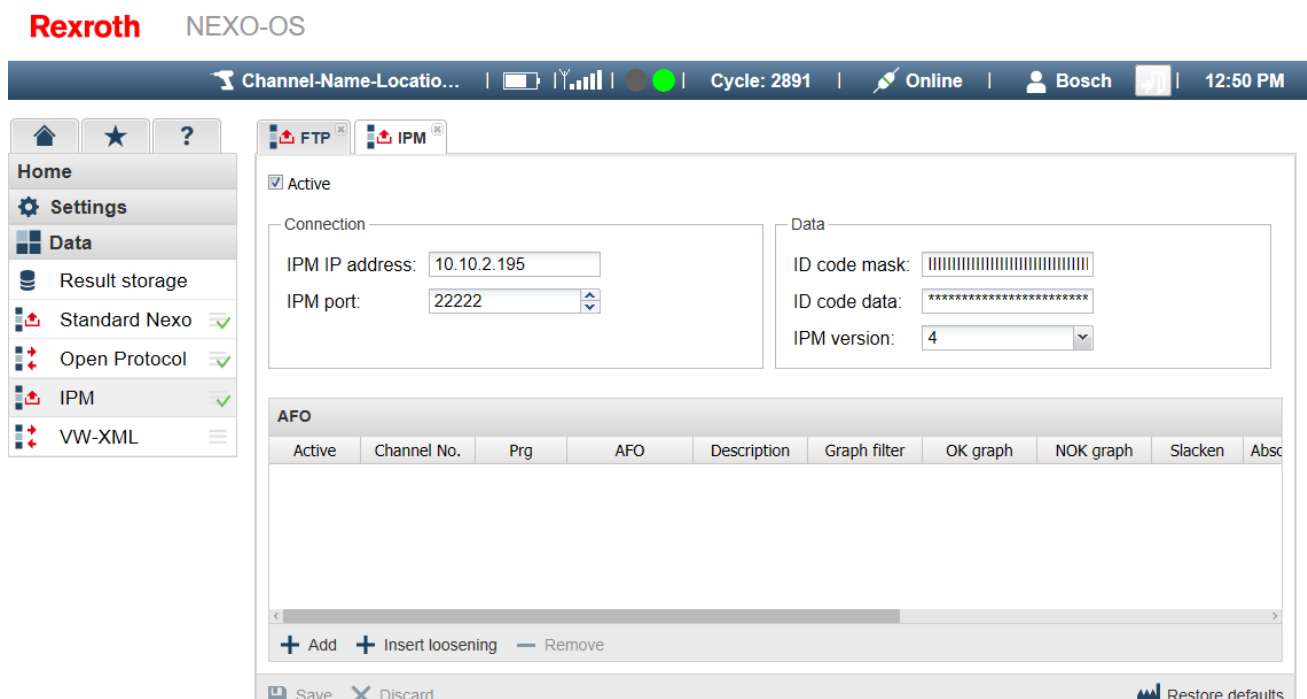
Notes:

- Step data output: The tool should be set to send out all data (as described above). However, by default only the last step data is actually processed by QTrans/Sys3xxGateway/QualityR.
- To send enable additional step data, change the step category to a non-zero value for each step you want to see in the result data.

1.1.5.2 Configure IPM

NOTE: The preferred data output is “Standard Nexo FTP” (see above). Make sure to only enable a single data output, else you will see double-data set on the servers end!

IPM should be configured as follows (change the IP-Address to match the servers IP and modify the ID-Code mask to include as much characters as needed):



Notes:

- Step data output: by default, only the final step data is sent out.
- To enable multi-step data output for IPM, AFOs need to be added (see the Nexo manual).

1.1.6 Tightening program setup

By default, only the last step data is recorded in the data collection package. Note that the various data output protocols behave differently regarding to the amount of data sent:

- OpenProtocol (heOGS) always sends out final angle and torque values.
- Nexo Standard (FTP) only sends out data values if monitoring is active in the tightening program
and the data output setting allows for sending out this data (enabled accordingly for angle/torque actual, limit, ...)

Regarding data from multiple steps there are also setup requirements:

- OpenProtocol: only supports final step data.
- Nexo Standard (FTP): Two requirements must be met:
 - o Enable step output in the driver configuration
 - o Change the step category in the tightening program setup to non-zero

Here is a screenshot on how to setup the tightening program correctly to enable data output for angle and torque for the Pre-Torque step:

The screenshot displays the Rexroth NEXO-OS software interface. The main window shows a tightening program setup for 'Prg 12: 12Nm'. The 'Stufe: Pre-Torque' configuration window is open, showing the following settings:

Einstellungen	Wert	Schaltend
Name	Pre-Torque	
Kategorie	2: Vorläufiges Drehmoment	
Kurvenauflösung	As start	
Ziel		
Drehzahl	140 1/min	
Anlaufunterdrück...	30 ms	
Schwellmoment	0 Nm	
Erste Zielfunktion	Moment	
Erstes Ziel	3 Nm	
(Oder) 2. Zielfu...	Keine	
2. Ziel	0	
Überwachung		
Drehmoment Mi...	2 Nm	☐
Moment	3 Nm	
Drehmoment M...	4 Nm	☐
Winkel Minwert	4 °	
Winkel	-	
Winkel Maxwert	7200 °	☐
Zeit	10 s	☒

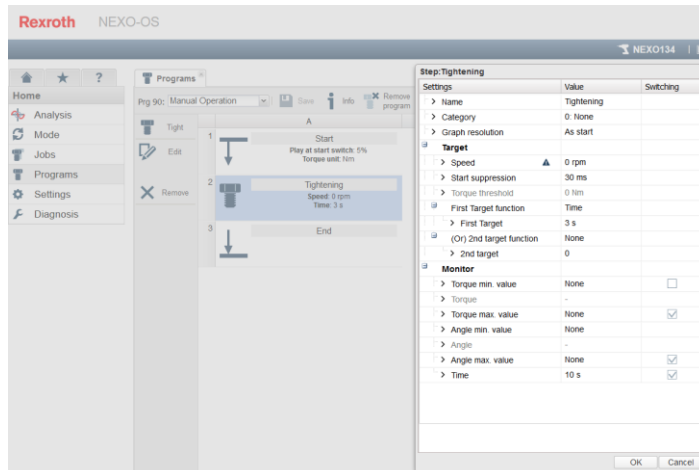
Note the following:

- Category is set to non-zero.

- Monitoring Angle is enabled and a minimum and maximum angle is given
- Monitoring Torque is enabled and a minimum and maximum torque is given

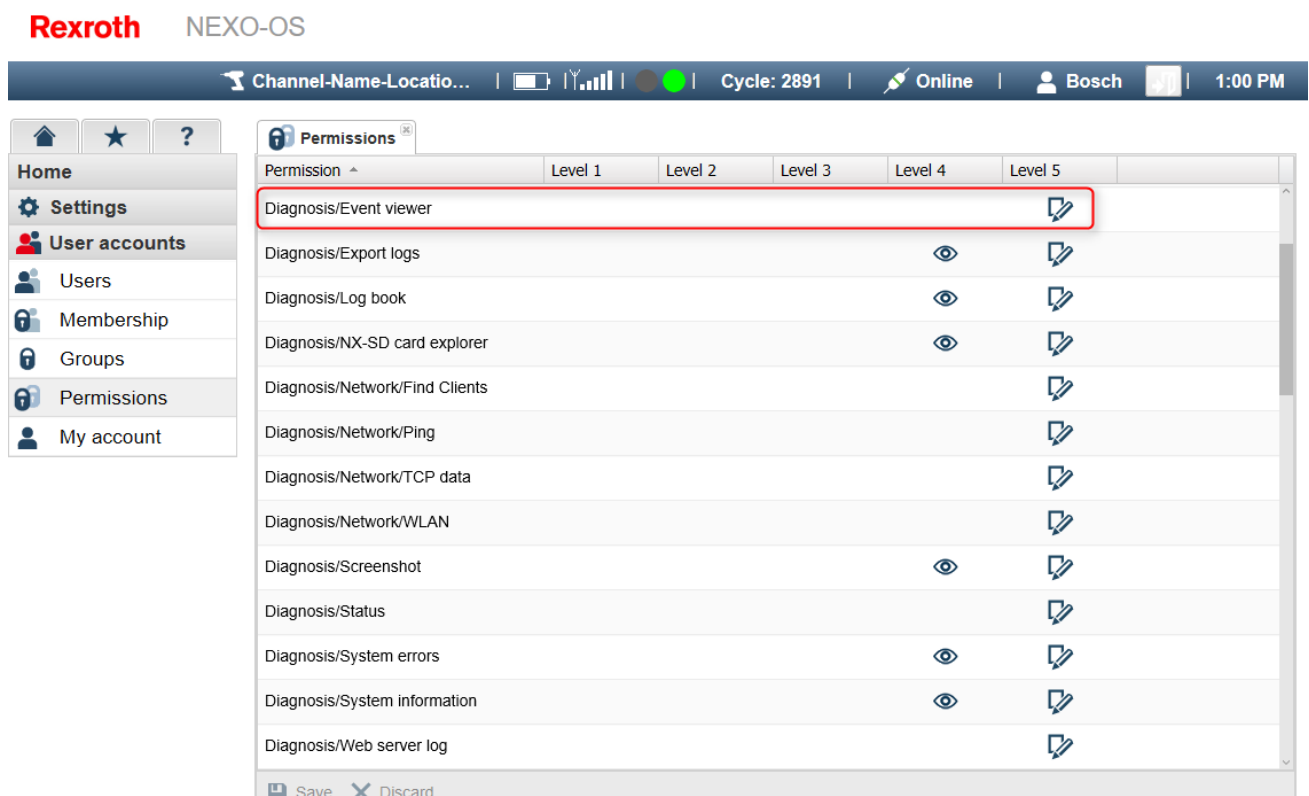
1.1.7 Setup for manual acknowledge

To use the tool as “manual acknowledge” input (see chapter 9.3), the tightening program should be setup similar to the following (simple wait step without the tool actually turning). An OK will then generated whenever the operator holds the tool button for more than the given waiting time.



1.1.8 Diagnostics

To get more detailed diagnostics, the relevant pages must be enabled in the user permissions:



1.2 heOGS configuration

1.2.1 Driver Setup

Rexroth Nexa uses the OPCConn.dll tool driver. To enable this driver, please make sure to setup the section [TOOL_DLL] as follows:

```
[TOOL_DLL]
OPConn.dll=1
;MBLightCli.dll=1
;EXACConnecT.dll=1
;heGWK.DLL=1
;ProfiFTP.dll=1
```

1.2.2 Driver Configuration

The heOGS tool driver is configured in the station.ini file as follows (make sure to set the_TYPE parameter to NEXO):

```
[OPENPROTO]
;
; _____
; Global Open Protocol parameters
; Port number
PORT=4545
; Poll cycle time in ms
Tact=10

;
; _____
; Configuration for Tool 1 (Channel 01)
; Define the IP address of the tool
CHANNEL_01=10.10.2.160
; Define the tool type (allowed values are NEXO, CS351, CRANE)
CHANNEL_01_TYPE=NEXO
; Define the program number used for manual acknowledge (not
; practically applicable for Crane wrenches).
CHANNEL_01_MANUAL_OPFRG=98

; Parameter 'CHECK_EXT_COND'
; 0 (default): do not check ext. conditions (socket tray) to enable
; or disable a tool if tightening process is already started
; 1: force to check external conditions even if the tool is running
CHANNEL_01_CHECK_EXT_COND=1

; optional settings:
; Port number used for tool communication (default 4545) or
; port number set through PORT parameter
CHANNEL_01_PORT=4545

;
; _____
; additional tools using OpenProtocol
; CHANNEL_02_...
```

To use the builtin barcode scanner with Nexa, please see chapter 1.1.4.3)

2 Rexroth CS351 with OpenProtocol

The CS351 system uses the open protocol interface to communicate with the heOGS software. In addition, data output can be enabled to send information out to the data collection package (Sys3xxGateway, QTrans or QualityR).

CS351 system requirements:

- CS351 Firmware >= V2.400

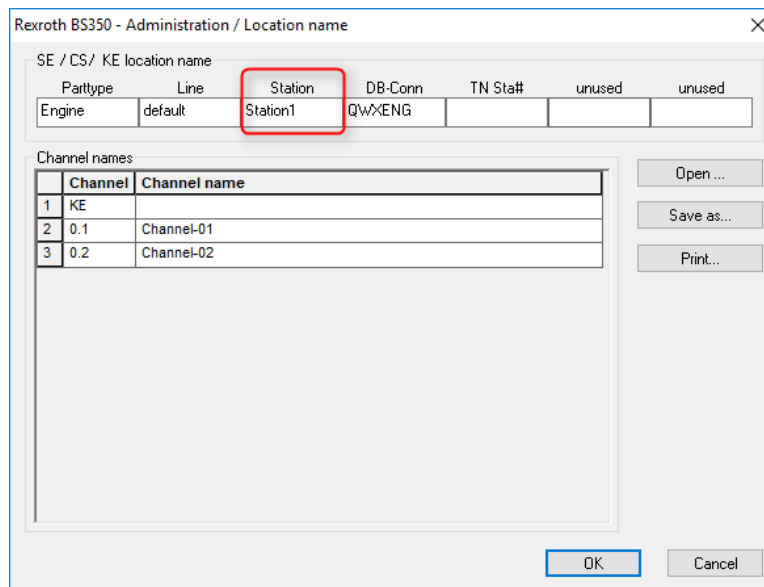
Note: this is an updated recommendation, using the "simple" OpenProtocol enable commands (not ActEn).

2.1 Tool configuration

2.1.1 System configuration

2.1.1.1 Location information

The system information is primarily used to setup location information for the tool. Make sure to setup at least a meaningful channel name in the Administration → Location Names settings:



SE / CS/ KE location name						
Parttype	Line	Station	DB-Conn	TN Sta#	unused	unused
Engine	default	Station1	QWXXENG			

Channel names	
Channel	Channel name
1	KE
2	0.1 Channel-01
3	0.2 Channel-02

used for data output (see chapter 2.1.3) as follows:

- field 1: assembly type (engine, transmission, ...)
- field 2: line name
- field 3: station name
- field 4: database alias (important for Sys3xxGateway)
- field 5: *not used*
- field 6: *not used*
- field 7: *not used*

2.1.1.2 Setup Batch Counters (OK/NOK Counters)

To make sure the OpenProtocol interface correctly runs single rundowns, setup OK/NOK counters (under System → OK/NOK counter configuration) as follows:

OK/NOK counter configuration: 0.1

0: n.a.		
1: TEST1		
2: P2_2_0Nm		
3: P3_3_0Nm		
4: P4_4_0Nm		
5: P5_5_0Nm		
6: n.a.		
7: P7_3_5Nm		

PLC signal selection
☒ Prog0 ... Prog7
☐ CntSel0 ... CntSel7

Counting direction
Sum OK: ☒ 0, 1, ..., Max OK ☐ Max OK, ..., 0
Sum NOK: ☒ 0, 1, ..., Max NOK ☐ Max NOK, ..., 0

Procedure if number of OK/NOK reached or for abort
☒ Block nutrunner
☐ Enable loosening program no. 99

Conditional blocking of loosening program
☐ Block loosening program after tightening OK
☐ Block loosening program after tightening NOK

Special functions
☐ All configured counters must be completed
☐ Set socket tray signals
☐ Allow socket tray signals to blink
☐ Decrement the OK counter after loosening

Start delay
OK tightening: 0 [ms] time delay
NOK tightening: 0 [ms] time delay

2.1.2 Open Protocol configuration

2.1.2.1 Setup and enable Open Protocol

The Open Protocol driver (under System → Tightening Cell Data → Open Protocol) should be set up as follows:

Rexroth BS350 - Tightening cell data/Open protocol

Interface
☒ Protocol activated TCP/IP port: 4545

Protocol
Version: Rexroth OP R1.0
Keep alive timeout: 15 s
Response timeout: 10 s
☒ Suppress echo from ID codes
☐ Also forward ID codes from non-selected sources
☐ Asynchronous mode
☐ Forward I/O signals to IL
☐ Auto-disable application
☐ MID31 provides Job list
☐ MID61 valuation and batch information from application result

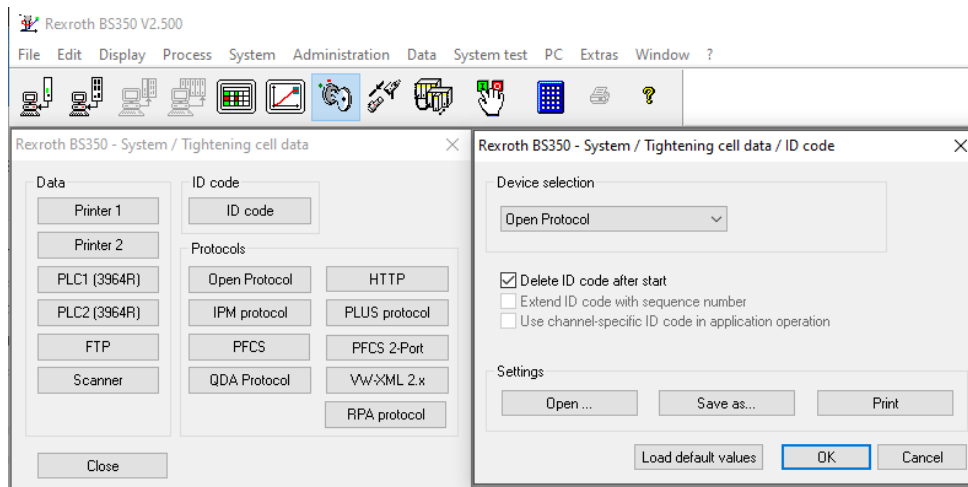
Slaves
☐ Slave 1 IP address: 0 . 0 . 0 . 0 Port: 4545
☐ Slave 2 IP address: 0 . 0 . 0 . 0 Port: 4545
☐ Slave 3 IP address: 0 . 0 . 0 . 0 Port: 4545
☐ Forward ID code to slaves Connection timeout: 10 s

Settings
Open ... Save as ... Print ...
Load default values OK Cancel

2.1.2.2 Setup ID Code source

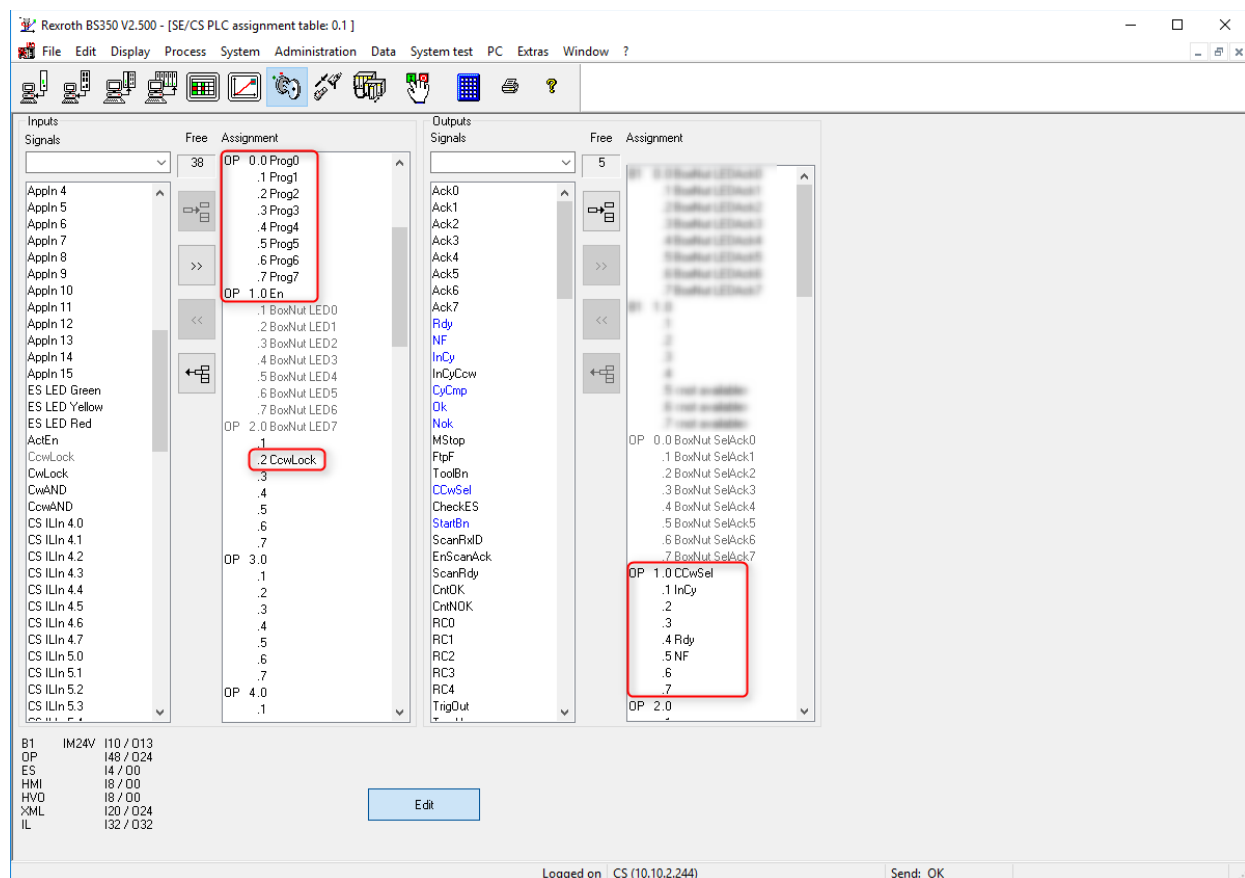
The ID Code

source (under System → Tightening Cell Data → ID Code) should be set up as follows:

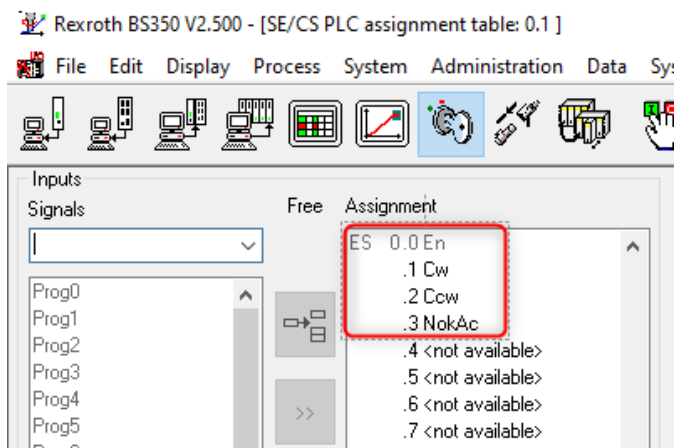


2.1.2.3 Setup PLC signals

To allow controlling the tool correctly, the PLC signals should be set up as follows (System → SE/CS PLC) assignment table:



2.1.2.4 Setup Ergospin PLC signals



2.1.2.5 Setup PLC signals for use with IM24V

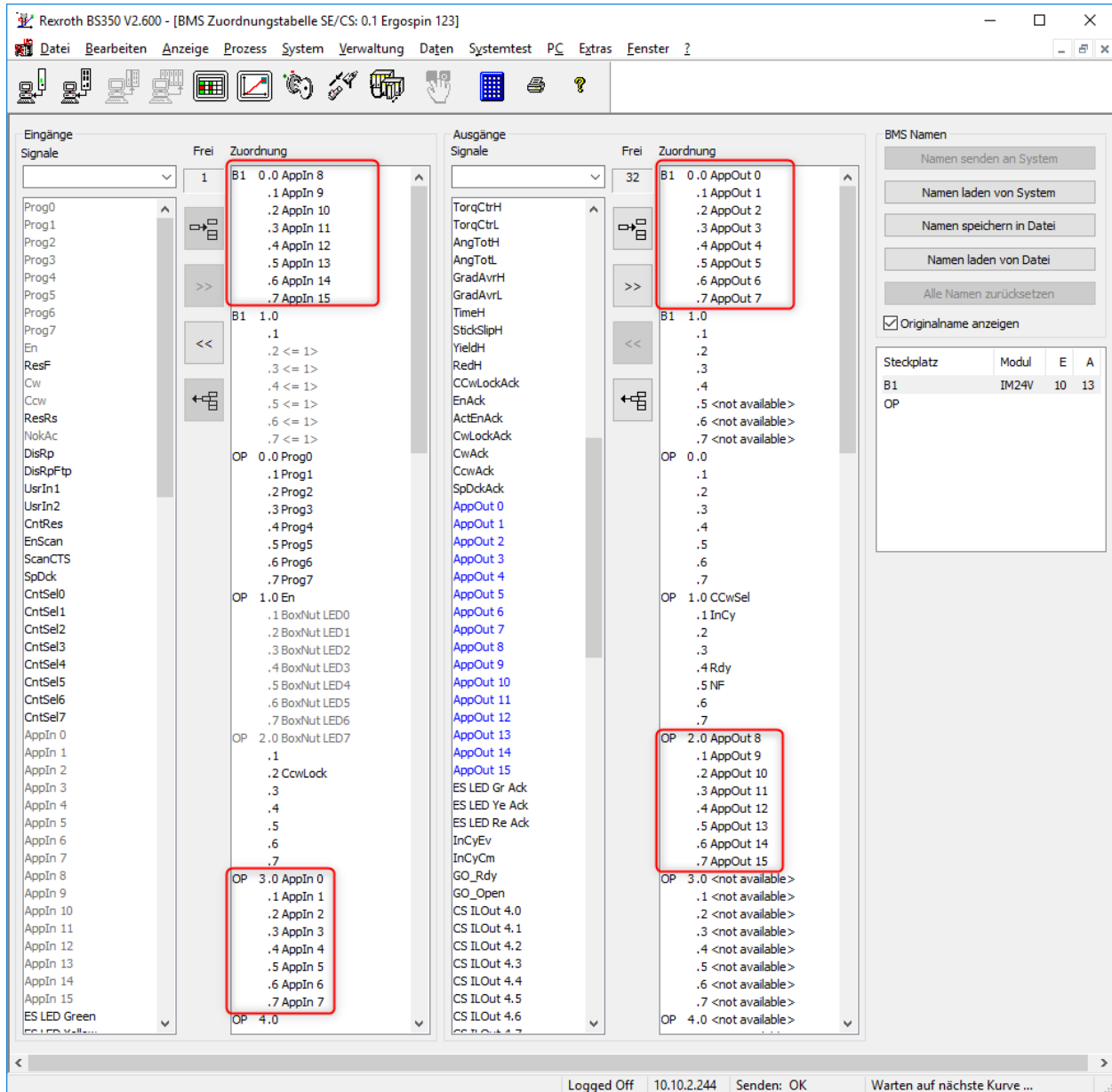
OpenProtocol supports connecting 8 userdefined I/O-signals (8 inputs, 8 outputs). This can be used to e.g. connect an external positioning system to the OGS software.

The following setup shows an example use – note that the Signals AppInX and AppOutX are a virtual bridge, i.e. the status of the AppInX signal is automatically reflected in the AppOutX status. This is used to route between the OpenProtocol signals and the actual hardware signals.

The user defined signals are available as follows:

- 8 output signals: In the “Inputs” section of the PLC assignment table, at module OP 3.0-3.7. These signals are written by OGS.
- 8 input signals: In the “Outputs” section of the PLC assignment table, at module OP 2.0-2.7. These signals are read by OGS.

The sample setup below connects the in- and outputs to an IM24V card at slot B1.



2.1.3 Data output configuration

Data reported to Sys3xxGateway will use the following mapping by default:

- CS351 IP address-->Default Sys3xxGateway station name
- CS351 channel name --> if non-empty is used as Sys3xxGateway station name
- CS351 channel number --> Sys3xxGateway channel number
- CS351 location names (see chapter 2.1.1.1):
- Tightening program name --> Used as operation name (QWX)

2.1.4 Tightening program setup

Tbd.

2.1.5 Diagnostics

In case there is no response from the tool or it indicates disconnect, check the System 350 Error log under System test → SE/CS error list...

2.2 heOGS configuration

2.2.1 Driver Setup

Rexroth Nexo uses the OPConn.dll tool driver. To enable this driver, please make sure to setup the section [TOOL_DLL] as follows:

```
[TOOL_DLL]
OPConn.dll=1
;MBLightCli.dll=1
;EXAConnecT.dll=1
;heGWK.DLL=1
;ProfiFTP.dll=1
```

2.2.2 Driver Configuration

The heOGS tool driver is configured in the station.ini file as follows (make sure to set the ..._TYPE parameter to CS351):

```
[OPENPROTO]
;
; _____
; Global Open Protocol parameters
; Port number
PORT=4545
; Poll cycle time in ms
Tact=10

;
; _____
; Configuration for Tool 1 (Channel 01)
; Define the IP address of the tool
CHANNEL_01=10.10.2.160
; Define the tool type (allowed values are NEXO, CS351, CRANE)
CHANNEL_01_TYPE=CS351
; Define the program number used for manual acknowledge.
CHANNEL_01_MANUAL_OPFRG=98

; optional settings:
; Port number used for tool communication (default 4545) or
; port number set through PORT parameter
CHANNEL_01_PORT=4545

;
; _____
; additional tools using OpenProtocol
;CHANNEL_02_...
```

3 Rexroth KE350 Applications

The KE350 system uses the “Ford” open protocol interface to communicate with the heOGS software. The “Ford” open protocol interface allows running tightening applications, i.e. multi-spindle tools. However, only a single “application” is supported, i.e. a single KE can only control a single (possibly multi-spindle) tool.

In addition, data output can be enabled to send information out to the data collection package (Sys3xx-Gateway, QTrans or QualityR).

KE350 system requirements:

- CS351 Firmware >= V2.400

Notes:

- This uses the “Ford multispindle” MIDs.
- Only a single application can be controlled, so this is typically used for a multi-spindle aggregate/tool. For each tool a single KE **must** be used, it is **not** possible to control multiple tools at the same time independently!

3.1 Tool configuration

3.1.1 System configuration

3.1.1.1 Location information

The system information is primarily used to setup location information for the tool(s). Make sure to setup at least a meaningful channel name in the Administration → Location Names settings:

SE / CS/ KE location name						
Parttype	Line	Station	DB-Conn	TN Sta#	unused	unused
Engine	default	Station1	QW&XENG			

Channel names	
Channel	Channel name
1	KE
2	0.1 Channel-01
3	0.2 Channel-02

3.1.1.2 Setup tightening applications and programs

Tbd.

3.1.2 Open Protocol configuration

3.1.2.1 Setup and enable Open Protocol

The Open Protocol driver (under System → Tightening Cell Data → Open Protocol) should be set up as follows:

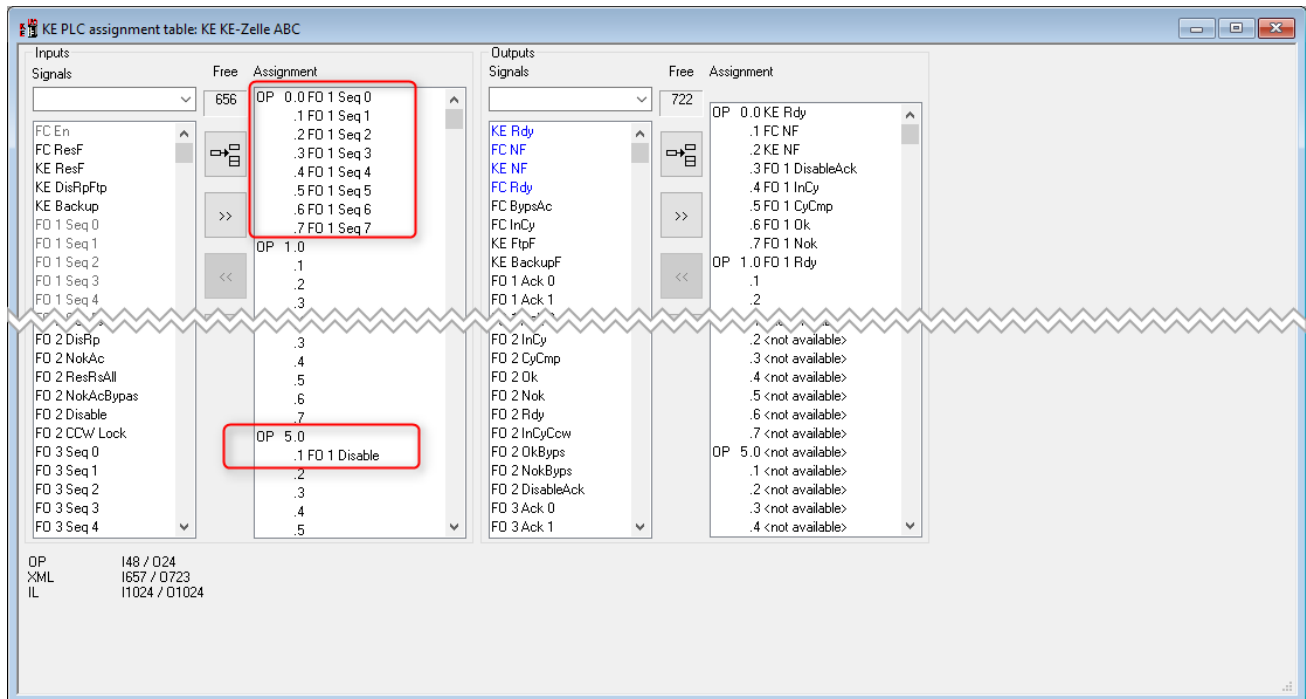
Please make sure to check “Auto-disable application”, this will ensure the tightening system disables the application after each rundown automatically.

Notes:

- For special cases (then also requires IndraLogic code running in the KE350) the “Forward I/O signals to IL” can be used. Control can then be chained – the IL gets controlled by the heOGS software, but can itself control the tightening system.

3.1.2.2 Setup PLC signals

To allow controlling the tool correctly, the PLC signals should be set up as follows (System → SE/CS PLC assignment table:



Make sure to set FO1 Disable to OP 5.1!

3.1.3 Data output configuration

Tbd.

3.1.4 Tightening program setup

Tbd.

3.1.5 Diagnostics

In case there is no response from the tool or it indicates disconnect, check the System 350 Error log under System test → SE/CS error list...

3.2 heOGS configuration

3.2.1 Driver Setup

Rexroth KE350 uses the OPCConn.dll tool driver. To enable this driver, please make sure to setup the section [TOOL_DLL] as follows:

```
[TOOL_DLL]
OPConn.dll=1
;MBLightCli.dll=1
;EXAConnecT.dll=1
;heGWK.DLL=1
;ProfiFTP.dll=1
```

3.2.2 Driver Configuration

The heOGS tool driver is configured in the station.ini file as follows:

[OPENPROTq]

```
CHANNEL_01=10.10.2.244  
CHANNEL_01_MANUAL_OPFRG=98
```

```
; Channel parameter code  
; Typical configuration values are:  
; 0 (default) - Nexo with text display enabled using ActEn signal  
; 3 - Crane WrenchStar  
; 5 - Nexo with text display  
; 6 - CS351 box (no text display)  
; 7 - Nexo without text display  
; NOTES:  
; - To go for maximum speed  
;   the text disappears  
;   However, this costs  
;   InCy, the last message  
; - do *not* assign any  
;   you want to have low
```

```
CHANNEL_01_PARAMS=7  
CHANNEL_01_ALIVEXMTT=5000  
CHANNEL_01_SHOWALIVE=0  
CHANNEL_01_RSPTIMEOUT=10000
```

```
PORT=4545  
; Poll cycle time in ms  
Tact=10
```

Nicht korrekt

Open Protocol Gegenstelle

Settings MIDs 0-99 MIDs 100-9999

main

MID0001 Communication Start (1)

Controller IP 10.10.2.243

Controller Port 4545

Version Rexroth OP-Ford R1.0

Revision 001 Send

MID0003 Communicati...

MID0003 Communication Stop (3)

Disconnect TCP Disconnect

Revision 001 Send

MID0018 Select Paramet...

MID0018 Select Parameter set (18)

Programm ID 1

Revision 001 Send

MID0042 Disable Too...

MID0042 Disable Tool (42)

Revision 001 Send

MID0043 Enable Tool...

MID0043 Enable Tool (43)

Revision 001 Send

MID0100 Subscribe Multispindle ...

MID0100 Subscribe Multispindle Results (100)

☐ No Ack needed

☐ Don't Send Ack

Revision 001 Send

MID0410 Get AutoDisable Set...

MID0410 Get AutoDisable Settings (410)

Revision 001 Send

log

```

INFO - sending: 0023|0018|00|001|00000000[001 ]
INFO - rcv-msg: 0024|0005|00|001|00000000[0018]
INFO - 0018      accepted by Controller
INFO - sending: 0020|0043|00|001|00000000[ ]
INFO - rcv-msg: 0024|0005|00|001|00000000[0043]
INFO - 0043      accepted by Controller
INFO - sending: 0020|0100|00|001|00000000[ ]
INFO - rcv-msg: 0024|0005|00|001|00000000[0100]
INFO - 0100      accepted by Controller
INFO - rcv-msg: 0192|0101|00|001|00000000[010102
                                03010400105    06    07 080004000900040010000000:
INFO - sending: 0020|0102|00|001|00000000[ ]
INFO - Länge: 192 ,MID: 101 ,Revision: 1 ,No Ack Flag: 0
01      01
02
03      01
04      001
05
06
07
08      000400
09      000400
10      000000
11      00370
12      00380
13      00375
14      2017-01-28:13:04:49
15      2017-10-02:10:21:51
16      00014
17      1
18:
    01: 01 1 1 000093 1 00376

INFO - sending: 0020|0410|00|001|00000000[ ]
INFO - rcv-msg: 0024|0411|00|001|00000000[0000]
INFO - Länge: 24 ,MID: 411 ,Revision: 1 ,No Ack Flag: 0
      OKs to disable tool: 00
      Current batch:      00
  
```

4 Crane TCI Multi

The TCI Multi is a RF ↔ Ethernet gateway for use with the Crane WrenchStar Multi electronic torque wrenches. It uses the open protocol interface to communicate with the heOGS software. In addition, data output can be enabled to send information out to the data collection package (Sys3xxGateway, QTrans or QualityR).



4.1 Crane TCI Multi configuration

The TCI Multi is configured through the Crane TCI Multi web page.

The sections below show the web pages for Firmware V: 184-2.05 / 162-5.0.

4.1.1 Communication settings

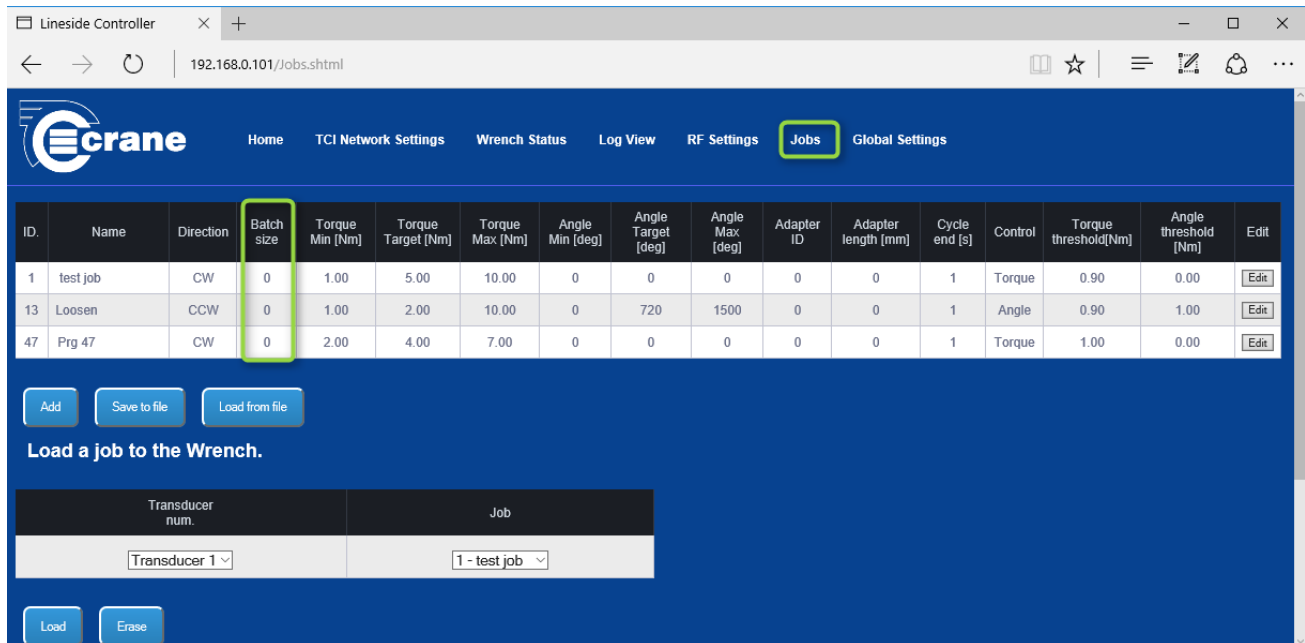
The screenshot shows the Crane TCI Multi web interface. The top navigation bar includes links for Home, TCI Network Settings, Wrench Status, Log View, RF Settings, Jobs, and Global Settings. The Wrench Status page displays a table with the following data:

Num.	Wrench status	Protocol status	Torque [Nm]	Angle [deg]	Serial No.	Battery Level	Wrench Ver.	Port No.	Settings
1	Paired	Connected	5.47	-	97808	98	182-KR-5.03 / 161-5.0	4545	Setup
2	Not Paired	Not connected	No results	No results	Unknown	Unknown	Unknown	0	Setup

A red arrow points from the 'Setup' button in the first row to the 'Change transducer settings' page. This page includes sections for 'Change transducer settings' and 'Change Open Protocol port'. The 'Change Open Protocol port' section shows a warning: 'Changing this setting will result in reboot of TCI, which will lead to connection drop. Please make sure to refresh your browser afterwards.' The current port is set to 4545.

4.1.2 Tightening program configuration

Performance: set batch count to 0!



ID.	Name	Direction	Batch size	Torque Min [Nm]	Torque Target [Nm]	Torque Max [Nm]	Angle Min [deg]	Angle Target [deg]	Angle Max [deg]	Adapter ID	Adapter length [mm]	Cycle end [s]	Control	Torque threshold[Nm]	Angle threshold [Nm]	Edit
1	test job	CW	0	1.00	5.00	10.00	0	0	0	0	0	1	Torque	0.90	0.00	Edit
13	Loosen	CCW	0	1.00	2.00	10.00	0	720	1500	0	0	1	Angle	0.90	1.00	Edit
47	Prg 47	CW	0	2.00	4.00	7.00	0	0	0	0	0	1	Torque	1.00	0.00	Edit

Buttons: Add, Save to file, Load from file

Load a job to the Wrench.

Transducer num.: Transducer 1 | Job: 1 - test job

Buttons: Load, Erase

4.1.3 Data output configuration

In Firmware V: 184-2.05 / 162-5.0 not yet available.

Data output is implemented through the heOGS ftp client driver, see chapter 15.1.

4.2 heOGS configuration

4.2.1 Driver Setup

Rexroth Crane TCI Multi uses the OPConn.dll tool driver. To enable this driver, please make sure to setup the section [TOOL_DLL] as follows:

```
[TOOL_DLL]
OPConn.dll=1
;MBLightCli.dll=1
;EXACConnecT.dll=1
;heGWK.DLL=1
;ProfiFTP.dll=1
```

4.2.2 Driver Configuration

The heOGS tool driver is configured in the station.ini file as follows (make sure to set the ..._TYPE parameter to CRANE):

```
[OPENPROTO]
;
; _____
; Global Open Protocol parameters
; Port number
PORT=4545
; Poll cycle time in ms
Tact=10

;
; _____
; Configuration for Tool 1 (Channel 01)
; Define the IP address of the tool
CHANNEL_01=10.10.2.160
; Define the tool type (allowed values are NEXO, CS351, CRANE)
CHANNEL_01_TYPE=CRANE
; Define the program number used for manual acknowledge (not
; practically applicable for Crane wrenches).
; CHANNEL_01_MANUAL_OPFRG=98

; optional settings:
; Port number used for tool communication (default 4545) or
; port number set through PORT parameter
CHANNEL_01_PORT=4545

;
; _____
; additional tools using OpenProtocol
;CHANNEL_02_...
```

5 GWK electronic torque wrench

The GWK Operator+ is an electronic torque wrench with builtin WiFi communications. It uses a special communication protocol to communicate with the heOGS software. It does not provide its own data output, but rundown results and curves can be transmitted by heOGS to the data collection package (Sys3xxGateway, QTrans or QualityR).



The device features an on-screen display to view enable state and rundown information.

5.1 GWK configuration

The tool settings are configured through a programming cable and a Windows based configuration software (ConnectBlue/uBlox "Serial Port Monitor").

5.1.1 Communication settings

Use uBlox to configure the tool. Note that the configuration differs between firmware versions. See the GWK reference manual for more info.

Basically the following settings are required to communicate with the heOGS software:

- Enable "TCP Server" mode
- Set the "listening port number" to the same number as defined in the heOGS settings.ini file (see below)

5.1.2 Tightening program configuration

Tightening programs are defined in station.ini and downloaded to the tool. See below for more info.

5.1.3 Data output configuration

The GWK tool is not able to directly send out runtime result data to an FTP server. However OPG provides this feature – please see chapter 15.1 for more info.

5.2 heOGS configuration

5.2.1 Driver Setup

GWK tools use the heGWK.dll tool driver. To enable this driver, please make sure to setup the section [TOOL_DLL] as follows:

```
[TOOL_DLL]
OPConn.dll=1
;MBLightCli.dll=1
;EXAConnect.dll=1
heGWK.DLL=1
;ProfiFTP.dll=1
```

5.2.2 Driver Configuration

The heOGS tool driver is configured in the station.ini file as follows:

```
[GWK]

; serial connection
;CHANNEL_04=COM11 9600,n,8|halfduplex
; TCP connection
;CHANNEL_12=10.23.232.105

CHANNEL_03=10.10.2.170
CHANNEL_03_PORT=4002

;Tightening parameter set
; defaults: Target=T; CW; Loosen=0.0;
PRG_01 = name=Prg1; Target=A; Snug=1.0; Loosen=12.0; A=45; A-=20.00; A+=60; T-=2.5; T+=10.2; CW; Curve;
PRG_02 = name=GETRIEBEKOPF_2; Target=T; Snug=0.5; A-=50; A+=1000; T-=1; T=2; T+=3; CW;
PRG_03 = name=GETRIEBEKOPF_3; Target=A; CW; Snug=5.0; A=60; A-=40; A+=80.00; T-=6.0; T+=10.0; Loosen=12; Curve;
PRG_04 = name=GETRIEBEKOPF_4; Target=A; CW; Snug=5.0; A=60; A-=40; A+=80.00; T-=6.0; T+=10.0; Loosen=12; Curve;
```

Note that the settings.ini also configured the tightening parameters for the tool. The following parameters are available:

- name: Descriptive name, will be shown on the tool when a tightening program is started
- Target: Defines the tightening mode. Allowed values are:
 - A: Angle controlled mode. Target value is angle. Note that in this mode T= (i.e. the target torque parameter) **must** be set to zero.
 - T: Torque controlled mode: Target value is torque. Note that in this mode A= (i.e. the target angle parameter) **must** be set to zero.
- Snug: Snug torque to start angle counting
- Loosen: Torque threshold to detect a loosen operation
- CW or CCW: Intended operation direction.
- A-, A+: lower and upper angle limit for OK-status
- T-, T+: lower and upper torque limit for OK-status
- A, T: Target values for angle, torque. Note that only one of these might be non-zero.
- Curve: If given, then the result angle/torque curve will be collected and sent out through the data channel (if configured).

6 SCS Freedom^{3/4} torque wrenches

The SCS Freedom³ and Freedom⁴ are Wifi connected electronic torque wrenches. They use a special communication protocol to communicate with the heOGS software. It does not provide its own data output, but rundown results and curves can be transmitted by heOGS to the data collection package (Sys3xxGateway, QTrans or QualityR).

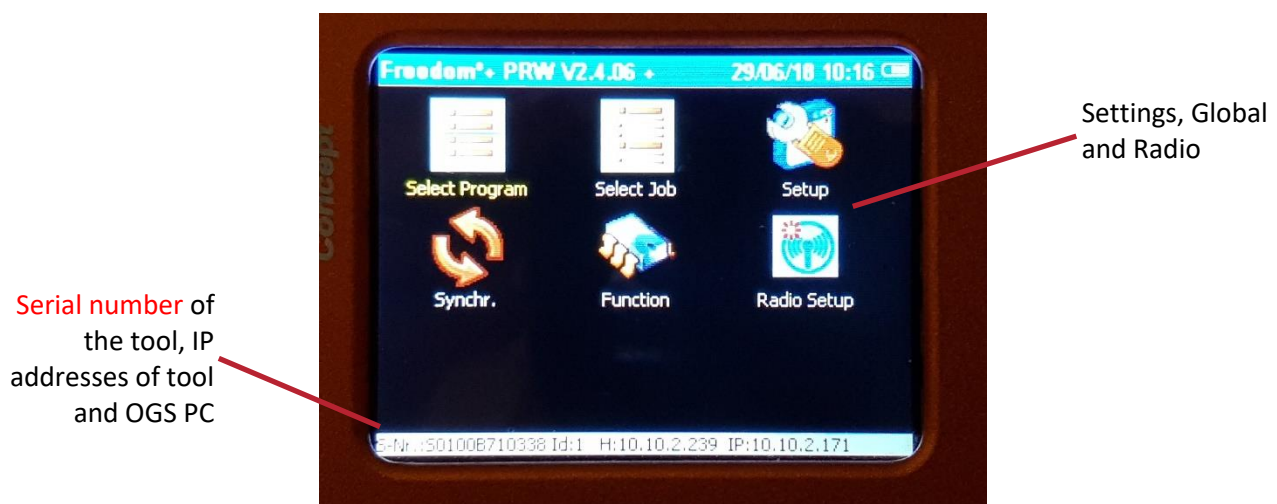
The device features an on-screen touch display to access status and parameters.



6.1 Wrench configuration

All settings can be edited through the on-screen display of the device.

Important: Communication with the SCS tool is linked by the tool serial number (see screenshot below). Changing tools will require not only changing the Wifi parameters, but also the heOPG settings (change tool serial number in settings.ini)!



6.1.1 Communication settings

Global settings (Menu "Setup", default password 1247):

- Server-Client Mode: CLIENT
- Must work online: checked
- Radio Type: Wifi
- Wait NOK Confirm: unchecked
- Comm. Fail strategy: Warn & Disable

Radio settings (Menu "Radio Setup", default password 7421):

- WIFI Host: Enter the IP address of the PC running heOGS
- WIFI Remote Port: Port number for the tool as setup in heOGS settings.ini (see below). By default it is using 3456. Note: even if multiple wrenches are used with a single station, only one port is used.

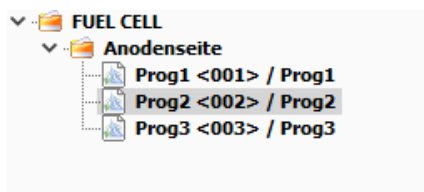
6.1.2 Tightening program configuration

Tightening programs must be defined through the SCS software SQnet+. To do so, a few steps are required:

1. Create a new test definition (each program/pset is represented as test):



2. Select a program/PSet:



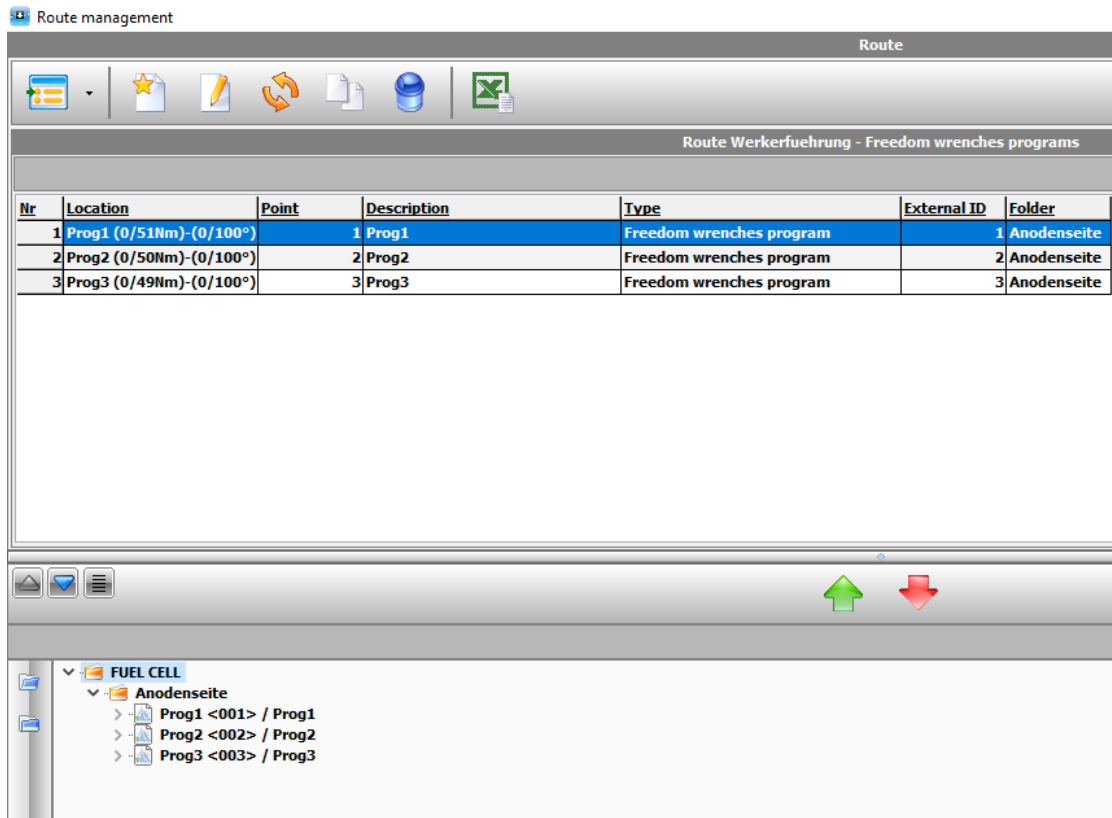
3. Setup the parameters:

Production	
Parameter	Value
Type	Torque with angle monitoring
Station ID	0
Torque	5 Nm
Limits	0/51 Nm
Angle	0,1°
Limits	0/100°
Threshold	2,5 Nm
Socket Nr.	0
External ID	1
Wrench type	1

4. Create a route through the "route management" function:



5. Add the previously configured tests to the list. Note that the PSet/Program number used to select a tightening parameter set selects the line number in the route list (Point and External ID are irrelevant):



Important: make sure to set the programs batch count to 1 when creating programs for the tool!

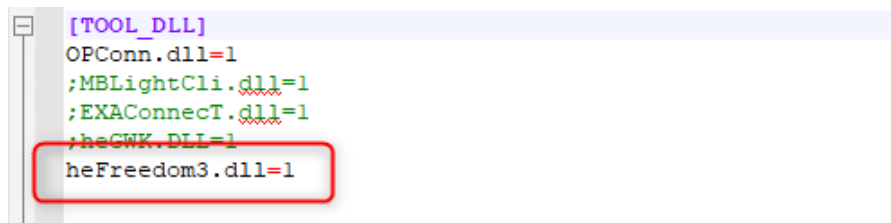
6.1.3 Data output configuration

The SCS tools are not able to directly send out runtime result data to an FTP server. However, OGS provides this feature – please see section 15.1 for more info.

6.2 heOGS configuration

6.2.1 Driver Setup

SCS tools use the heFreedom3.dll tool driver (for both Freedom³ and Freedom⁴). To enable this driver, please make sure to setup the section [TOOL_DLL] as follows:



6.2.2 Driver Configuration

The heOGS tool driver is configured in the station.ini file as follows:

```
[FREEDOM]
; Define the local listening port for SCS communication. Note that
; this setting must be also configured in the tool (Wifi Remote Port).
; The tool communication mode must be set to "CLIENT".
PORT=3456

; Define a SCS Freedom3 tool by assigning a new channel (number 6) and the
; tool serial number.
; More tools can be added by adding more channel_xx=serialno entries.
CHANNEL_06=S0100B710338
```

The tools are identified by their serial numbers (that's how the SCS tools work), so you have to read out the serial number of the device and enter it into the station.ini configuration to map it to a OGS tool number (channel number).

6.2.3 OGS editor tool registration and usage

To use a tool in the OGS editor, you must register it first. To do so, use the tools tab in the editor and add a new tool under the Torque wrenches (GWK) category:

ERGOSPIN				
GWK		Program number	Program name	MODUS
4	SCS 30Nm	1	Prog1	0
6	SCS 50Nm	2	Prog2	0
7	SCS 300Nm	3	Prog3	0

Then add the PSet names and number on the right (so you can choose later when assigning the tool to tasks and operations).

To use the tool, create a new operation and select it. Then choose one of the previously defined PSets:

The screenshot shows the OGS editor interface with a 3D model of a torque wrench. The left panel shows the 'bolting configuration: SCS12345' with a task list (S1, S2). The right panel shows the 'Operation' list with S1 and S2. The bottom panel shows the 'Property' and 'Value' fields for the selected operation, including 'Name', 'Number', 'dest_task', 'Position sensor(PS)', and 'new-torque sen'. The 'Operation' table at the bottom shows the following data:

Operation	T(-)	Torque	T(+)	A(-)
SCS				
Tool	Type	Prq	Socket	
SCS 30Nm	process	1	0 (don't use socket)	
	Prq	DATE/AME	MODUS	
	1		0	
	2		0	
	3		0	

6.3 Firewall configuration

SCS tools use TCP connections and actively connect to the PC (PC is acting as “server”, the SCS tool as “client”). From the PC’s point of view, SCS tools use “incoming” connections which are normally blocked by the Windows firewall.

To allow the SCS tools to communicate with the PC, a firewall exception must be added to the Windows Firewall. The following options are possible:

- Allow all traffic for the Monitor.exe application (e.g. allow OGS runtime to accept all traffic)
- Allow incoming TCP connections for the TCP port used by the SCS tools (see chapter 6.2.2, listening PORT setting)

7 BOSCH EXAConnect II

The EXAConnect II is a Bluetooth ↔ Ethernet gateway for use with the BOSCH BT-EXACT 6 bluetooth battery production tools. It uses the open protocol interface to communicate with the heOGS software. In addition, data output can be enabled to send information out to the data collection package (Sys3xxGateway, QTrans or QualityR).

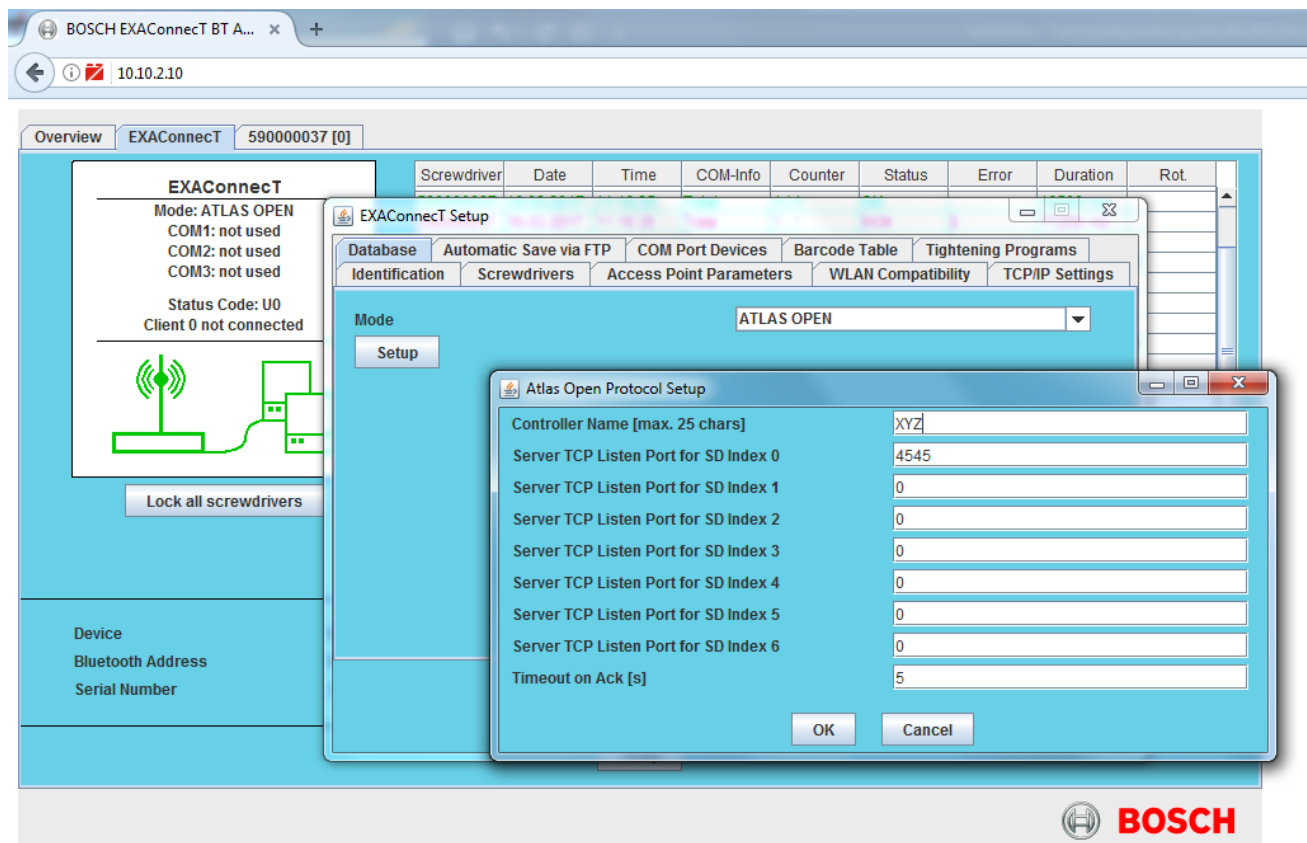
7.1 EXAConnect configuration

The Bluetooth access point and Ethernet gateway is configured through the EXAConnect II web page (java applet).

7.1.1 Communication settings

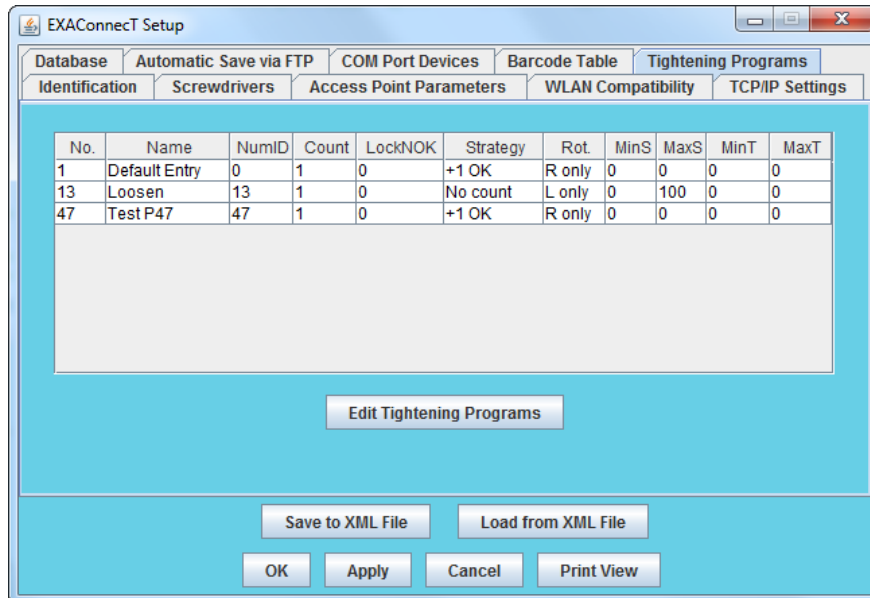
To make heOGS work with the BT-EXACT tools connected to the EXAConnect II, use the following settings:

- Database Mode: ATLAS OPEN
- Mode settings: define a port number for each BT-EXACT tool (4545 is the default port used by heOGS)



7.1.2 Tightening program configuration

Setup the tightening programs according to your requirement in the EXAConnect GUI:



The screenshot shows the 'EXAConnect Setup' window with the 'Tightening Programs' tab selected. It contains a table with the following data:

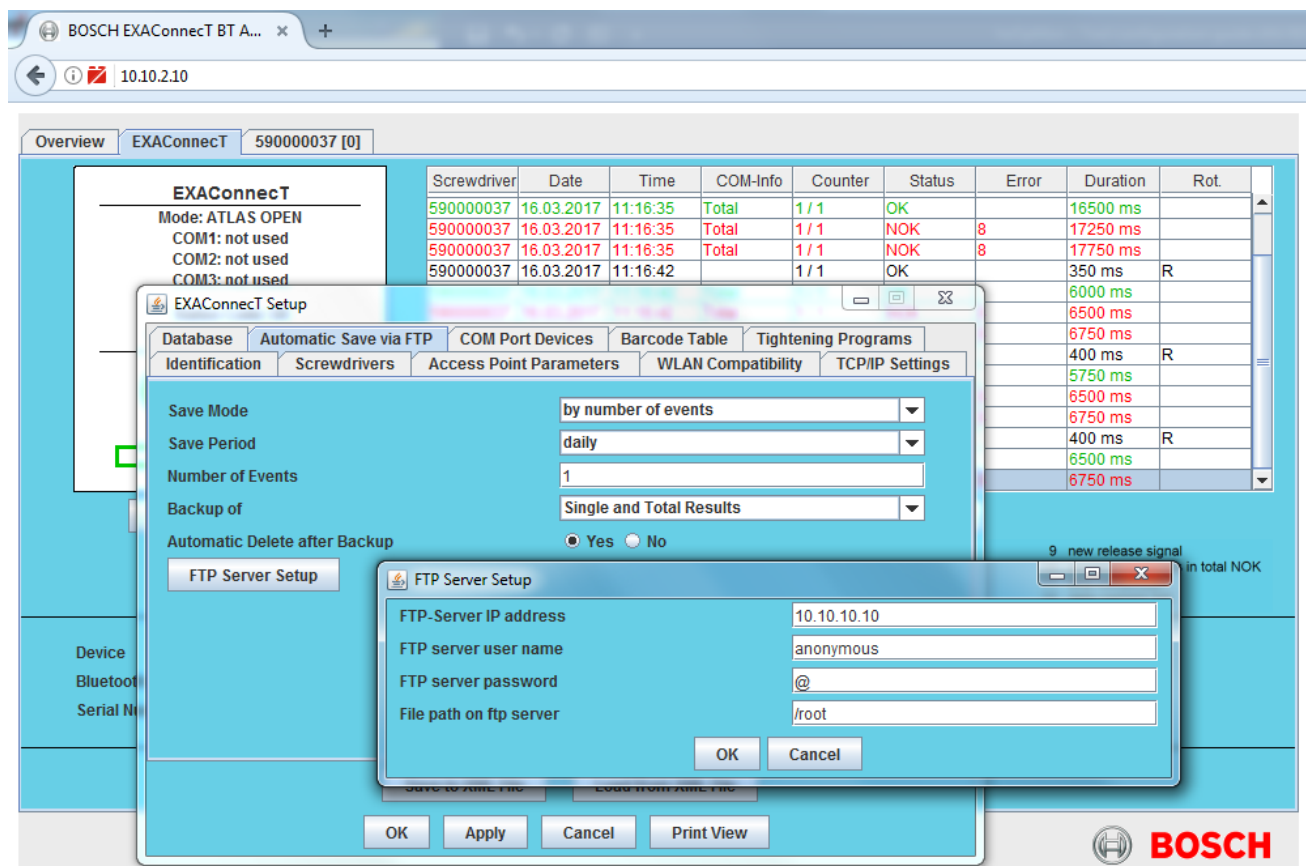
No.	Name	NumID	Count	LockNOK	Strategy	Rot.	MinS	MaxS	MinT	MaxT
1	Default Entry	0	1	0	+1 OK	R only	0	0	0	0
13	Loosen	13	1	0	No count	L only	0	100	0	0
47	Test P47	47	1	0	+1 OK	R only	0	0	0	0

Below the table is an 'Edit Tightening Programs' button. At the bottom of the window are buttons for 'Save to XML File', 'Load from XML File', 'OK', 'Apply', 'Cancel', and 'Print View'.

7.1.3 Data output configuration

Data output is implemented through the heOGS ftp client driver, see chapter 15.1.

To send data out to a central QTrans/Sys3xxGateway/QualityR server, enable the FTP data output of the EXAConnect gateway as follows:



The screenshot shows the EXAConnect GUI with the 'Overview' tab selected. The 'EXAConnect' section shows 'Mode: ATLAS OPEN' and 'COM1: not used', 'COM2: not used', 'COM3: not used'. A table displays data for screwdriver 590000037, including Date, Time, COM-Info, Counter, Status, Error, Duration, and Rot. An 'EXAConnect Setup' window is open, showing the 'FTP Server Setup' tab. The 'FTP Server Setup' window has the following fields:

- FTP-Server IP address: 10.10.10.10
- FTP server user name: anonymous
- FTP server password: @
- File path on ftp server: /root

The 'EXAConnect Setup' window also shows 'Save Mode' set to 'by number of events', 'Save Period' set to 'daily', 'Number of Events' set to '1', and 'Backup of' set to 'Single and Total Results'. The 'Automatic Delete after Backup' option is set to 'Yes'.

7.2 heOGS configuration

7.2.1 Driver Setup

Bosch EXAConnectT II uses the tool driver. To enable this driver, please make sure to setup the section [TOOL_DLL] as follows:

```
[TOOL_DLL]
;OPConn.dll=1
;MBLightCli.dll=1
EXAConnect.dll=1
;heGWK.DLL=1
;ProfiFTP.dll=1
```

7.2.2 Driver Configuration

The heOGS tool driver is configured in the station.ini file as follows:

```
[OPENPROTO]

CHANNEL_01=192.168.0.101
CHANNEL_01_MANUAL_OPFRG=98

; Channel parameter code
; Typical configuration values are:
; 0 (default) - Nexo with text display enabled using ActEn signal
; 3 - Crane WrenchStar Multi with TCI Multi
; 5 - Nexo with text display enabled, using Enable/disable tool.
; 7 - Nexo without text display, using Enable/disable tool.
; CS351 box (no text display)
; NOTES:
; - To go for maximum speed with Nexo, use firmware > V1200SP1 and mode 0/2.
; For Nexo firmware < V1200SP1 consider a firmware update, if this is not
; possible, use mode 6 or 7.
; Note that to make the text disappear you *must* assign InCy to Bit 0.2
; of the PLC table. However, this costs about 1/2 second of cycle time -
; if you do not assign InCy, the last message will stay on screen and must
; be acknowledged manually.
; - For Nexo firmware < V1200SP1: do *not* assign any per-cycle changing
; I/O-signal to the PLC table if you want to have low cycle times.
; Each signal change costs ~500ms.

CHANNEL_01_PARAMS=3
CHANNEL_01_ALIVEXMTT=5000
CHANNEL_01_SHOWALIVE=0
CHANNEL_01_RSPTIMEOUT=10000

PORT=4545
; Poll cycle time in ms
Tact=10
```

8 MSTKN socket trays

The MSTKN socket trays are available with different interfaces. Depending on the interface, the setup is a bit different.

8.1 General

There is a special section in each station.ini to configure socket trays to be used with the Operator Guidance. The configuration of each parameter is described in detail in the following sections.

Please see the screenshot below for an example configuration of the socket tray section. This is a basic configuration for one MSTKN-H-S24-HM-ETH-M12 socket tray.

```
[SocketTray]
ENABLED = 1
Groups = 8
;Map = 1 2 3 4 5 6 7 8
IP=10.10.2.65
PORT=502
```

In this section, the following keys can be used:

- **ENABLED** (int, default = 0): *all types, mandatory*
 Enable or disable the socket tray. If set to non-zero, then the socket tray driver is enabled.
- **Groups** (list of int, default = 4): *all types, mandatory*
 This option allows you to divide the sockets available at one station into groups, so one tool from each group can be removed from the sockets without causing a “false pick” error. This can be helpful in some scenarios, e.g. when two workers are using the same socket tray.
 As grouping works differently for serial trays than for wireless or Ethernet trays, there is a special section for serial trays (8.4.3) and one section is for wireless and Ethernet trays (8.2.3).
- **Map** (list of int, default = 1 2 3 4 (...)) *only Serial, optional*
 This parameter allows remapping socket numbers (e.g. to renumber from right to left). As this option is only available for serial (USB) socket trays, please see the corresponding chapter 0 in the “Serial socket trays (USB)” section.
- **Group_to_Channel** (list of int, default = none): *all types, optional*
 This parameter works in combination with the “Groups” parameter. This allows assigning tool channels to the socket groups that were defined by the “Groups” parameter. By default (missing), a single group is defined automatically and all channels are associated with it.
 As grouping works differently for serial trays than for wireless or Ethernet trays, there is a special section for serial trays (8.4.7) and for Wireless and Ethernet trays (0).
- **IP** (IP address), **PORT** (int): *only Ethernet & wireless, mandatory*
 Only used for wireless and Ethernet trays, see chapter 8.2.3.
 Serial trays (USB) use the connection defined in the [SERIAL] section, see 8.4.3.

8.2 Wireless socket trays

8.2.1 Overview

This is the most basic configuration required to get a MSTKN-H-S14-HM-WLAN socket tray running:

```
[SocketTray]
;socket tray(s) ON / OFF
ENABLED = 1
;groups
Groups = 4
;Ethernet trays use port 502
;IP=10.10.2.65
;PORT=502
;WIFI Tray uses port 5003
IP=10.10.2.88
PORT=5003
```

8.2.2 Notes

Only a single wireless main module (-S14/-S24/-P14/-T18) is allowed. If you need more than 8 sockets, then add (an) extension module(s).

8.2.3 IP address & port number

- **IP** (IP address): Please enter the IP address of the socket tray that you want to use. The IP address of Ethernet socket trays can be read (and changed) with the free tool “Anybus IPconfig” by HMS. The IP address of Wireless socket trays can be read (and changed) with the free “MSTKN WLAN Configurator” available at the Google Play Store.
- **PORT** (int): Please enter the port you want to use for communicating with the socket tray.
Careful: Port 5003 is the only port that will work for wireless socket trays. Assigning any other port number is telling the operator guidance software that it's dealing with an Ethernet socket tray.

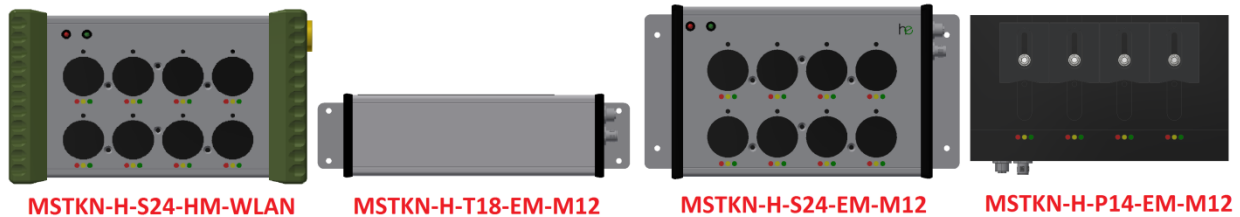
8.2.4 Grouping

Grouping works similarly for all socket tray chains that begin with a main module. All combinations of groups are possible. Just check how many sockets are connected to the Ethernet or Wireless module and split them up as you wish.

Please see the following examples to understand how the sockets of this chain of MSTKN could be grouped.

Ethernet and Wireless modules are exchanged in the examples as it makes no difference which main module is used – at least in terms of how the grouping feature works.

Example MSTKN chain:



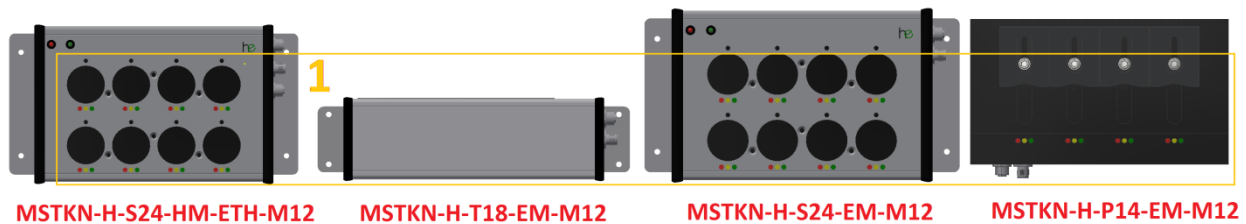
Syntax:

Groups = <no. of sockets grp 1> | <no. of sockets grp 2> | <no. of sockets grp 3> ...

Example 1:

Groups = 28 (or not defined at all)

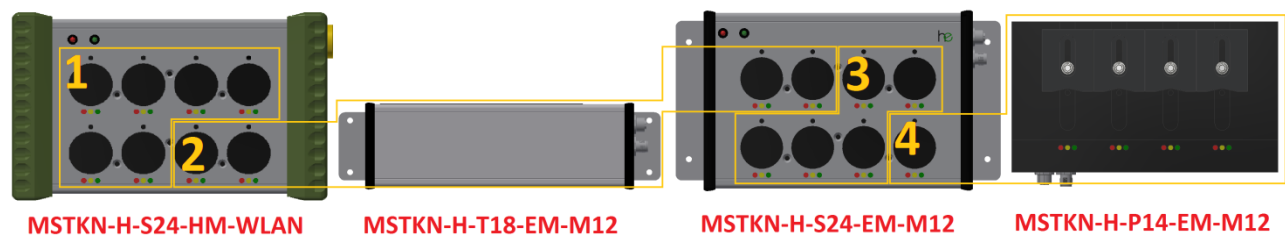
OK, One group with 28 sockets



Example 2:

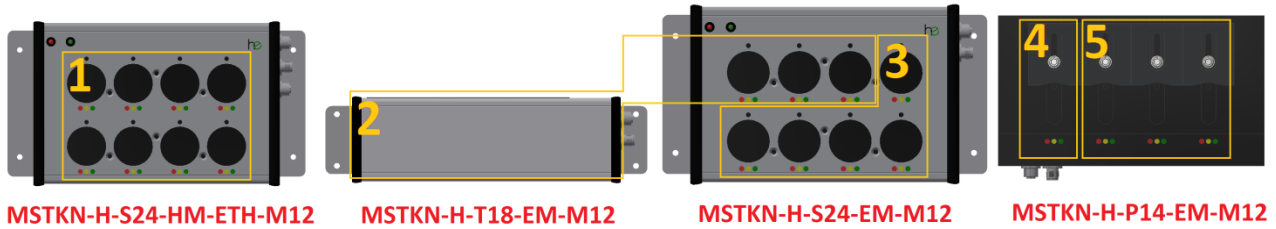
Groups = 6 | 12 | 5 | 5

OK, Four groups with 6, 12, 5 and 5 sockets



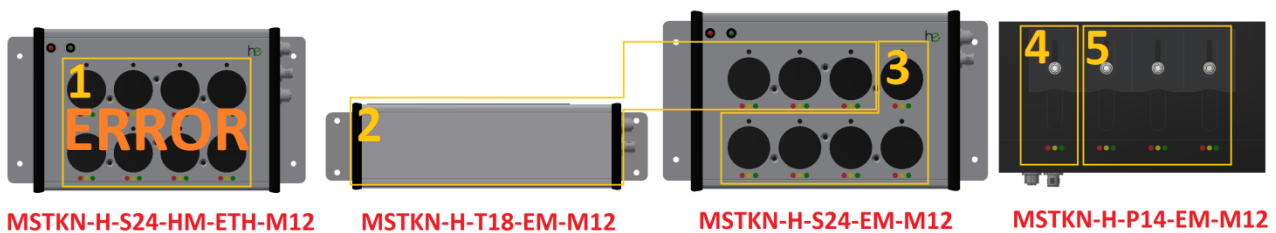
Example 3:

Groups = 8 | 11 | 5 | 1 | 3 **OK, Five groups with 8, 11, 5, 1 and 3 sockets**



Example 4:

Groups = 4 4 | 11 | 5 | 1 | 3 **ERROR, it makes no sense to use this syntax for non-serial socket trays.
Syntax should be like in Example 3. (Only 1 number per group)**

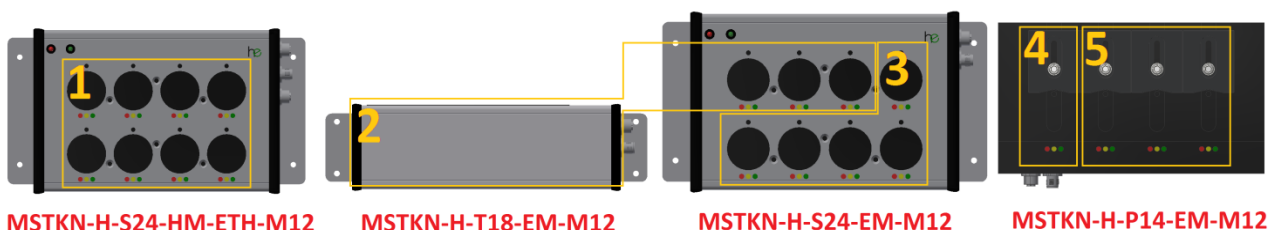


8.2.5 Group to Channel

The “Group_to_Channel” parameter allows you to assign tools to the socket groups that you created. The syntax is similar to the “Group” syntax. Just list all the numbers of the channels that you want to assign to group 1 with a space in between, then write a | and start listing the channel numbers for group 2 etc. Leading zeros in the channel numbers can be left out.

Please see the following example to understand the syntax:

This is the group assignment from the last “Grouping” example:



Group_to_Channel = 4 | 1 5 | 2 | 3 | 6

This “Group_to_Channel” assignment assigns the tool channels to the groups as follows:

Group	1	2	3	4	5
Tool Channel(s)	4	1 & 5	2	3	6

8.3 Ethernet socket trays

8.3.1 Overview

This is the most basic configuration required to get a MSTKN-H-S14-HM-ETH-M12 socket tray running:

```
[SocketTray]
;socket tray(s) ON / OFF
ENABLED = 1
;groups
Groups = 4
;Ethernet trays use port 502
IP=10.10.2.65
PORT=502
;WIFI Tray uses port 5003
;IP=10.10.2.88
;PORT=5003
```

8.3.2 Notes

Only a single Ethernet socket tray (-S14/-S24/-P14/-T18) is allowed. If you need more than 8 sockets, then add an extension module.

8.3.3 Grouping

Grouping works similarly for Ethernet and Wireless socket trays. Please see section 8.2.4.

8.3.4 Group to Channel

The Group_to_Channel parameter works similarly for Ethernet and Wireless socket trays. Please see section 8.2.5.

8.4 Serial socket trays (USB)

When configuring a serial socket tray, the procedure slightly differs from the Ethernet or Wireless tray configuration.

8.4.1 Overview

This is the most basic configuration required to get a MSTKN-H-S14-EM-M12 socket tray running:

```
[SERIAL]
;Profibus    =    COM4 187500,e,8|halbduplex
;Barcode     =    COM4 115200,n,8|halbduplex
SocketTray  =    COM8 115200,e,8|halbduplex
;Barcode_Rfid =    COM41 9600,n,8
```

```
[SocketTray]
;socket tray(s) ON / OFF
ENABLED = 1
;groups
Groups = 4
```

8.4.2 Notes

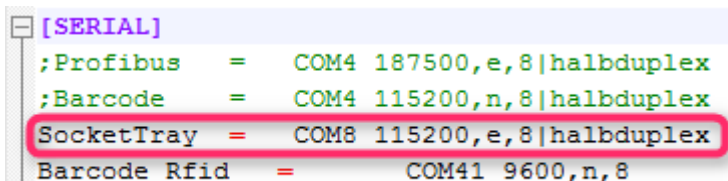
The serial trays require two settings in station.ini and the installation of a special USB driver:

- Configure the serial port under [SERIAL] (see 8.4.7)
- Configure the settings under [SocketTray] (see 8.1)
- Install the FTDI driver. The serial trays are connected through the H+E “USB -> M12 Programming Cable” (order no. 222252) which requires a driver from <http://www.ftdichip.com/>. The driver is also contained on the CD that came with your socket tray.

WARNING: If using an 8-way tray (type -S24-), then it needs to be configured as two trays having 4 sockets each (see below on how to configure multiple trays).

8.4.3 Serial parameters

The serial socket trays are connected via a special USB cable. The serial communication parameters need to be defined in the [SERIAL] section of the station.ini file. Please see the screenshot below for an example configuration.



```
[SERIAL]
;Profibus    =    COM4 187500,e,8|halbduplex
;Barcode     =    COM4 115200,n,8|halbduplex
SocketTray  =    COM8 115200,e,8|halbduplex
Barcode_Rfid =    COM41 9600,n,8
```

- **COM8** Number of the serial port
- **115200** Baud rate
- **e** Parity (e=even, n=none)
- **8** Data Bits
- **|halbduplex** Half Duplex communication, completely leave out this parameter for full duplex

8.4.4 Configuring multiple trays

When more than 8 sockets are needed on a station, it is necessary to chain two (or more) serial trays together. To have several serial MSTKN work in a chain, you need to configure their internal addresses to tell the master (Monitor software) which socket tray is which. Please follow the steps below to make sure everything is configured correctly.

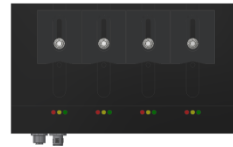
- Picture the MSTKN chain that you want to create. For example (left to right):



MSTKN-H-S24-EM-M12



MSTKN-H-S14-EM-M12

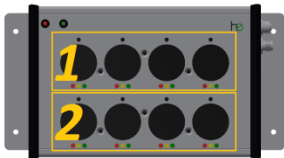


MSTKN-H-P14-EM-M12

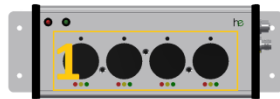


MSTKN-H-T18-EM-M12

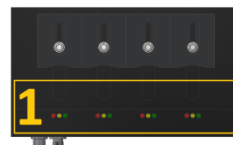
- Every sensor board in these devices has an internal address that tells the master with which board it is communicating right now. When the socket trays are shipped, they are configured to be used standalone and out-of-the-box, so their board addresses will most likely be assigned like this:



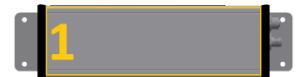
MSTKN-H-S24-EM-M12



MSTKN-H-S14-EM-M12



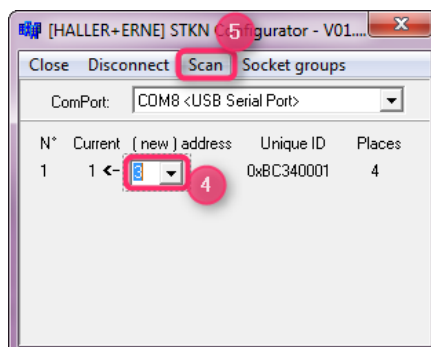
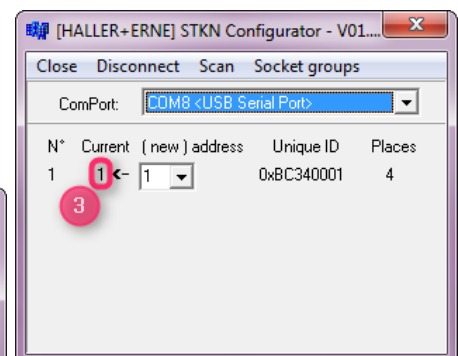
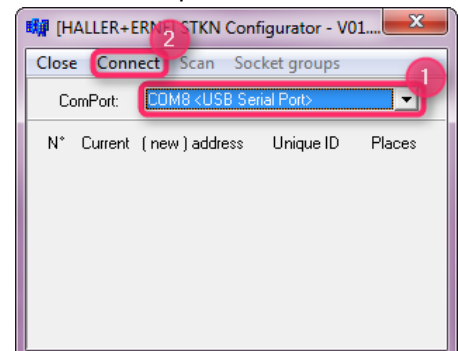
MSTKN-H-P14-EM-M12



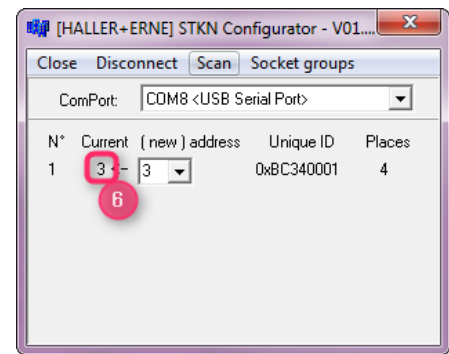
MSTKN-H-T18-EM-M12

- To read and change the board addresses, please connect a socket tray to a computer using the H+E “USB -> M12 Programming Cable” (order no. 222252) and start the “MSTKN-HM-Config.exe” that can be found on the CD that came with your socket trays and follow these steps:

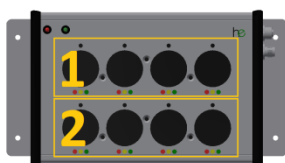
- 1.) Set the correct COM-Port used by your “USB -> M12 Programming Cable”
- 2.) Click “Connect”
- 3.) The address in the column “Current” is the one that is currently active on your MSTKN. In this example, the address read is “1”.
- 4.) If you wish to change the address, simply choose the desired address from the dropdown that says “(new) address”
- 5.) Click “Scan” to write the new address and read the new (changed) address from the socket tray.



- 6.) The new address (3) has been correctly adapted by the socket tray.



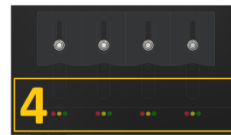
- To configure the example MSTKN chain correctly, this procedure would have to be repeated until the devices have the following addresses:



MSTKN-H-S24-EM-M12



MSTKN-H-S14-EM-M12



MSTKN-H-P14-EM-M12



MSTKN-H-T18-EM-M12

- Now the chain of serial socket trays is ready to use with Monitor. The (default) addresses of their sockets would now be:



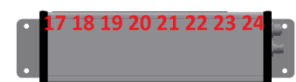
MSTKN-H-S24-EM-M12



MSTKN-H-S14-EM-M12



MSTKN-H-P14-EM-M12



MSTKN-H-T18-EM-M12

8.4.5 Changing the socket mapping

The configuration in the station.ini makes it possible to change the mapping of the sockets of serial trays.

The screenshots below show a standard configuration for a MSTKN-H-S24-EM-M12 module. Please note the parameter "Map = 1 2 3 4 5 6 7 8" on the left hand side which makes it possible to map a different number to each socket. The default configuration is 1 2 3 4 5 6 7 8. This is the standard mapping that is used, even if the Map parameter is uncommented and the MSTKN sockets are assigned as in the picture to the right. Any mapping is possible, as long as all available socket numbers are in the list and you put a space between every

```
[SERIAL]
;Profibus    = COM4 187500,e,8|halbduplex
;Barcode     = COM4 115200,n,8|halbduplex
SocketTray  = COM8 115200,e,8|halbduplex
;Barcode_Rfid = COM41 9600,n,8

[SocketTray]
ENABLED = 1
Groups  = 4 4
Map      = 1 2 3 4 5 6 7 8
```



number! E.g.: If you configure "Map = 3 1 2 4" for a tray with 8 sockets, the software will not recognize your settings and use the default mapping.

8.4.6 Grouping

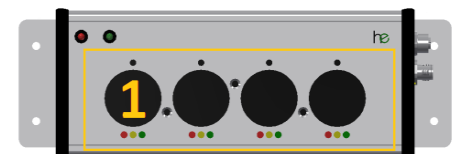
Serial trays: For chains of serial trays of types S14 and P14 it is only possible to divide the available sockets into groups of 4. The S24 trays have to be assigned as if they were two S14 trays with 4 sockets each. The T18 trays are the only ones that cannot be logically divided into groups of 4. They have to be seen as a unit of 8 sockets. Writing a | between two numbers divides them into two groups.

Please see the following examples to understand the syntax:

Example 1:

Groups = 4

One S14 (1x4 sockets) as 1 group of 4 sockets

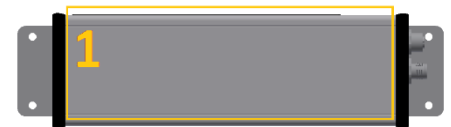


MSTKN-H-S14-EM-M12

Example 2:

Groups = 8

One T18 (1x8 sockets) as 1 group of 8 sockets

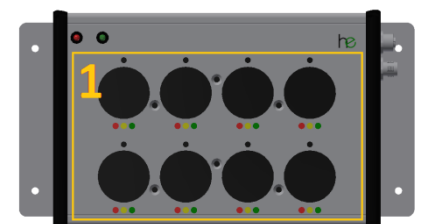


MSTKN-H-T18-EM-M12

Example 3:

Groups = 4 4

One S24 (2x4 sockets) as 1 group of 8 sockets

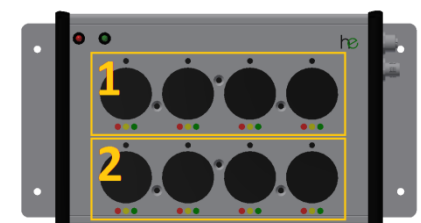


MSTKN-H-S24-EM-M12

Example 4:

Groups = 4 | 4

One S24 (2x4 sockets) as 2 groups of 4 sockets each

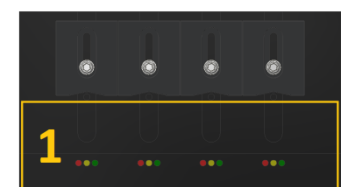


MSTKN-H-S24-EM-M12

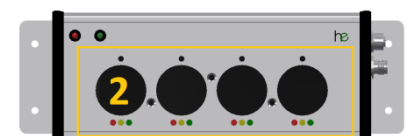
OR

Groups = 4 | 4

Two S14 or P14 (2x4 sockets) as 2 groups of 4 sockets each



MSTKN-H-P14-EM-M12



MSTKN-H-S14-EM-M12

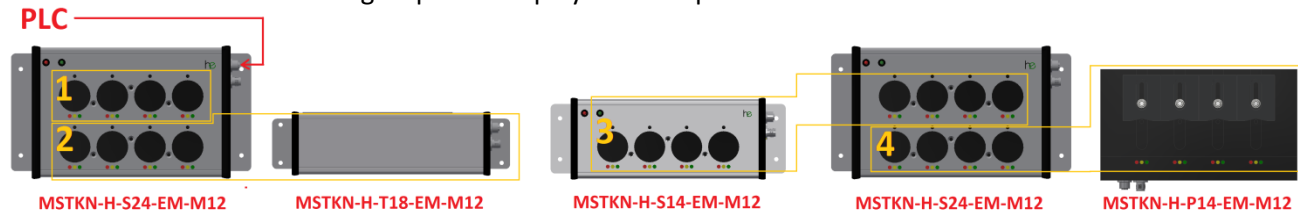
Final Example:



This chain of 5 different types of MSTKN gets the following group assignment:

Group = 4 | 4 8 | 4 4 | 4 4

The devices' sockets are now grouped as displayed in the picture below.

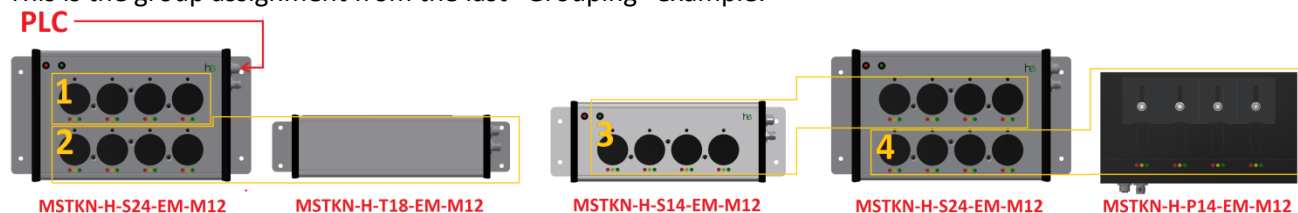


8.4.7 Group to Channel

The "Group_to_Channel" parameter allows you to assign tools to the socket groups that you created. The syntax is similar to the "Group" syntax. Just list all the numbers of the channels that you want to assign to group 1 with a space in between, then write a | and start listing the channel numbers for group 2 etc. Leading zeros in the channel numbers can be left out.

Please see the following example to understand the syntax:

This is the group assignment from the last "Grouping" example:



Group_to_Channel = 1 | 3 4 | 2 | 5

This "Group_to_Channel" assignment assigns the tool channels to the groups as follows:

Group	1	2	3	4
Tool Channel(s)	1	3 & 4	2	5

9 Manual Operation

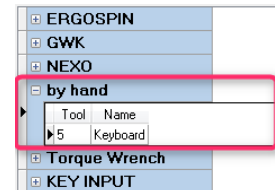
Currently, three different manual input/acknowledge tools available. The following sections describe how to setup and use them.

9.1 GUI popup

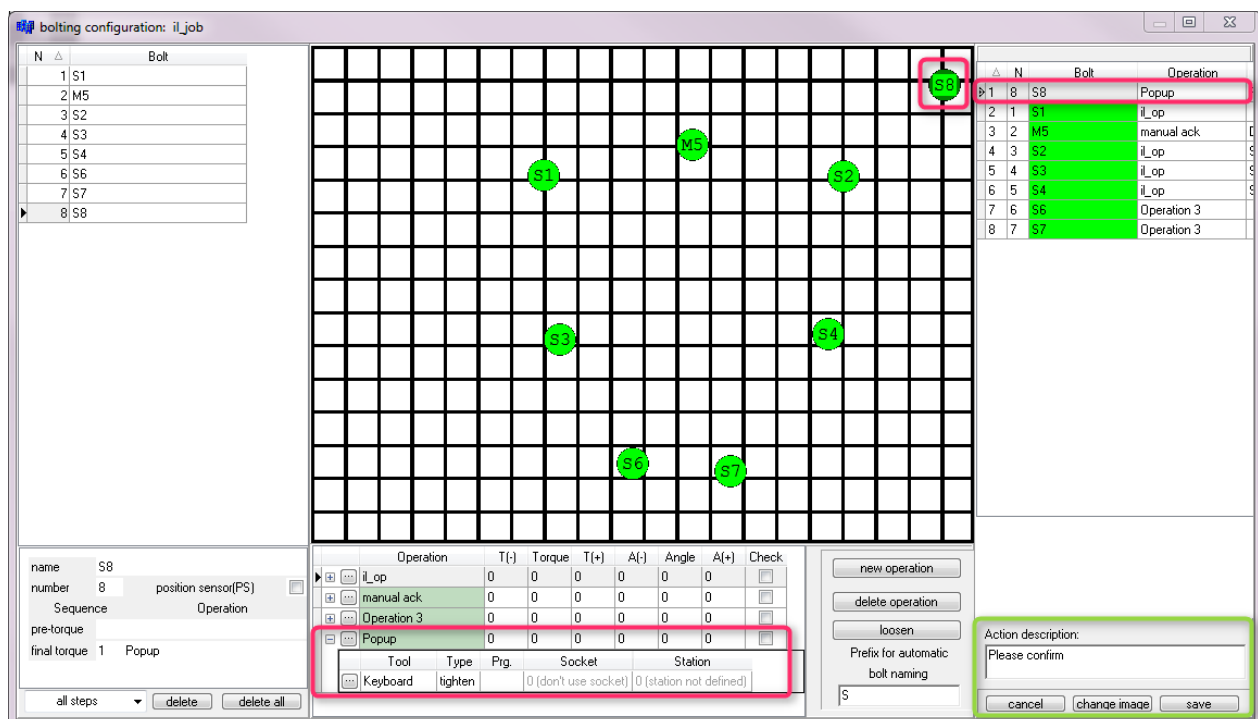
Shows a popup with custom text and waits for manual user input. With this tool type, a mouse or touchscreen is needed to click the “OK” button in the popup menu.

9.1.1 Configure GUI popup

Open the tab “Tools” and add a new tool with type “by hand”. This tool can be used in the job/operation configuration exactly as other tools like NEXO, Torque Wrench etc.

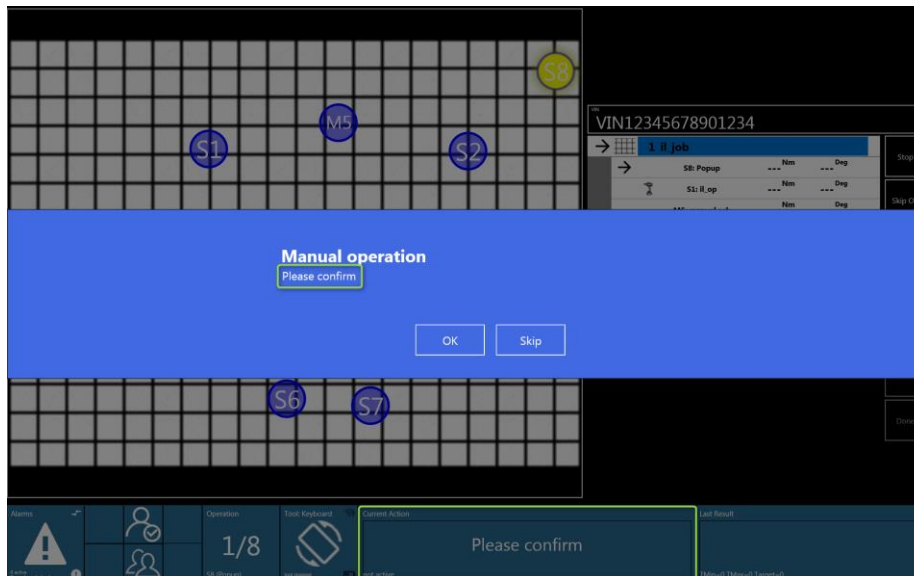


In the second step, this tool can be used in a job/operation definition (see screenshot below). Please note the bottom right window “Action description” (green box in screenshot below): Here, the custom message to be shown can be configured. This message will be displayed in the popup itself and in the bottom of the GUI.



9.1.2 Display of GUI popup

The message entered as “Action description” will be displayed in both the popup and the “Current Action” field in the GUI:



9.1.3 Data output configuration

The “GUI popup” tool is not able to directly send out runtime result data to an FTP server. However OPG provides this feature – please see section... for more info.

Data output is implemented through the heOGS ftp client driver, see chapter 15.1.

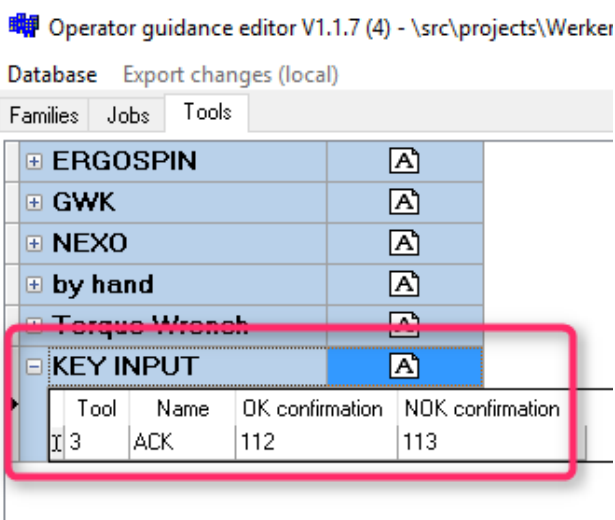
9.2 KEY input

Shows a custom text at the bottom of the screen and waits for manual user input. With this tool type, either a mouse, a keyboard or a touchscreen is needed to click the “OK” button in the popup menu.

9.2.1 Check Database

Please check if your database contains a tool type “KEY INPUT”:

- Open configuration database in “heOpCfg.exe”.
- Open the tab ‘Tools’ and check a tool type “KEY INPUT”:

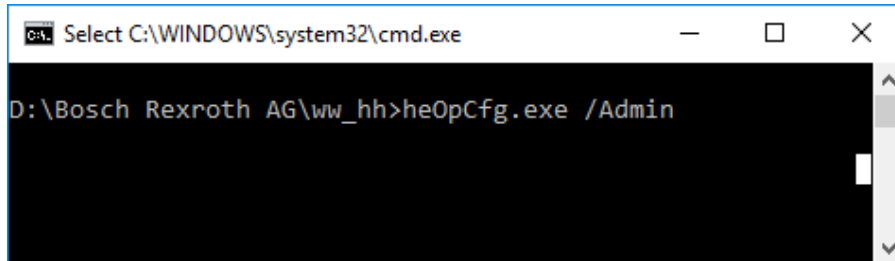


- If this type is already available, please skip the following section (9.2.2).

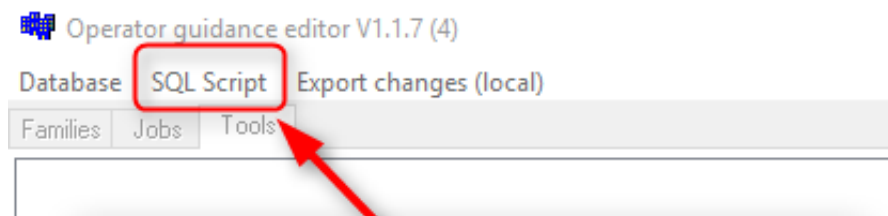
9.2.2 Update Database

WARNING!!! To update the configuration database for using of tool type “KEY INPUT” You need a “heOpCfg.exe” equal or greater than version 1.1.7-4 and an SQL script “Manual acknowledge.sql”.

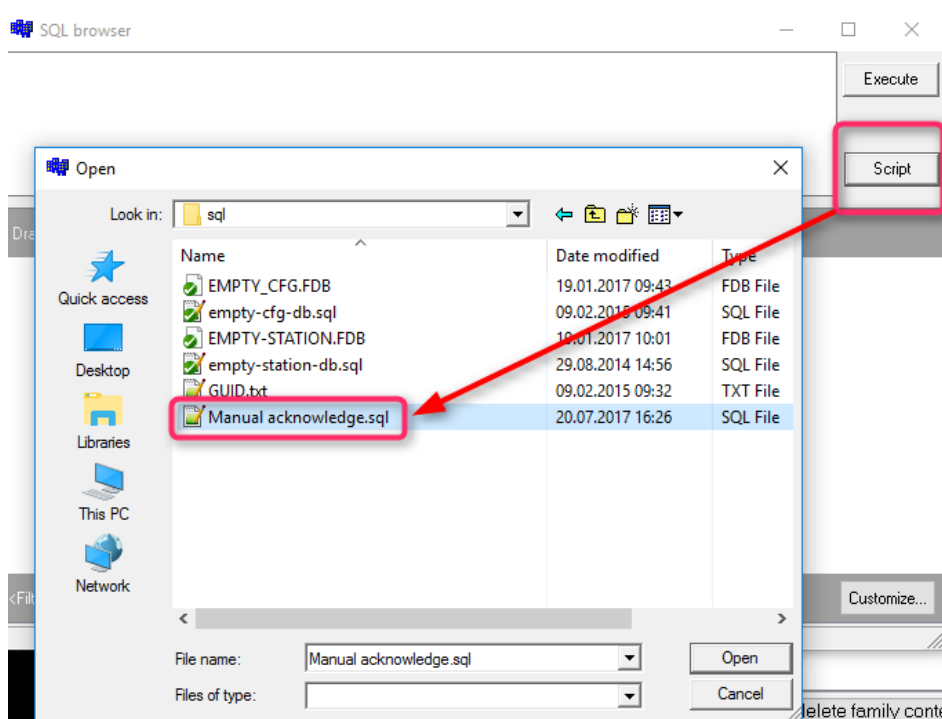
- Start “heOpCfg.exe” from the working directory with parameter /Admin:



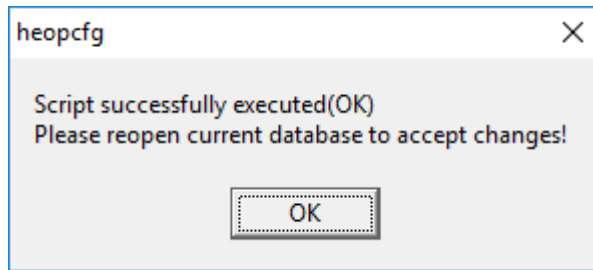
- Open database and select 'SQL Script' in main menu. This menu Item is available only after starting “heOpCfg.exe” with parameter /Admin:



- Press the 'Script' button and select “Manual acknowledge.sql” file in Open Dialog:



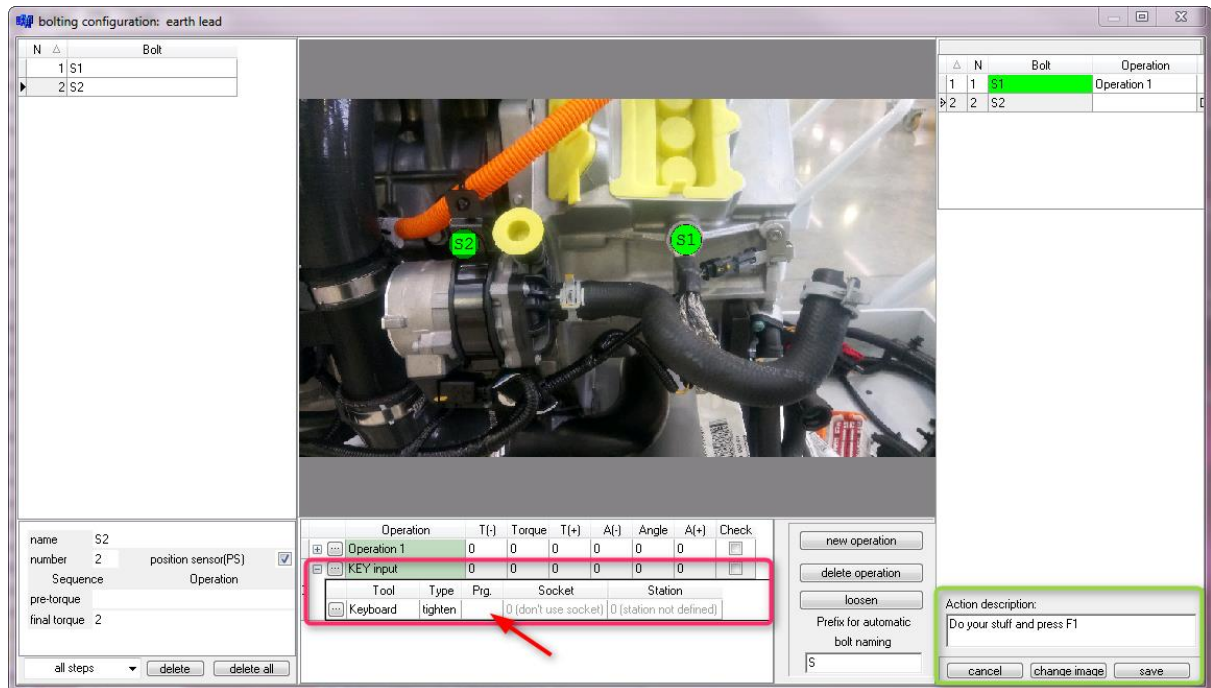
- After successful script execution, the following message appears:



- To accept the changes, please close the application and start it again without “/Admin” parameter.

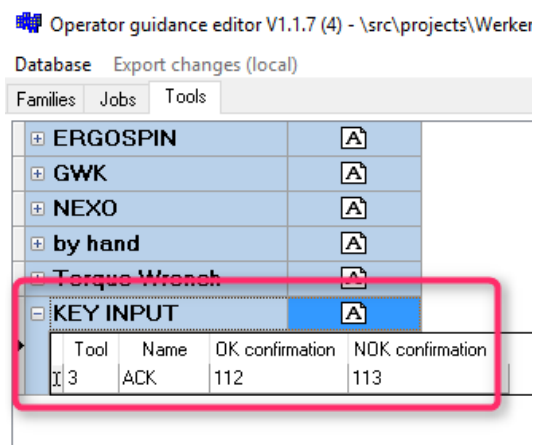
9.2.3 Configure KEY input

- Open the tab “Jobs” and add the “KEY input” tool type to an operation. You can configure a custom message to be displayed in the GUI when configuring this tool (green box). This tool can be used for job configuration exactly as other conventional tools like NEXO, Torque Wrench or Ergospin, but without the input of a program number (this field has to be empty), like in the sample below (red box):



The screenshot below shows an example configuration for the “KEY input” tool.

- Tool (Channel) number: 3
- Tool name: ACK
- OK Code: 112 (key code for F1 on keyboard)
- NOK Code: 113 (key code for F2 on keyboard)



Some key codes are:

Key	Code	Key	Code	Key	Code
F3	114	F4	115	F5	116
ESC	27	Enter (Return)	13	Space	32

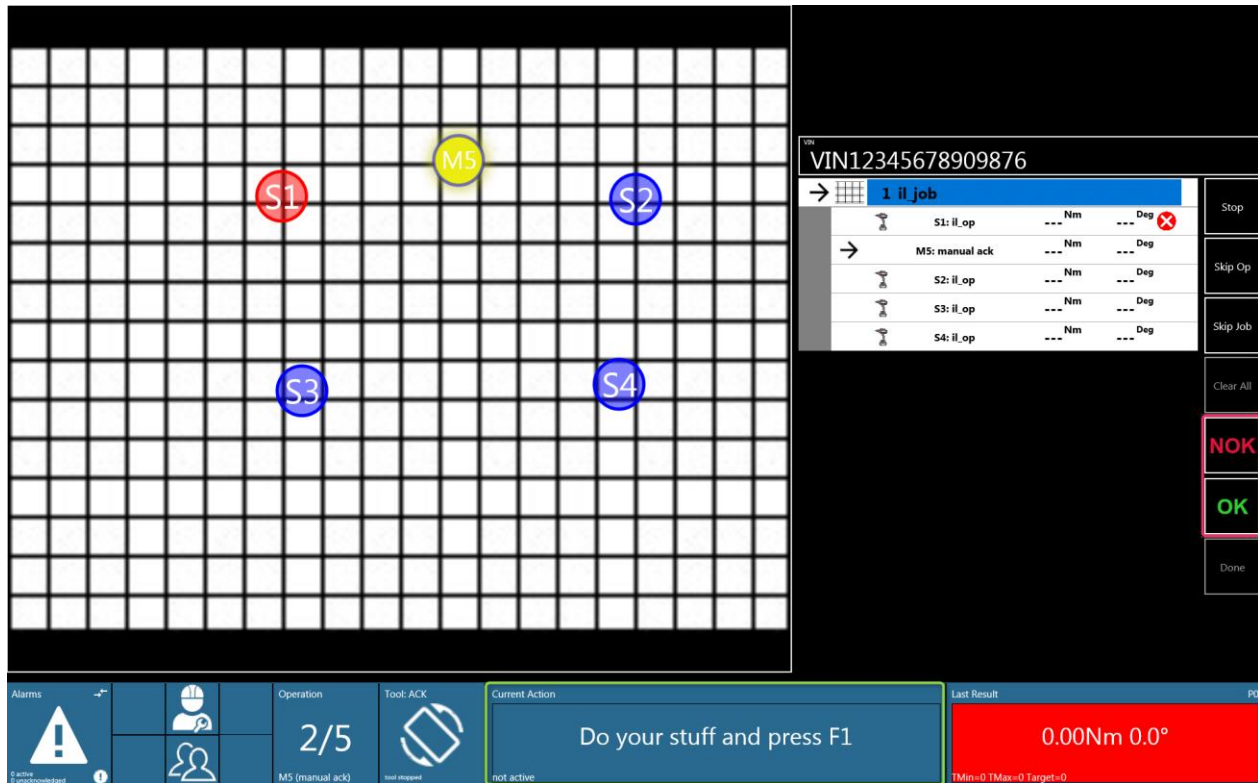
For a full list of key codes please check:

<https://www.cambiaresearch.com/articles/15/javascript-char-codes-key-codes>

9.2.4 Display of KEY input

The screenshot below shows how a “KEY input” tool type is displayed in the GUI.

If a mouse or touchscreen is available, it is also possible to use the buttons on the right hand side to input OK or NOK (red box). Please also note the custom message displayed at the bottom of the screen (green box):



9.2.5 Data output configuration

The “GUI popup” tool is not able to directly send out runtime result data to an FTP server. However OPG provides this feature – please see section... for more info.

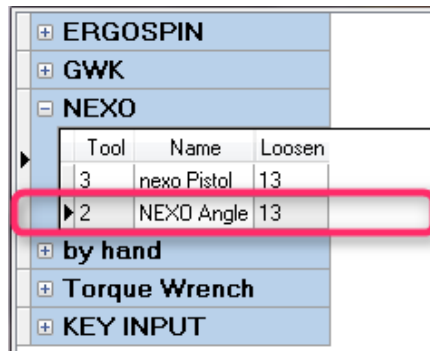
Data output is implemented through the heOGS ftp client driver, see chapter 15.1.

9.3 TOOL acknowledge

Shows a popup with custom text and waits for manual user input. This tool type is similar to the “GUI popup”, except you don’t need a mouse or touchscreen to confirm the operation, because a button on the worker’s tool (like Bosch Rexroth Nexo) can be configured as “OK” button. The worker simply has to hold this button down until the rundown is accepted as “OK”.

9.3.1 Configure TOOL acknowledge

- Open the tab “Jobs” and add a new job with the tool type for which you want to configure the tool acknowledge operation. For this example, we will use a Rexroth Nexo.



- Open the station.ini and find the [GENERAL] section. Here can you disable the popup for the TOOL acknowledge operation. Setting the “DISABLE_MANUALOP_POPUP=” to “1” will disable the popup, setting it to “0” will enable the popup.

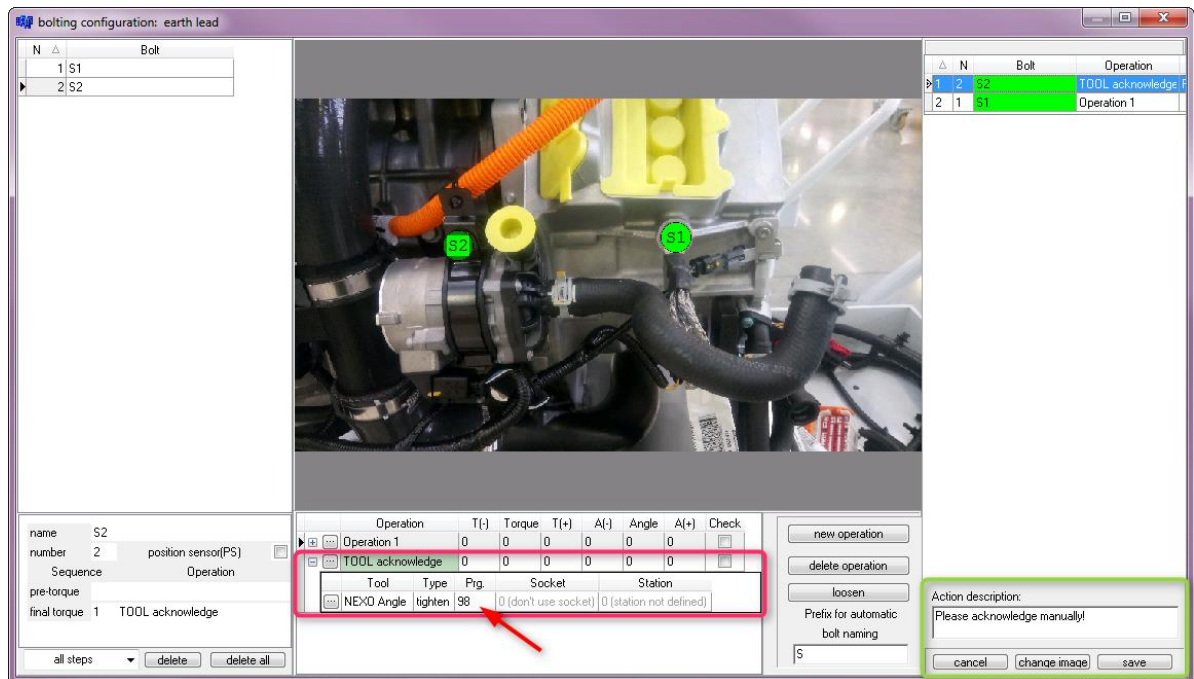
```
[GENERAL]
Version=101
; Show popup for manual operations or not
; =1 - Do not show popups. This can be overridden on a per operation base by prefixing
;       the action text with a plus sign ('+').
; =0 - (default) show the popup.
DISABLE_MANUALOP_POPUP=1
```

- Then scroll down in the station.ini and find the [OPENPROTO] section. Then find the channel of the tool that you want to configure. In this example, the Rexroth Nexo that we want to configure is on channel 02. Now you can select a program number which will turn the operation into a “TOOL acknowledge” when used. Simply add the line:

CHANNEL_02_MANUAL_OPFRG=<program number>

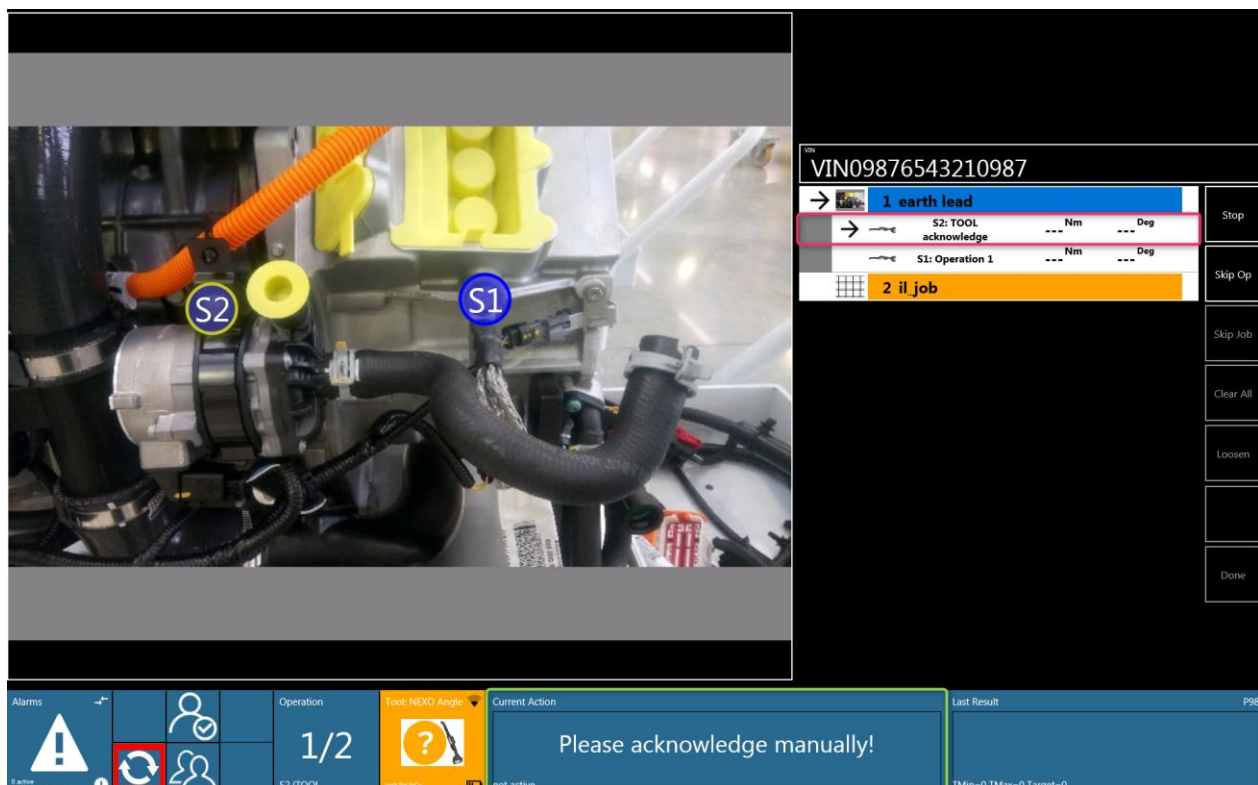
```
[OPENPROTO]
; Example NEXO config entry
CHANNEL_02=10.10.2.160
CHANNEL_02_MANUAL_OPFRG=98
PORT=4545
; Poll cycle time in ms
Tact=10
```

- After selecting the desired tool in the HeOpCfg.exe's "Jobs" tab, please input the channel number that you specified in the station.ini before. (e.g. 98)

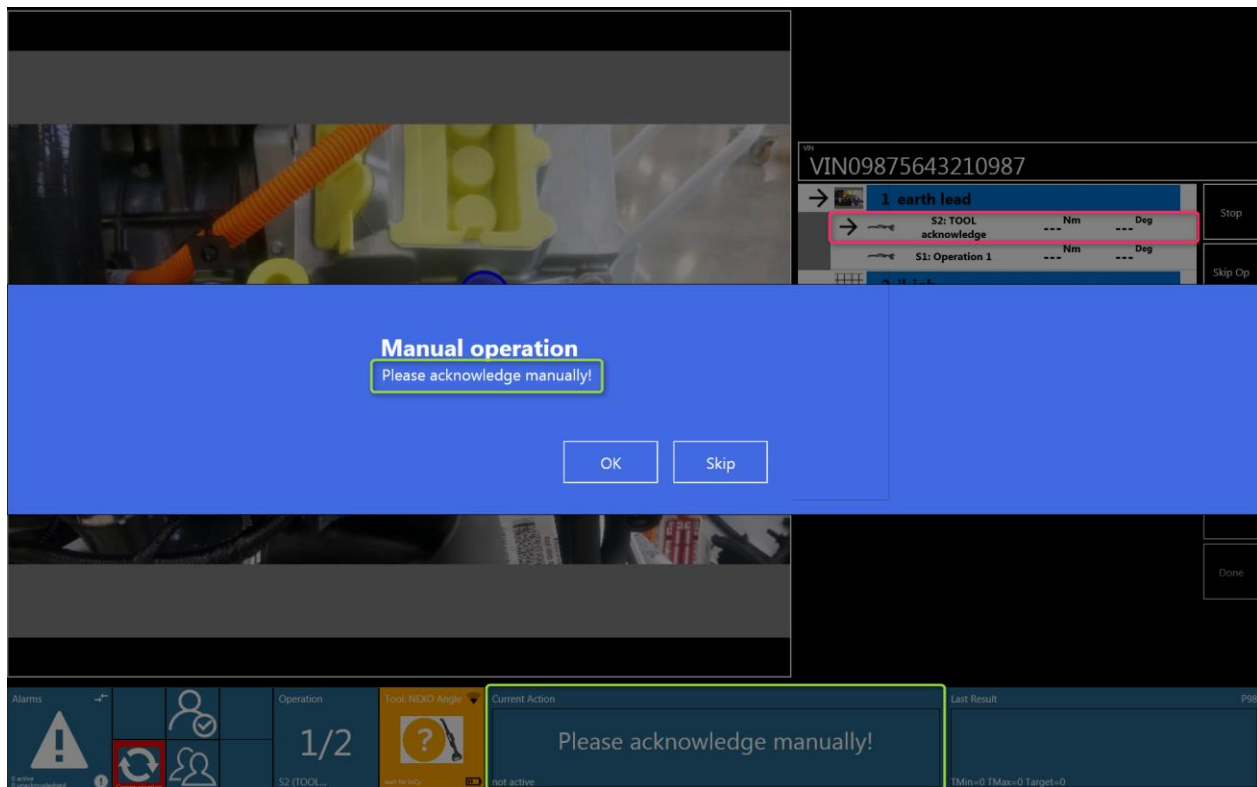


9.3.2 Display of TOOL acknowledge

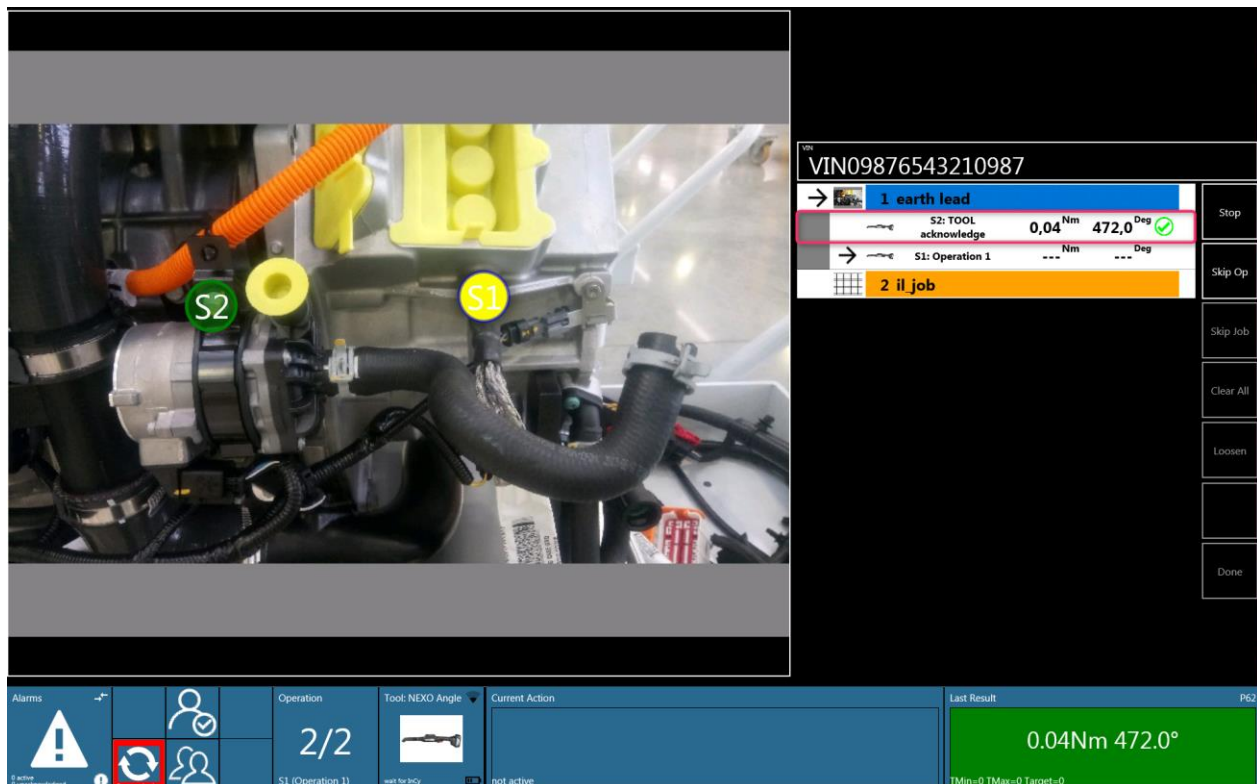
The screenshot below shows how the TOOL acknowledge is displayed in GUI with popup disabled. The "Action description" is only displayed in the "Current Action" box at the bottom of the screen.



The screenshot below shows how the TOOL acknowledge is displayed in GUI with popup enabled. The “Action description” is displayed in both the “Current Action” box and in the popup window.



The screenshot below shows what a successful rundown with “TOOL acknowledge” looks like. The operator guidance system will create a dummy result and the operation is sent to the database as a successful rundown with that dummy value.



9.3.3 Data output configuration

The “GUI popup” tool is not able to directly send out runtime result data to an FTP server. However OPG provides this feature – please see section... for more info.

Data output is implemented through the heOGS ftp client driver, see chapter 15.1.

10 Modbus Tool Interface

The Modbus Tool Interface allows interfacing 3rd party hardware with the OGS system.

The Modbus communication Protocol and the register layout of the modbus tool interface (MBToolCli) is described at <https://gogs.haller-erne.de/he/heOpGui/wiki/MBToolCli+-+Modbus+Tool+interface>.

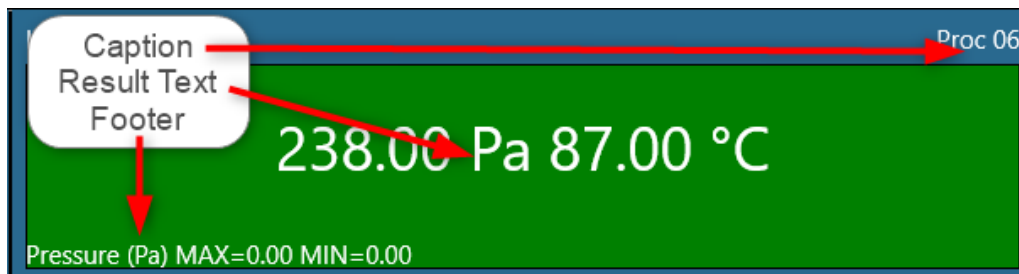
10.1 heOGS GUI

If the Modbus Tool Interface is configured, 3rd party hardware is integrated seamlessly into the OGS interface as shown in the following screenshots:

Result list in OGS screen (note the “Pa” and “°C” units):

Model (use Barcode Sheet)	Serial (3-digit alnum)
TYP7	002
→ 	1 MDB
S1: Press 1	1,23 Pa 45,6 °C ✓
S2: Press 1	1,23 Pa 45,6 °C ✓
S3: Press 1	--- Pa --- °C
S4: Press 2	--- Pa --- °C
→ 	
	Stop
	Skip Op

Last result detail view:



10.2 heOGS configuration

10.2.1 Station configuration

10.2.1.1 Driver setup

The custom Modbus tool is implemented in the MBToolCli.dll tool driver. To enable the tool driver, add it to the [TOOL.DLL] section in station.ini as follows:

```

103 [TOOL_DLL]
104 OPCConn.dll=1
105 ;MBLightCli.dll=1
106 ;EXAConnect.dll=1
107 ;heGWK.DLL=1
108 ;ProfiFTP.dll=1
109 ;heFreedom3.dll=1
110 MBToolCli.dll=1
111

```

10.2.1.2 Add a tool

Add Modbus tool Channel number into [CHANNELS] section:

Use channel number as configured in the workflow editor (see below)

```
112 ; =====
113 ; Channel definitions
114 ; This links drivers, channels and tools
115 ;
116 [CHANNELS]
117
118 7=ModbusTool
119
```

10.2.1.3 Tool configuration

The tool parameters are set up in the [ModbusTool] section in station.ini. See <https://gogs.haller-erne.de/he/heOpGui/wiki/MBToolCli+-+Modbus+Tool+interface> for all settings.

10.2.1.4 Configuring units and data output

The tool visuals and data output is defined in the modbus_gui.lua file in the custom folder. To make this available, import the modbus_gui.lua from the config.lua (or from the monitor.lua) file as follows:

```
1 requires = {
2     "barcode",
3     "user_manager",
4     "modbus_gui",
5 }
```

LUA script "modbus_gui" is necessary to present tool specific measurement values on the OGS GUI and in output JSON file.

Although the Modbus Result Data Registers contain 6 measurement values, but only the first two can be shown in OGS GUI on the place of Torque and Angle accordingly. JSON output, against that, contains a full value set.

Lua script implements the following functions:

- provides data mapping from Modbus Result Data Registers to JSON data structure.
- provides tool specific measurement units to show two first measurement values in Result View
- provides tool specific scale factors to show two first measurement values in Result View.

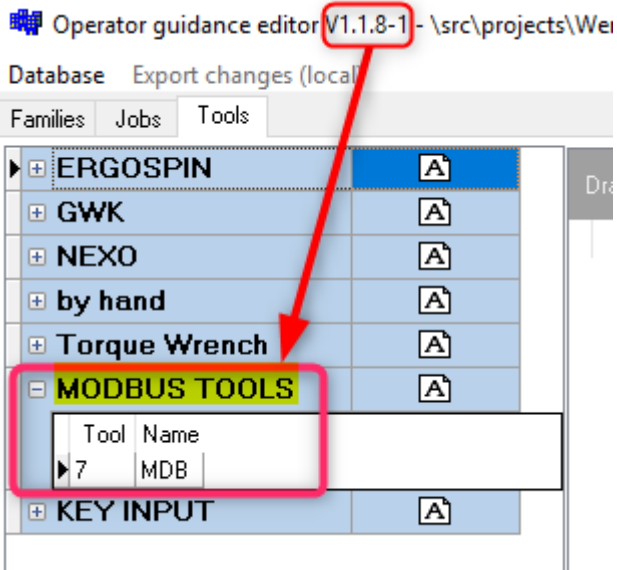
10.2.2 Workflow Editor

The workflow editor supports the Modbus Tool Interface starting from version V1.1.8-1.

10.2.2.1 One-time configuration: prepare database

To make the Modbus tool work, the configuration database must be set up correctly. The configuration database is upgraded automatically to include the "Modbus Tools" tool category if opened using the heOpCfg.exe >= V1.1.8-1.

To verify that the database supports the barcode tool, open it, then switch to the "Tools" tab and look for the tool type "MODBUS TOOLS":



As with other tools, add a new tool of this type to use it in an operation and add it to a job/task.

10.2.2.2 Create an operation to use the Modbus tool

If you want to pass parameters to the Modbus tool, then define these in the operation as follows (note that currently the measurement units are not shown correctly in the heOpCfg):

Operation			T(-)	Torque	T(+)	A(-)	Angle	A(+)	Check
...	Press 1		0	0	0	0	0	0	☐
	Tool	Type	Prg.	Socket					
...	MDB	process	6	0 (don't use socket)					
...	Press 2		3	0	20	0			
	Tool	Type	Prg.	Socket					
...	MDB	process	10	0 (don't use socket)					

To do!
measurement
units

11 Euchner EKS RFID key

The Euchner key allows using RFID tokens to logon to the OGS system. Supported are EKS readers providing USB or serial interfaces.

The Euchner key reader is configured and used similar to a barcode scanner by the OGS system. Each incoming message (token inserted, token removed, etc.) is routed through the barcode.lua barcode handler. By default, the messages are then forwarded to the user_manager.lua user management.



11.1 heOGS configuration

11.1.1 Station configuration

11.1.1.1 Configure heOGS – station.ini

The Euchner key reader is configured in the [SERIAL] section in station.ini similar to a barcode reader. The name of the entry must be set to “Euchner” to automatically make the OGS system detect the hardware and select the correct communication protocol.

See the screenshot below for an example:

```
[SERIAL]
;Barcode      = COM8 115200,n,8
SocketTray    = COM5 115200,e,8|halbduplex
Barcode_Rfid  = COM6 9600,n,8
Euchner       = COM7 9600,e,8,1
```

11.1.1.2 Setup users and access rights

The users are setup as default, just using the corresponding key IDs (in station.ini):

```
[USER]
; Define logins
; Format:
;   <username>=<password>[,<userlevel>]
; If <userlevel> is omitted, 1 is used as default.
; Allowed userleves are 1=Operator, 2=Supervisor, 3=Administrator
; The <username> "autologon" is specially treated, and must follow
; the format:
;   autologon=<username>
; If autologon is available and a non-empty username is given, then an
; operator with the given <username> is logged on (userlevel is always
; = 1).
;
Joe Bloggs=password,1,U40003ACC4D
Stuart Williams=password,3,UA1008DB131
{YELLOW:"green"}=yellow,1,U40001AC01D
TRIUMPH=triumph,1,HAFA6D310
```

Note that detailed user rights (mapping between user level and allowed actions) can be set through user_manager.lua.

12 Jäger positioning system

The Jäger positioning system “HandyTrack 200” is used to locate exact positions on a part. The mounted tool is only enabled, if the operator moves the tool to the correct position.

heOGS supports using the HandyTrack system similar to a socket tray. The socket number (configured in the workflow) is used as a position number for the HandyTrack. The HandyTrack uses this number to pull out a 3D-position from its configuration and returns an “In Position” signal if the tool is within the configured position. heOGS uses this signal to enable the tool, so overall an operation can only be started, if the tool is in the correct operating position.

NOTE: Positioning system support is available starting with V2.0.6.5 of heOGS.



12.1 System configuration

The HandyTrack provides a 24V I/O interface. heOGS supports connecting to the HandyTrack using a CS351 over OpenProtocol with an IM24V card.

The PLC assignment table in the CS351 box should be set up as follows:

Rexroth BS350 V2.600 - [SE/CS PLC assignment table: 0.1 Ergospin 123]

File Edit Display Process System Administration Data System test PC Extras Window ?

Inputs Signals Free Assignment

Prog0
Prog1
Prog2
Prog3
Prog4
Prog5
Prog6
Prog7
En
ResF
Cw
Cw
ResRs
NokAc
DisRp
DisRpFtp
UsrIn1
UsrIn2
CntRes
EnScan
ScanCTS
SpDck
CntSel0
CntSel1
CntSel2
CntSel3
CntSel4
CntSel5
CntSel6
CntSel7
AppIn0_PosReq_0
AppIn1_PosReq_1
AppIn2_PosReq_2
AppIn3_PosReq_3
AppIn4_PosReq_4
AppIn5_PosReq_5
AppIn6_PosReq_6
AppIn7_PosReq_7
AppIn8_PosOk
AppIn9
AppIn10
AppIn11
AppIn12
AppIn13
AppIn14
AppIn15
ES LED Green

Free
B1 0.0
B1 1.0
OP 0.0
OP 1.0
OP 2.0
OP 3.0
OP 4.0

Assignment
0.0 AppIn8_PosOk
.1
.2
.3
.4
.5
.6
.7
1.0
.1
.2 <= 1>
.3 <= 1>
.4 <= 1>
.5 <= 1>
.6 <= 1>
.7 <= 1>
0.0 Prog0
.1 Prog1
.2 Prog2
.3 Prog3
.4 Prog4
.5 Prog5
.6 Prog6
.7 Prog7
1.0 En
.1 BoxNut_LED0
.2 BoxNut_LED1
.3 BoxNut_LED2
.4 BoxNut_LED3
.5 BoxNut_LED4
.6 BoxNut_LED5
.7 BoxNut_LED6
2.0 BoxNut_LED7
.1
.2 CwLock
.3
.4
.5
.6
.7
3.0 AppIn0_PosReq_0
.1 AppIn1_PosReq_1
.2 AppIn2_PosReq_2
.3 AppIn3_PosReq_3
.4 AppIn4_PosReq_4
.5 AppIn5_PosReq_5
.6 AppIn6_PosReq_6
.7 AppIn7_PosReq_7
4.0

Signals
Ack0
Ack1
Ack2
Ack3
Ack4
Ack5
Ack6
Ack7
Rdy
NF
InCy
InCyCw
CyCmp
Ok
Nok
Mstop
FtpF
ToolBn
CCwSel
MntTool
StartBn
ScanRxID
EnScanAck
ScanRdy
CntOK
CntNOK
RC0
RC1
RC2
RC3
RC4
InfoTool
TrigOut
TorqH
TorqL
AngH
AngL
GradH
GradL
TorqRelH
TorqRelL
TorqMaxH
TorqMinL
TorqAvrH
TorqAvrL
TorqRgeH
TorqRgeL

Free
B1 0.0
B1 1.0
OP 0.0
OP 1.0
OP 2.0
OP 3.0
OP 4.0

Assignment
0.0 AppOut0_PosReq_0
.1 AppOut1_PosReq_1
.2 AppOut2_PosReq_2
.3 AppOut3_PosReq_3
.4 AppOut4_PosReq_4
.5 AppOut5_PosReq_5
.6 AppOut6_PosReq_6
.7 AppOut7_PosReq_7
1.0
.1
.2
.3
.4
.5 <not available>
.6 <not available>
.7 <not available>
0.0 BoxNut_SelAck0
.1 BoxNut_SelAck1
.2 BoxNut_SelAck2
.3 BoxNut_SelAck3
.4 BoxNut_SelAck4
.5 BoxNut_SelAck5
.6 BoxNut_SelAck6
.7 BoxNut_SelAck7
1.0 CCwSel
.1 InCy
.2
.3
.4 Rdy
.5 NF
.6
.7
2.0 AppOut8_PosOk
.1
.2
.3
.4
.5
.6
.7
3.0 <not available>
.1 <not available>
.2 <not available>
.3 <not available>
.4 <not available>
.5 <not available>
.6 <not available>
.7 <not available>
4.0 <not available>

Send names to system
Load names from system
Save names to file
Load names from file
Reset all names
☐ Show original name

Card-slot	Module	I	O
B1	IM24V	10	13
OP		48	24
ES		4	0
HMI		8	0
HVO		8	0
IL		32	32
XML		20	24

Bosch Logged on 10.10.2.244 Send: OK

Here the following signals are set up:

- IM24V B1 IO.0: connected to the “InPosition” signal output of the HandyTrack.
- IM24V B1 Q0.0-0.7: connected to the position number input (“Job number” in HandyTrack terms)

Note that the driver automatically switches the position number back to 0 whenever a new task is started, so the HandyTrack always sees a change in its job number.

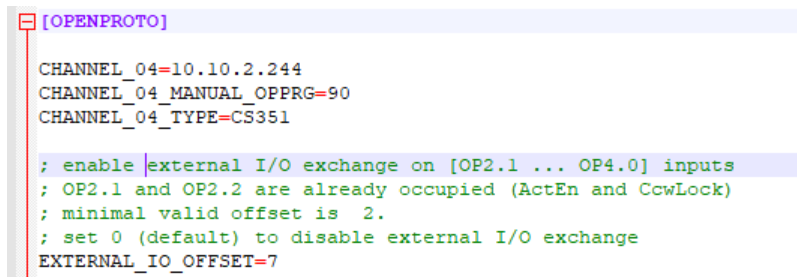
See also chapter 2.1.2.5 for more information.

12.2 heOGS configuration

12.2.1 Driver configuration

The HandyTrack positioning system works in conjunction with the Rexroth CS351 tool. See chapter 2.2 for more information about the generic setup.

To correctly map the IM24 I/O signals to the internal processing, the `EXTERNAL_IO_OFFSET` must be set to 7 (this basically maps OpenProtocol signal 3.0-3.7 as position number), see the following screenshot:



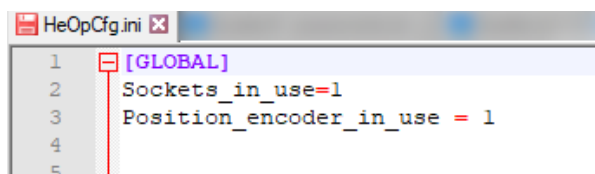
```
[OPENPROTO]
CHANNEL_04=10.10.2.244
CHANNEL_04_MANUAL_OPFRG=90
CHANNEL_04_TYPE=CS351

; enable external I/O exchange on [OP2.1 ... OP4.0] inputs
; OP2.1 and OP2.2 are already occupied (ActEn and CcwLock)
; minimal valid offset is 2.
; set 0 (default) to disable external I/O exchange
EXTERNAL_IO_OFFSET=7
```

Note that the offset is 7 (bits), as the CS351 OpenProtocol implementation assigns the first externally controllable signal to OP2.1 (and internally signals 2.1-2.7 are used for tool control, see Nexio interface).

12.2.2 Workflow editor configuration

To enable the positioning mode in the heOpCfg workflow editor, the following settings must be configured in `heOpCfg.ini`:



```
HeOpCfg.ini
1 [GLOBAL]
2 Sockets_in_use=1
3 Position_encoder_in_use = 1
4
5
```

NOTE: Positioning system support in the workflow editor is available for `heOpCfg.exe` starting with V1.2.0.0.

12.2.3 Workflow setup

To enable the HandyTrack positioning, one must enable the position option in the job screen and add the position number (as defined in the HandyTrack settings) into the “PS” column next to the bolt definition:

bolting configuration: POS

N	Task	PS
1	S1	1
2	S2	2
3	S3	3
4	S4	4
5	S5	5
6	S6	6
7	S7	7
8	S8	8

name: S8
number: 8
position sensor(PS): 8
Sequence: Operation
pre-torque: 8
final torque: 8
CheckPos

Operation	T(-)	Torque	T(+)	A(-)	Ar
CheckPos	0	0	0	0	0

Tool	Type	Prg.	Socket
CS.351	process	40	0 (don't use socket)

new operation
delete operation
add rework
Prefix for automatic task naming: S

Action description:

cancel change image save

NOTE: To disable positioning for a specific bolt, set the “PS” column value to zero.

12.2.4 LUA configuration notes

TBD.

13 Barcode entry

There are three main areas in the system where barcode input gets processed:

- Workflow start
- Job selection and start
- Barcode tool

This section describes the “Barcode tool” configuration and use.

13.1 Barcode tool

The Barcode tool allows adding a scanning action similar to a tightening action in the normal workflow. If the task with a scanning action is reached, then the operator is required to scan (or enter) a barcode.

NOTE: The barcode tool is available starting with version >= V2.1.

13.1.1 Station configuration

13.1.1.1 Configure heOGS – station.ini

Usually, no additional configuration in station.ini is required, as the barcode tool uses the globally defined scanner.

The barcode scanners serial communication parameters are defined in the [SERIAL] section of the station.ini file. The entry “Barcode” defines the parameters used, see the screenshot below for an example:

```
; Zuordnung und Schnittstellenparameter für die
; seriellen Schnittstellen. Die Syntax entspricht den
; Regeln des Kommandos 'MODE' , "Windows Kommandozeile" },

[SERIAL]
Barcode          = COM13 9600,n,8
Barcode_Rfid     = COM9 9600,n,8
SocketTray       = COM9 115200,e,8|halbduplex
```

13.1.2 Workflow editor

13.1.2.1 One-time configuration: heOpCfg.ini

To enable editing the barcode masks, a new custom property is added to the action properties.

Please check, if the entry “barcode mask=string” in Section [ACTION PROPERTY NAMES] in heOpCfg.ini is available. If not, then add it (see screenshot on the right for an example).

```
[GLOBAL]
Level=0
Sockets_in_use=1
Multiple_stations=1
Position_encoder_in_use = 1
Nachanzug=0

[JOB PROPERTY NAMES]
;description=string

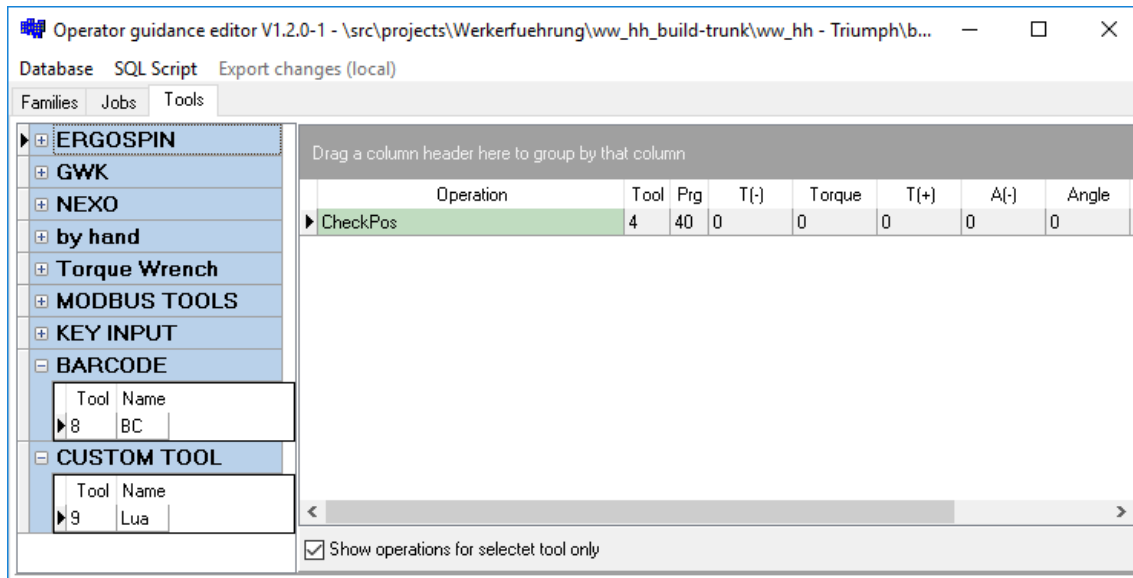
[TASK PROPERTY NAMES]

[ACTION PROPERTY NAMES]
barcode mask=string
```

13.1.2.2 One-time configuration: prepare database

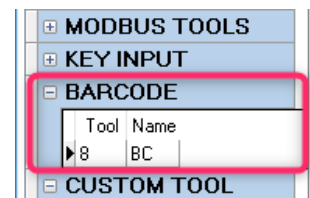
To make the barcode tool work, the configuration database must be set up correctly. The configuration database is upgraded automatically to include the “BARCODE” tool if opened using the heOpCfg.exe >= V1.2.0-1.

To verify that the database supports the barcode tool, open it, then switch to the “Tools” tab and look for the tool type “BARCODE”:



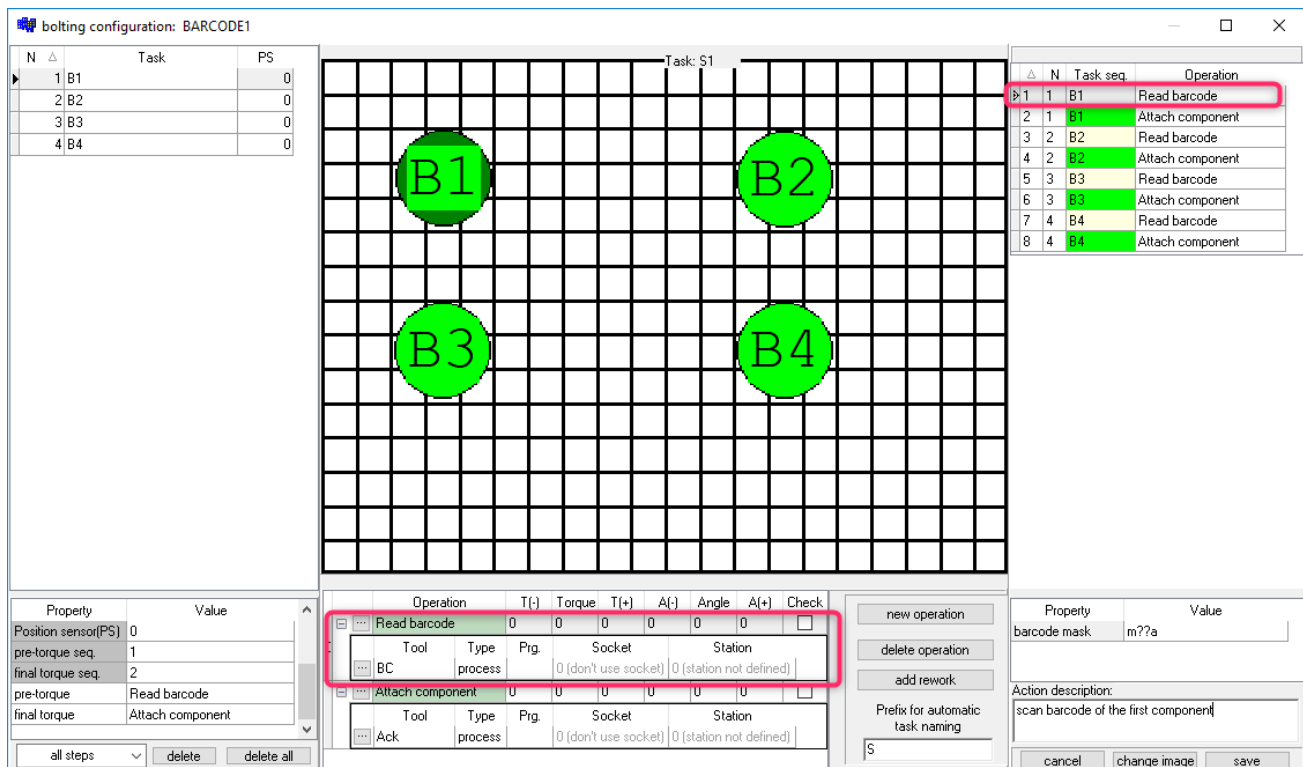
13.1.2.3 Create a new tool for barcode input

Open the tab “Tools” and add a new tool in the section “BARCODE”. This tool can be used in the job/operation configuration exactly as other tools like NEXO, Torque Wrench etc.



13.1.2.4 Create a task for barcode input

To add a task in the workflow job, open the job editor and add a new task. Then add a new operation which uses the barcode tool (Tool “BC”, see screenshot below). Finally assign the operation to the task.



NOTE: The barcode tool requires entering a barcode mask to define the allowed barcodes.

The screenshot on the right shows how to enter the barcode mask.
Allowed values for the barcode mask are:

- ?: Wildcard character. Matches exactly one character but does not care about its value
- Other characters: exact match required

Note also that the action description can be set. The description will be displayed in the barcode popup (if enabled) and in the task bar in the operator main GUI.

In addition to setting the barcode mask for barcode validation, more complex validations can be setup through a custom LUA function. The relevant function is `BarCode_GetActionCode()`, usually located in "barcode.lua".

△	N	Task seq.	Operation
1	1	B1	Read barcode
2	1	B1	Attach component
3	2	B2	Read barcode
4	2	B2	Attach component
5	3	B3	Read barcode
6	3	B3	Attach component
7	4	B4	Read barcode
8	4	B4	Attach component

Property	Value
barcode mask	m??a

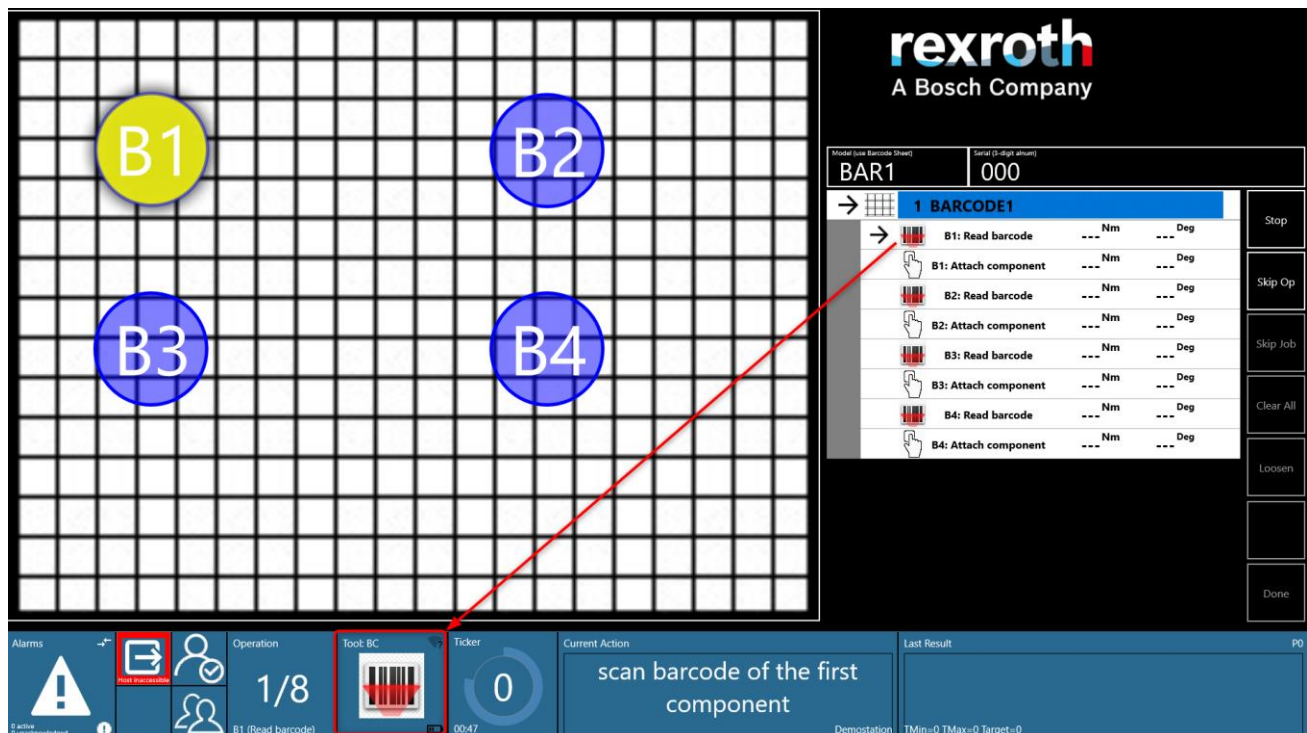
Action description:
scan barcode of the first component

cancel change image save

13.1.3 Usage

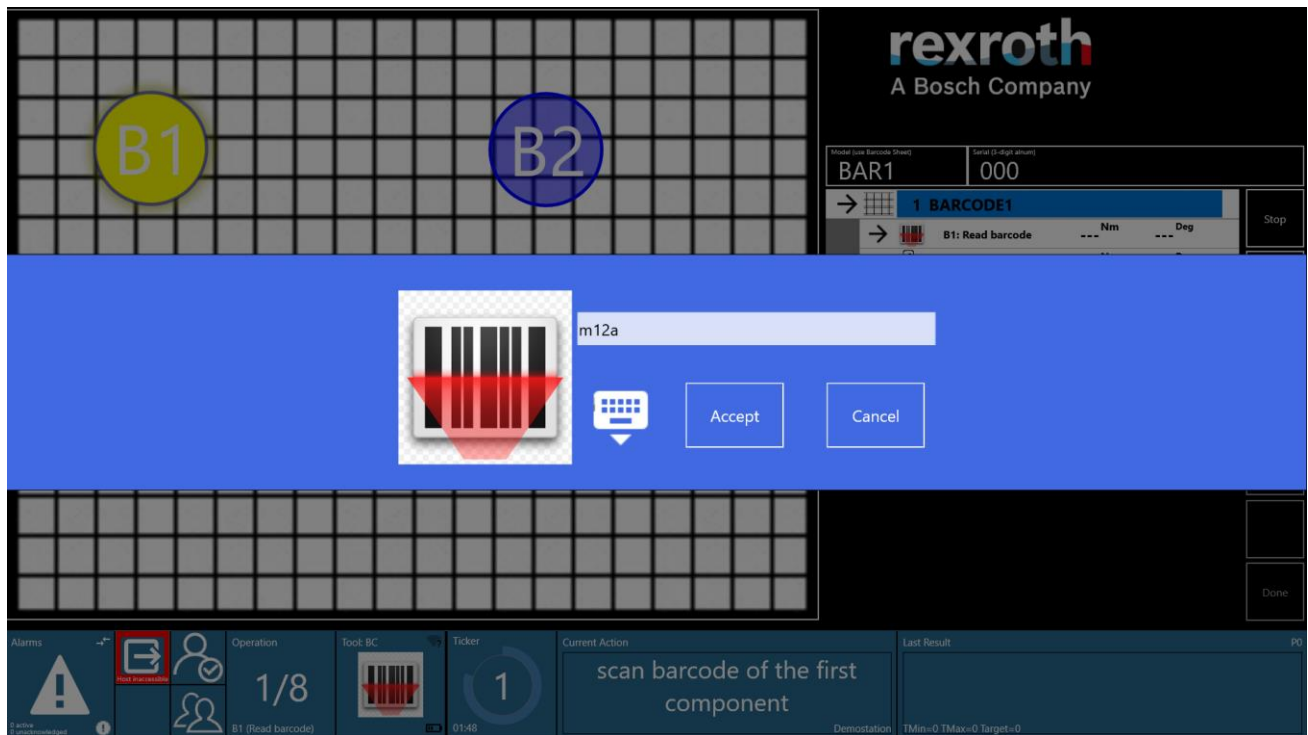
13.1.3.1 General usage and manual input

If a barcode tool gets active, the display looks similar to the following screenshot:



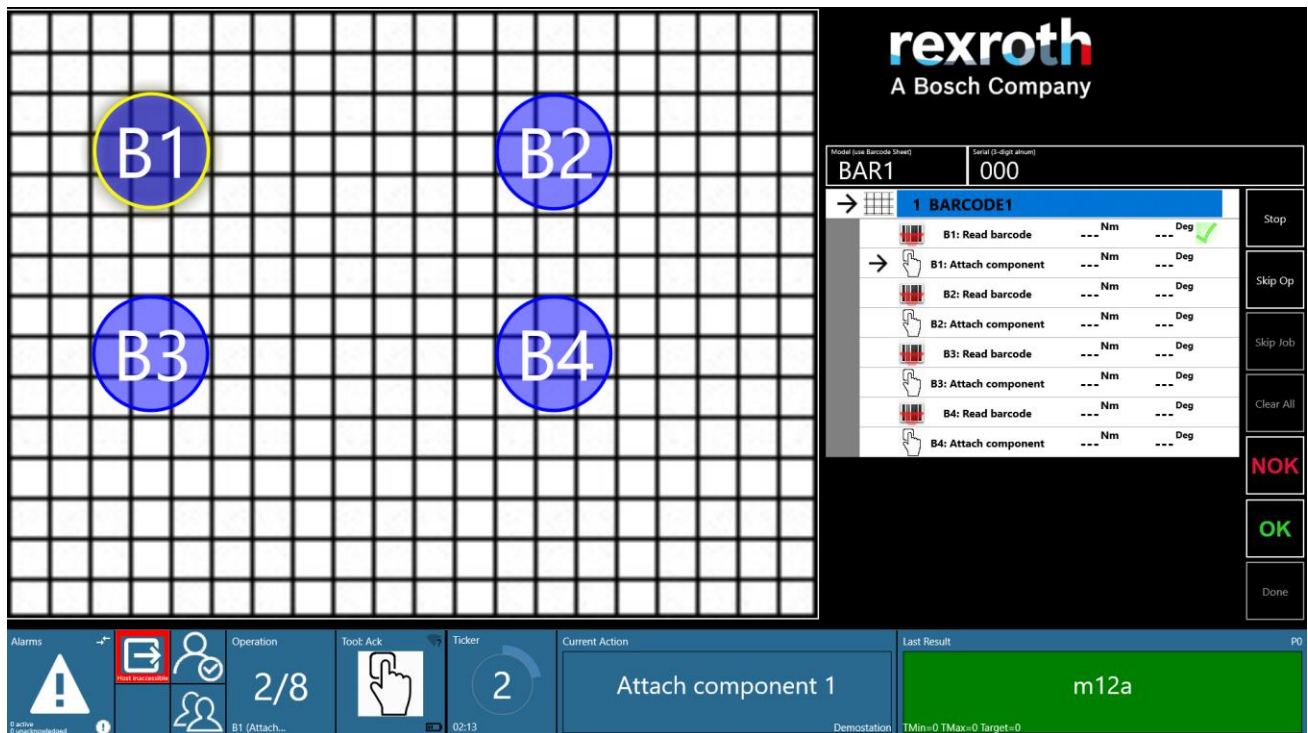
Note that the barcode tool supports manual input by clicking the tool icon in the taskbar.

This will then show the keyboard input popup as follows (note that you can also bring up the on-screen keyboard by clicking the keyboard icon):

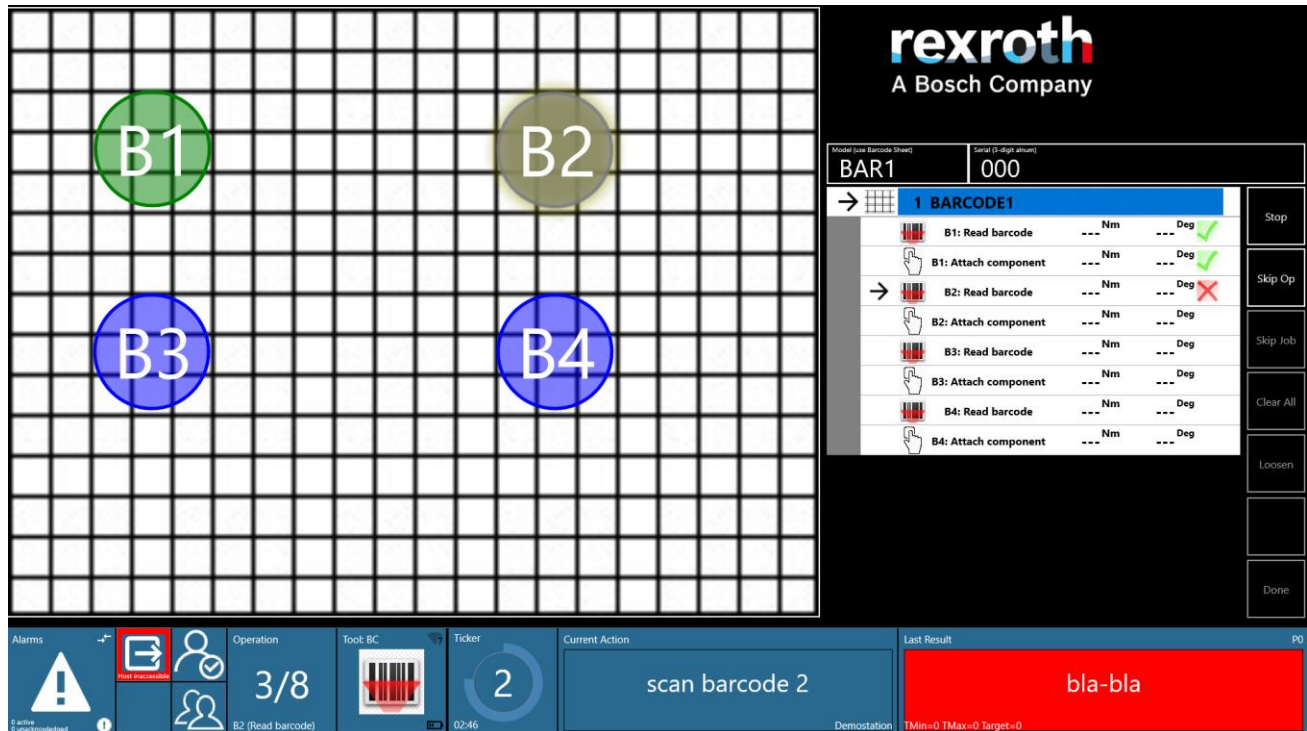


13.1.3.2 Scanned barcode result display

If a valid barcode was scanned, the workflow continues to the next step and the “Last Result” shows the previously scanned barcode (on green background):



If an invalid barcode was scanned, the workflow does not continue, but stays in the current task. The falsely scanned barcode is displayed in the “Last Result” box (with a red background). The operator is required to scan again:



13.1.4 Data output configuration

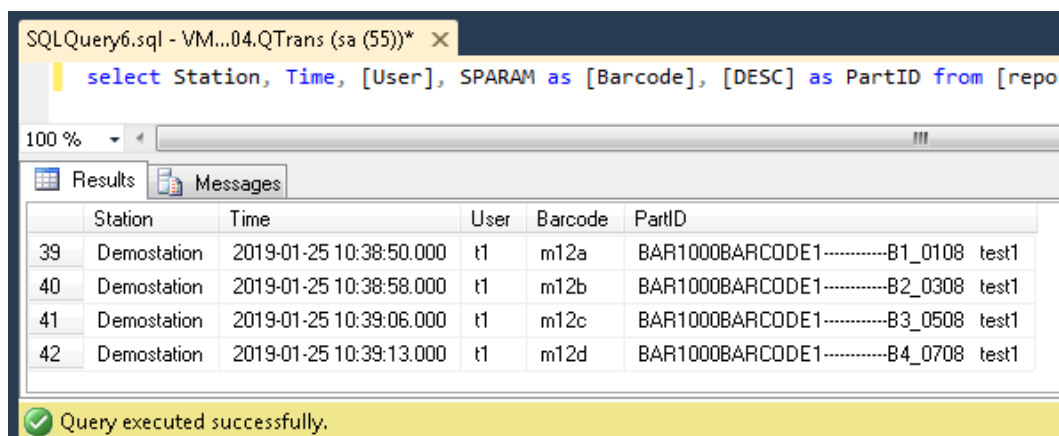
Data output is implemented through the heOGS ftp client driver, see chapter 15.1. Data is transmitted as “event” data to the database.

13.1.4.1 Configuration in station.ini

The barcode tool uses the INFO_CHANNEL to send data to the server. See chapter 15.1.2 on how to configure this.

13.1.4.2 Data mapping using Sys3xxGateway/SQL server

Barcode data is available in on the SQL server in the [report].[events] table:



14 Custom (LUA) tool

14.1 Overview

The “custom tool” provides an interface between the workflow configuration, the monitor GUI/runtime and custom code. The idea is to provide a generic interface to add custom tools and/or behaviors to the heOGS system.

14.1.1 Station configuration

14.1.1.1 Driver Setup

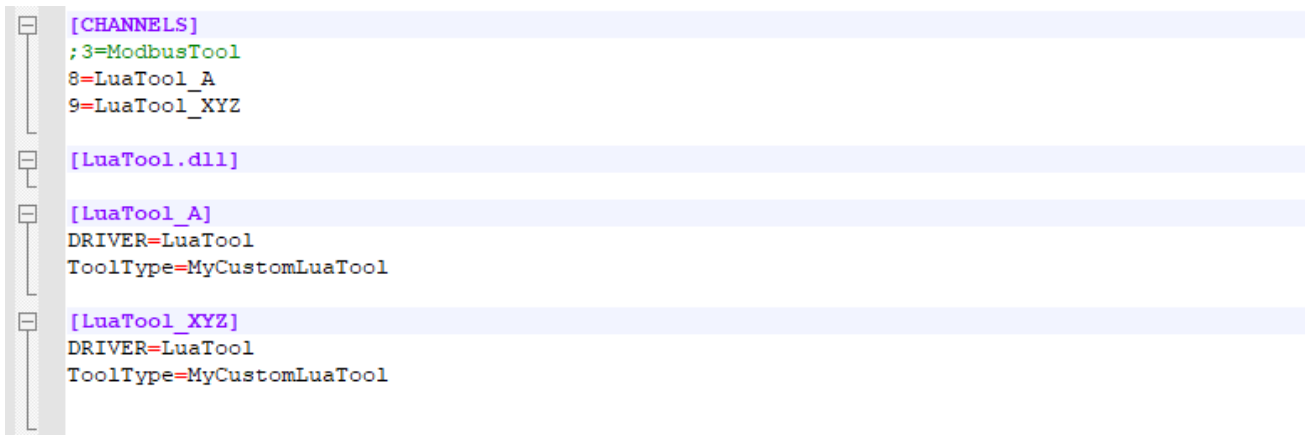
The custom LUA tool is implemented in the LuaTool.dll tool driver. The tool driver properties are defined on the [LuaTool.dll] section in the station.ini file.

Adding a custom tool requires the following:

- Add a new channel in the [CHANNELS] section. The channel number (left to the = sign) must match the tool ID defined in the workflow editor (see below). The “tool name” (right to the = sign) links the parameters to the channel.
- Add a new tool parameter section [“tool name”]. The “tool name” must match a “tool name” defined in the [CHANNELS] section. In this section, the following keys are required:
 1. DRIVER=LuaTool: The DRIVER key must be equal to “LuaTool”. This basically links the tool to the [LuaTool.dll] driver.
 2. ToolType: This is a custom text, which is passed to the LUA script code during initialization of the tool. This can be used to implement a set of “custom tool” classes, e.g. a “Press” and a “Meter”. Multiple tool-instances can then reference a single LUA script driver.

Note that this section may contain other keys/values which define the tools behavior.

The following screenshot shows a typical setup for a tool (ids = 8 and 9, tool names = “LuaTool_A” and “LuaTool_XYZ”, both of ToolType “MyCustomLuaTool”):



```

[CHANNELS]
;3=ModbusTool
8=LuaTool_A
9=LuaTool_XYZ

[LuaTool.dll]

[LuaTool_A]
DRIVER=LuaTool
ToolType=MyCustomLuaTool

[LuaTool_XYZ]
DRIVER=LuaTool
ToolType=MyCustomLuaTool

```

14.1.1.2 Driver Parameters

The driver parameters are set in the [LuaTool.dll] section.

14.1.1.3 Tool Parameters

The tool parameters are set in the [“tool name”] section. The following keys are required:

- DRIVER=LuaTool: The DRIVER key must be equal to “LuaTool”. This basically links the tool to the [LuaTool.dll] driver.
- ToolType: This is a custom text, which is passed to the LUA script code during initialization of the tool. This is typically used to implement a set of “custom tool” classes, e.g. a “Press” and a “Meter”. Multiple tool-instances can then reference a single LUA script driver. Please see the following sections (starting with chapter 14.2) for specific tool implementations.

14.1.2 LUA scripts

14.1.2.1 Overview

The actual “custom tool” behavior is implemented through a LUA script. A typical custom tool implementation consists of the following files:

File name	Notes
lua/lib/lua_tool.lua	Interface between the OGS system and the custom LUA tool implementations. Do not change!
<custom>/lua_tool_<tool>.lua	The custom tool implementation. This file interfaces between the lua/lib/lua_tool.lua and should implement the basic tool interface.
<custom>/<tool>_driver.lua	Not specifically required, but recommended to factor out communication specifics related to the tool communication)
<custom>/config.lua	OGS standard file to load customer specific data/code. The <custom>/lua_tool_<tool>.lua must be referenced here.

There might be more files needed depending on the actual tool, e.g. for the Sylvac BLE caliper interface, a Bluetooth LE library is needed (luabluegiga.dll, placed into the installation folder), see 14.3.

14.1.2.2 OGS data view

The LUA tool implementation may present the onscreen values shown in a flexible way. To do so, the following functions should be implemented in the LUA script tool interface:

- *function <lua_tool>.get_tool_units(tool)*: Returns two strings to indicate the primary measurement units (e.g. ‘Nm’, ‘°’). These units are used to render the process view list result values (in the right pane).
- *function <lua_tool>.get_tool_result_string(tool)*: Returns a string shown in the bottom right result pane (as result text, large font, centered). Typically, this returns a formatted string in the format “xxNm yy°”.
- *function <lua_tool>.get_footer_string(tool)*: Returns a string shown in the bottom right result pane (as limit values, small text in the bottom line). Typically, this shows a string indicating the lower/upper limits.

14.1.2.3 FTP/JSON data output

The LUA tool implementation may present the onscreen values shown in a flexible way. To do so, the following functions should be implemented in the LUA script tool interface:

- *function <lua_tool>.default_get_tags(tool)*: Returns the system number (see <https://gogs.haller-erne.de/he/heOpGui/wiki/Data+output+format+-+Sys3xxGateway+custom+units>) and an array of tags (max. 6) for each measurement returned, so that Sys3xxGateway/QualityR can correctly identify the data types.
- *(optional) <lua_tool>.JsonFmt*: If this variable is non-nil, then it is used as a JSON template to generate the JSON file. See the lua/lib/json_ftp.lua for more details.

14.1.2.4 TODO: more detailed description of the LUA interface.

14.1.3 Workflow editor

14.1.3.1 One-time configuration: heOpCfg.ini

The custom tools usually require additional parameters in the workflow editor. These parameters must currently be added to the heOpCfg.ini, so the GUI is able to present them for editing by the user.

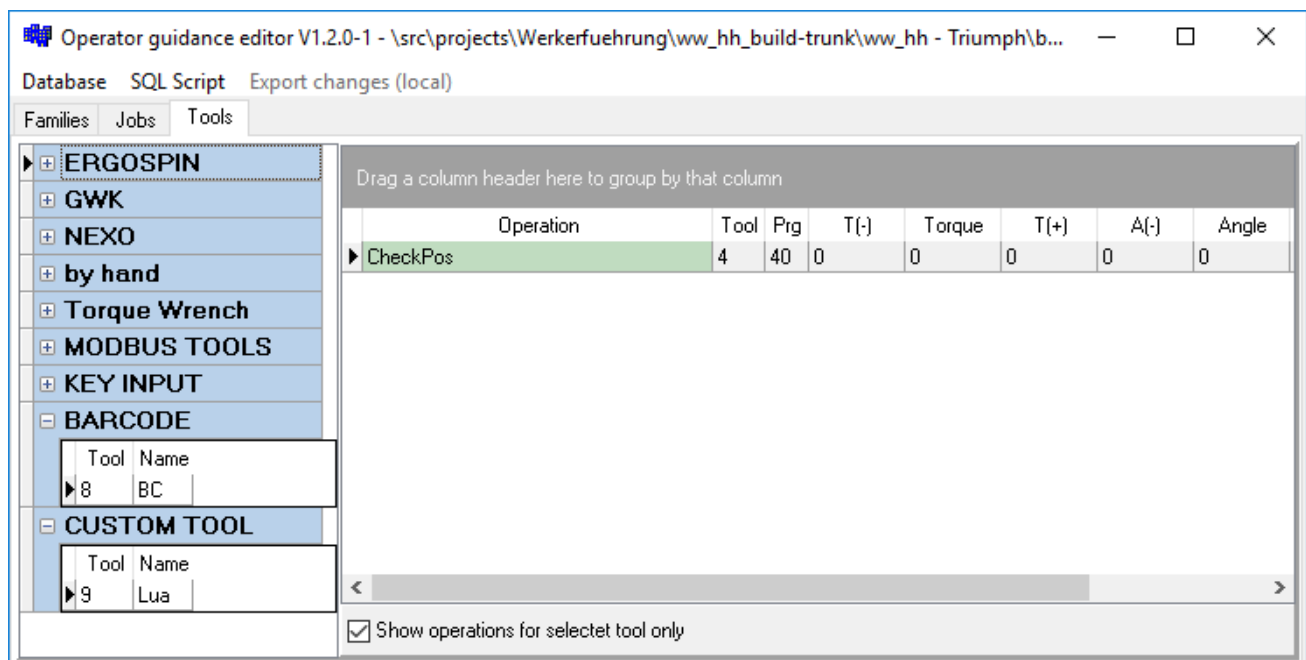
All properties must be added into section [ACTION PROPERTY NAMES] in heOpCfg.ini.

Note that the properties should match the required properties in the LUA tool implementation.

14.1.3.2 One-time configuration: prepare database

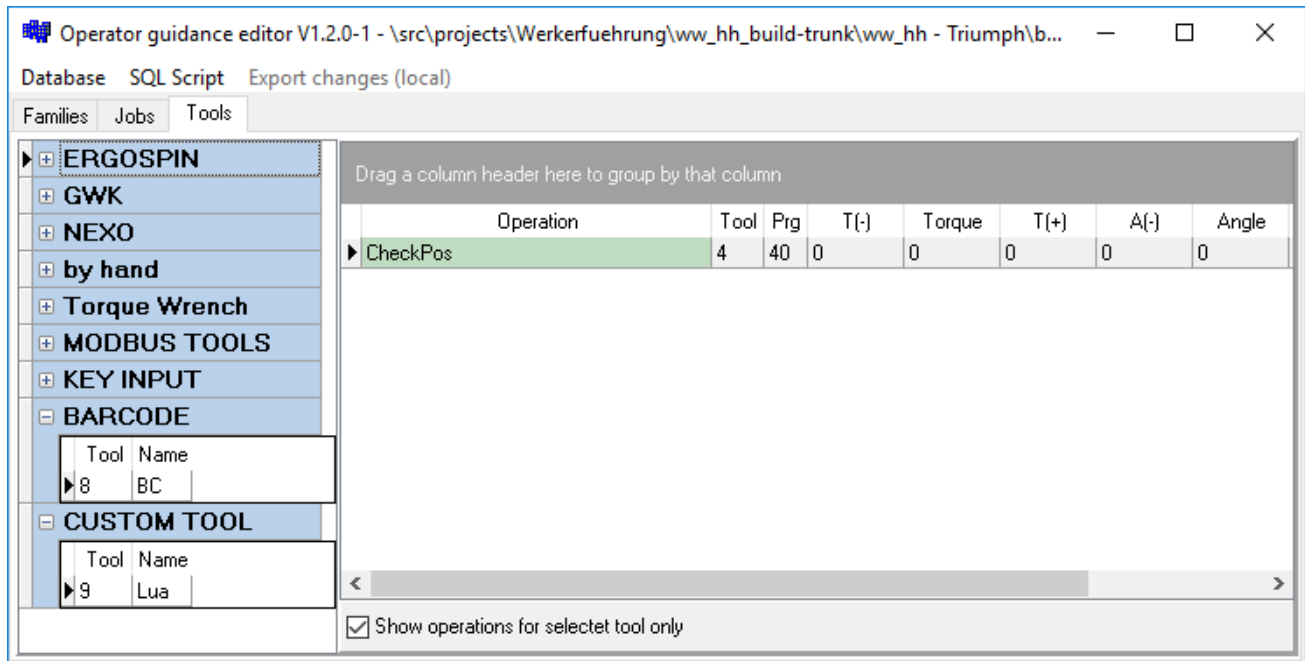
To make the custom tool work, the configuration database must be set up correctly. The configuration database is upgraded automatically to include the “CUSTOM TOOL” tool if opened using the heOpCfg.exe >= V1.2.0-1.

To verify that the database supports the “custom” tool, open it, then switch to the “Tools” tab and look for the tool type “CUSTOM TOOL”:



14.1.3.3 Create a new custom tool

Open the tab “Tools” and add a new tool in the section “CUSTOM TOOL”. This tool can be used in the job/operation configuration exactly as other tools like NEXO, Torque Wrench etc.



NOTE: A custom tool requires a supporting LUA script for its functionality!

NOTE: Running heOpCfg with the admin switch allows changing the tool custom fields!

NOTE: The mapping between a specific tool type is done in station.ini (not in heOpCfg)

14.2 Custom tool with onscreen input

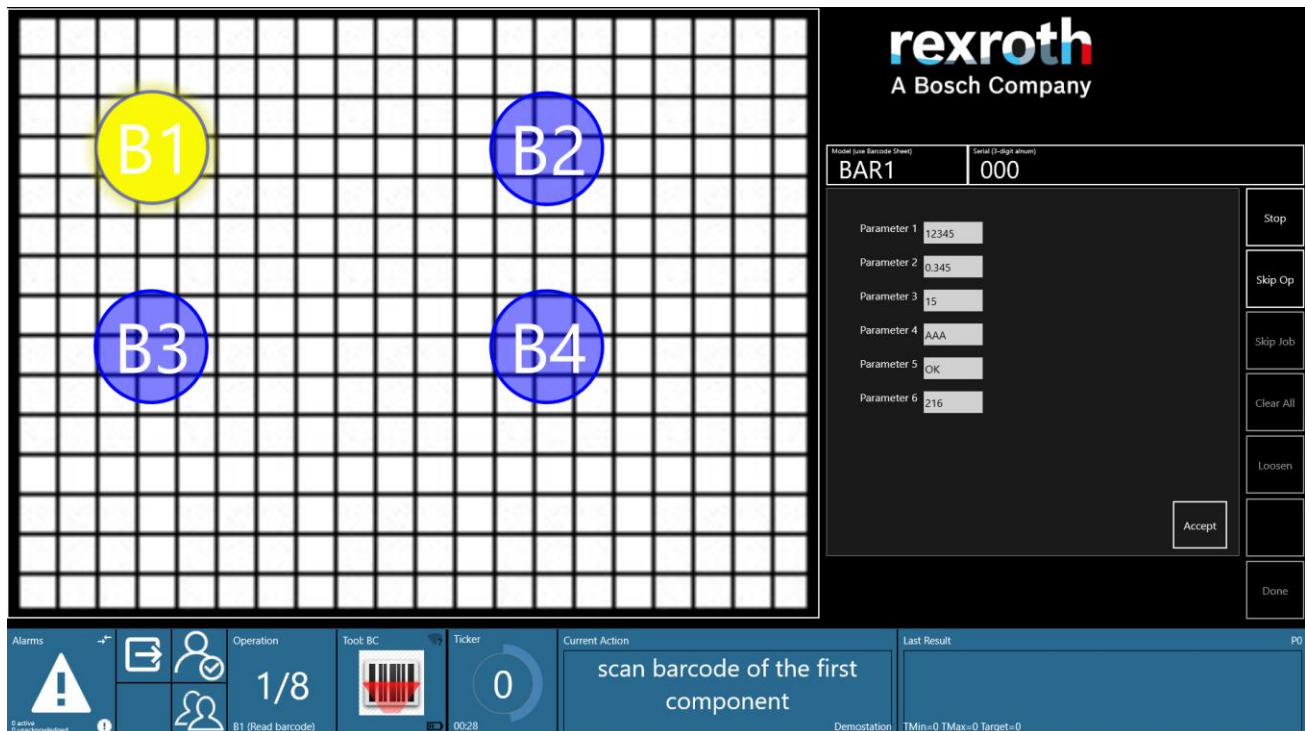
14.2.1 Station configuration

See chapter 14.1.1 on how to setup a custom tool in station.ini.

14.2.2 Workflow editor

See chapter 14.1.3 on how to setup a custom tool in station.ini.

14.2.3 Usage



14.2.4 Data output configuration

See Modbus tool.

14.3 Sylvac BLE devices

14.3.1 Overview

The provided implementation for the Sylvac BLE devices use the Sylvac Bluetooth Smart Dongle (see <https://www.sylvac.ch/download-en/softwares/drivers-and-firmwares> for drivers and firmware updates) to communicate with the Sylvac meters. The dongle is actually a brand labeled SiLabs Bluegiga BLED112 and emulates a serial COM port interface.

The following files are required to use the Sylvac BLE devices:

- ble_sylvac_driver.lua (usually placed in the <custom> folder). This implements the interface between the generic custom OGS LUA tool driver and the Sylvac devices – on a logical level.
- lua_tool_ble_sylvac.lua (usually placed in the <custom> folder). This implements the caliper communication and handles tool connection and data readout over Bluetooth using the bluegiga dongle.
- LuaBluegiga.dll (usually placed in the software installation folder). This implements the low-level LUA interface for the BLED112 dongle.

The Sylvac devices only measure a single value, this is mapped to the first measurement in OGS (including minimum and maximum values), which is sometimes indicated as the “Torque” value. For the caliper (or other distance/length measurements), OGS uses the following measurements type codes (see 14.3.3 to 14.3.6 for more information):

- LM: length measurement (in millimeters)

- LM+: upper length limit. The upper limit is configured through heOpCfg. The tool uses the T+ specification limit parameter in the heOpCfg operation settings.
- LM-: lower length limit. The lower limit is configured heOpCfg. The tool uses the T- specification limit parameter in the heOpCfg operation settings.

NOTE: As the OK/NOK-limits are bound to the operation, different limits require setting up a different operation, it is not possible to define limits independent of the operation!

14.3.2 Device configuration

14.3.2.1 S-CalEvo Caliper

To enable Bluetooth on the device, follow the procedure outlined below:

1. Choose the Bluetooth "PAIRED" profile as follows (SIMPLE will also work for the "old" hardware revision, but "new" hardware should use PAIRED):
 1. Click and hold the "mode" button for > 2s, then click the "mode button" again (you should now see "bt Cfg" on the display)
 2. Click the "set" button until the display shows "PAIR". Then click the "data" button (on the top right) to activate the profile
2. Enable Bluetooth:
 1. Click the "mode" button on the tool until "bt" is shown in the display.
 2. Click the "set" button once. The display should now change to "On"
 3. Click the "mode" button again to accept. The display should change back to the main display, but should now show a Bluetooth icon (in the top-right corner). The icon will blink until a connection is established, then will show a steady Bluetooth icon.



To transmit a meter reading to the OGS system, click the top-right button. If no Bluetooth connection to the OGS system is active, an error message will be shown on the tools display.



To readout the MAC address (the unique tool identifier used by OGS to connect to the tool), see the usage sheet or run a Bluetooth LE scan on the laptop (Bluetooth Smart Software and SDK can be downloaded from the SiLabs site at <https://www.silabs.com/products/wireless/bluetooth/bluetooth-low-energy-modules/ble112-bluetooth-smart-dongle>).

For more information, see the usage sheet at <https://www.sylvac.ch/download-en/manuals/instruction-for-use-caliper-s-cal-evo-smart/download>.

Troubleshooting:

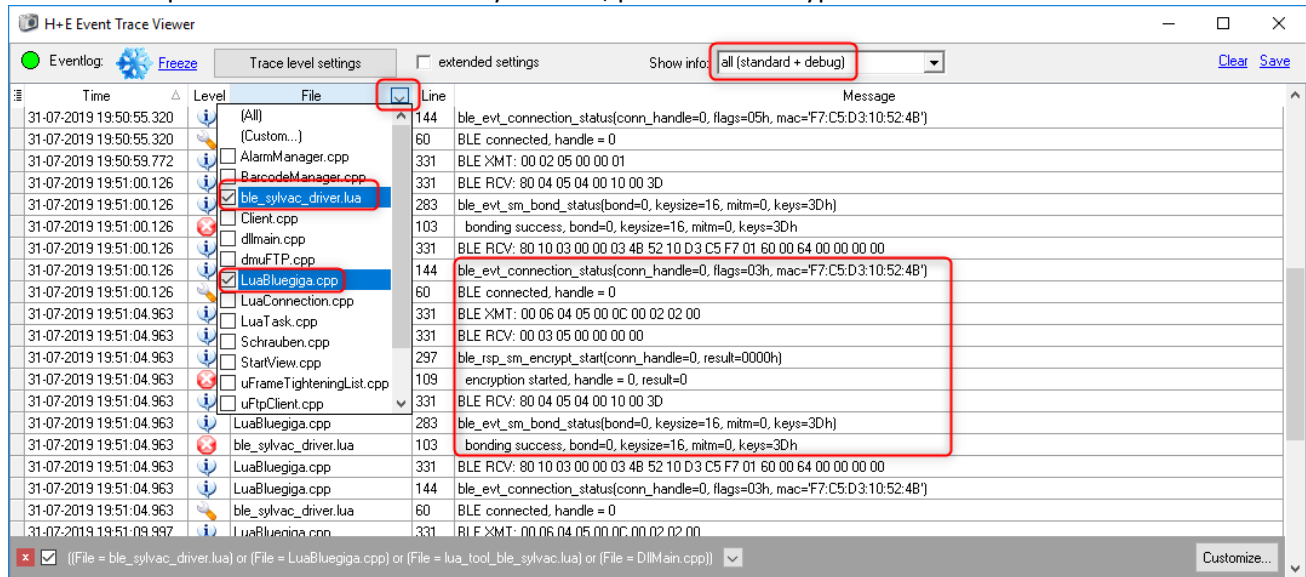
If the meter shows the error message "no Data" after clicking the "data" button, then either the device is not connected (you can verify this with the blinking Bluetooth icon on the device) or it is not correctly paired. If the connection is correctly established (steady Bluetooth icon), then check the ETWTraceviewer log (set level to "all" and filter for file "ble_sylvac_driver.lua" and the "LuaBluegiga.cpp" files).

In the log (see below), look for the lines following the "ble_evt_connection_status" line:

- Check, the "flags" parameter in the "ble_evt_connection_status"-line:
 1. If flags=3 then the connection is established and encrypted (correct)
 2. If flags=5 then the connection is established but not yet connected. It is expected, that the driver now requests an encrypted connection (indicated by a line reading "ble_rsp_sm_encrypt_start" line)
- After the "ble_rsp_sm_encrypt_start", one of the following should occur:
 1. "ble_evt_sm_bond_status" → successfully started the encryption, it is expected, that another "ble_evt_connection_status" will follow (with flags=5)

2. “ble_evt_bond_failed” → error. The error code indicates the reason for the problem (see below)

Here is a sample screenshot of a correctly bonded, paired and encrypted connection:



Typical error codes reported are:

- 518 (0x0206): Pairing failed because of missing PIN, or authentication failed because of missing Key. The most likely reason is that the Caliper was reset and therefore the pairing key was deleted in the caliper. The BLE dongle still remembers the key in its pairing/bonding setup. To fix this issue, the “delete bonding” command should be executed on the BLE dongle (Command in hex: 00 01 05 02 <handle>). The <handle> is the bonding handle as shown in the “ble_evt_connection_status”, parameter “bond_handle”. Newer versions of the driver try to reset this automatically, but might require a disconnect/reconnect of the tool.
- 1027 (0x0403): The attribute cannot be written (write permitted): The most likely reason is that the connection is not secured. Make sure to set the parameter BLE_ENCRYPT in the station.ini file as follows:

```
[LuaTool_BLE_SYLVAC]
DRIVER=heLuaTool
; NOTE: for custom LUA tools implemented through "heLuaTool", the LUA script
;       file used to provide the implementation of the tool interface is
;       identified through the "TYPE" set here.
; To use the TYPE=BLE_SYLVAC, you should also add "lua_tool_ble_sylvac" in config.lua
TYPE=BLE_SYLVAC
BLE_PORT=COM17
BLE_MAC=f7:c5:d3:10:52:4b
; NOTE: for "new" Sylvac calipers, devices in "paired" mode required "encrypted" communication
BLE_ENCRYPT=1
; 0 = public, 1 = (default) random
;BLE_MAC_TYPE=1
```

14.3.3 Station configuration

See chapter 14.1.1 on how to generally setup a custom tool in station.ini. The default Sylvac BLE tool LUA implementation (see files listed above) defines the following parameters:

- DRIVER: must be set to heLuaTool
- TYPE: must be set to BLE_SYLVAC

Bluetooth specific parameters are:

- BLE_PORT: Serial port used to communicate with the SYLVAC Bluetooth Smart Dongle (in the form “COM17”)

- BLE_MAC: Bluetooth MAC address of the meter in hexadecimal format (e.g. 12:34:56:78:90:AB). See 14.3.2.1 on how to get the device MAC address
- BLE_ENCRYPT (default = 0): Set to 1 for BLE devices requiring encrypted communication (like the "new" Sylvac tools in PAIRED mode)
- BLE_MAC_TYPE (default = 1): Set to the correct Bluetooth address type for the device. Allowed values are: 0 = Public, 1 = Random (default)

Here is the relevant part of station.ini:

```

325 ; The channel specific parameters are then defined in the [tool_section_name]
326 ; section of this file.
327 [CHANNELS]
328 7=LuaTool_SYNC
329 8=LuaTool_GUI2
330 10=LuaTool_GUI
331 11=LuaTool_MBCLI_NIDEC
332 13=LuaTool_BLE_SYLVAC
333
334 ; =====
335 ; Tool parameters
336 ; =====
337
338 [LuaTool_BLE_SYLVAC]
339 DRIVER=heLuaTool
340 ; NOTE: for custom LUA tools implemented through "heLuaTool", the LUA script
341 ; file used to provide the implementation of the tool interface is
342 ; identified through the "TYPE" set here.
343 ; To use the TYPE=BLE_SYLVAC, you should also add "lua_tool_ble_sylvac" in config.lua
344 TYPE=BLE_SYLVAC
345 BLE_PORT=COM17
346 BLE_MAC=DC:53:6A:1D:86:99
347

```



To load the scripts (see 14.1.2 and 14.3.1 for the list of files and their locations) correctly, the following entry must be added to the <custom>/config.lua file:

```

1  requires = {
2      "barcode",
3      "user_manager",
4      "lua_tool_gui",
5      "lua_tool_gui2",
6      "lua_tool_mbcli",
7      "lua_tool_sync",
8      "lua_tool_ble_sylvac",
9  }
10  current_project.logo_file = 'NIDEC.png'
11  current_project.billboard = 'local://index.html'
12
13

```

14.3.4 Workflow editor

See chapter 14.1.3 on how to generally setup a custom tool in station.ini. Note that the Sylvac BLE tool does not use any additional properties, so changing the heOpCfg.ini is not required.

14.3.5 Usage

To use the SYLVAC BLE tool, complete the one-time configuration (as outlined in chapters 14.3.3 and 14.3.4). Then follow (as with the other tools) these steps:

- Add an operation and choose the previously defined tool. You should use a speaking name for the operation (e.g. 5mm +/-1)
- Setup the specification limits for a good measurement in the newly created operation. Use the T+ and T- fields to enter the upper and lower limit.

- Finally add a task and assign the previously created operation.

Here is a sample setup:

After saving the database, the SYLVAC tools can be used similar to other tools.

NOTES:

- The SYLVAC tools do not support “enable” tool.
- If the tool goes to sleep, it sometimes does not reconnect automatically. To reconnect the tool, follow the procedure:
 1. Click the “mode” button on the tool until “bt” is shown in the display.
 2. Click the “set” button once. The display should now change to “On”
 3. Click the “mode” button again to accept. The display should change back to the main display, but should now show a Bluetooth icon (in the top-right corner). The icon will blink until a connection is established, then will show a steady Bluetooth icon.
- See 14.3.2.1 on how to get the device MAC address

14.3.6 Data output configuration

For the basic configuration, see the Modbus tool.

Note that the following measurements type codes are used:

- LM: length measurement (in millimeters)
- LM+: upper length limit (in millimeters)
- LM-: lower length limit (in millimeters)

To make this work with the Sys3xxGateway software, the type codes must be added to the Sys3xxGateway configuration and to the server database. Please see the wiki page at <https://gogs.haller-erne.de/he/heOpGui/wiki/Data+output+format+-+Sys3xxGateway+custom+units> for more information.

15 Additional configuration

15.1 FTP output driver

15.1.1 Overview

Some tools (e.g. GWK electronic torque wrench) cannot send FTP telegrams by themselves and some operations are done completely without a tool (e.g. GUI popup). For these cases, heOGS can act as an FTP client and send the FTP telegrams to the server.

```
[FTP_CLIENT]
ENABLED=0

HostIP=172.23.56.110
HostPort=21

; In case of Sys3xxGateway(Qtrans) as FTP server
; use "Username=sys3xx" and "Password=sys3xx"
; TargetFolderOnHost is not needed (ignored)

; In case of conventional FTP Server the "TargetFolderOnHost" parameter is obligatory
Username=sys3xx
Password=sys3xx
;TargetFolderOnHost=FTP-RESULTS

; intermediate buffer for FTP telegrams on FTP Client site
DIRECTORY=C:\FTP

; Channel info in JSON format
; IP address for manual operations or for tools connected via serial port/USB can be station IP:
CHANNEL_03_INFO={ "IP": "172.23.56.112", "ChannelName": "OGS-PICK2LIGHT", "location name": ["OGS_Demo", "", "STATION NAME", "", "", "", ""] }
CHANNEL_04_INFO={ "IP": "172.23.56.44", "ChannelName": "OGS-CRANE", "location name": ["OGS_Demo", "", "STATION NAME", "", "", "", ""] }
CHANNEL_05_INFO={ "IP": "172.23.56.101", "ChannelName": "OGS-STATION", "location name": ["OGS_Demo", "", "STATION NAME", "", "", "", ""] }
;
```

In this section, the following keys can be used:

- **ENABLED** (int): Enable or disable the FTP. If set to non-zero, then the FTP driver is enabled.
- **HostIP** (ip address): IP address of the FTP server.
- **HostPort** (int): The port used for FTP on the FTP server. It is usually 21.
- **Username** (string): The username to log into the FTP server. If you are using Sys3xxGateway, this has to be "sys3xx".
- **Password** (string): The password to log into the FTP server. If you are using Sys3xxGateway, this has to be "sys3xx".
- **Directory** (string): Defines a directory on the client side that is used as an intermediate buffer for the FTP telegrams.
- **TargetFolderOnHost** (string): Optional, Defines a directory on the server side that is used to store the FTP telegrams. This option is not needed (and thus should not be configured) when using Sys3xxGateway.
- **Channel info**: The channel info uses an JSON encoded string to define channel properties.

Syntax:

```
CHANNEL_<CHANNELNUMBER>_INFO={ "IP": "TOOL IP", "ChannelName": "EXAMPLE-CHANNEL",
"location name": ["OGS_Demo", "", "STATION NAME", "", "", "", ""] }
```

This information has to be provided for the FTP server to build a correct FTP telegram:

- **CHANNEL_<CHANNELNUMBER>_INFO**:
Enter the channel number here (e.g. **CHANNEL_02_INFO**)
- **"IP": "TOOL IP"**

Enter the IP address of the tool that sent the result. This IP address will be used to replace the IP address of the machine that is running HeOGS, because this is the IP address that will be in the header of the FTP telegram, but the final database entry should contain the real IP address of the tool that sent the result.

- **"ChannelName":** "EXAMPLE-CHANNEL"

Enter the channel name here (e.g. **"ChannelName":** "OGS-CRANE ")

- **"location name":** ["OGS_Demo", "", "STATION NAME", "", "", "", ""]

15.1.2 Station info channel 99

The station info channel provides station specific info (logon/logoff and takt times). It is active by default, but the settings can be overridden through the station.ini file.

Default settings are:

- Channel name: "INFO_CHANNEL"
- Station name (location[3]): station name, as defined in the [GENERAL] section.

To override (e.g. the station name), add a line similar to the following:

```
CHANNEL_99_INFO={ "location name": [ "", "", "MyStationName", "", "", "", "" ] }
```

15.2 Pick2Light configuration

15.2.1 Driver Setup

Pick2Light requires the Modbus driver. To enable this driver, please make sure to setup the section [TOOL_DLL] as follows:

```
99  [TOOL_DLL]
100  OPConn.dll=1
101  MBLightCli.dll=1
102  ;EXAConnecT.dll=1
103  ;heGWK.DLL=1
104  ;ProfiFTP.dll=1
```

15.2.2 Driver Configuration

The modbus tool driver is configured in the station.ini file as follows:

```

106 ; =====
107 ; Channel definitions
108 ; This links drivers, channels and tools
109 ;
110 [CHANNELS]
111 3=Pick2Light
112
113 ; =====
114 ; Driver parameters - global
115 ;
116
117 [MBLightCli.dll]
118
119 ; =====
120 ; Tool parameters - PickToLight
121 ;
122 [Pick2Light]
123 DRIVER=MBLightCli
124 CONN_ADDR=10.10.2.179
125 CONN_PORT=502
126 ;;;;LUA_PREFIX=P2L_
127 ; This is the initialisation 4x02006 register for the Modbus device.
128 ; It needs to be written to once when first contact is established or
129 ; once on re-acquisition of contact. This clears any latched watchdog
130 ; fault on the I/O block and allows it autonomously to set outputs
131 ; to 0 when comms is lost.
132 ; NOTE: Init is *always* done through FC "Write single register"
133 ; NOTE: if INIT_REG is missing or = 0, then no init is done.
134 INIT_REG=2006
135 ; On Rexroth this clears the latched fault and resets the block. Other vendors
136 ; use other settings.
137 INIT_VAL=6
138 ; Cyclic I/O
139 ; Note that the following registers (holding registers, 4x) are cyclically
140 ; read/written through FC23 (read/write multiple registers). Note also, that
141 ; the IOs are bit-mapped, i.e. Bit 0 = Reg[0].0, Bit 17 = Reg[1].1, ...
142 ; The tool parameter (in the config) is a "Program number". The "program number"
143 ; actually is the "bin number" of the pick 2 light system. Bits start a 0,
144 ; so do the bins - Prg 0 --> Bin 0 --> Modbus reg[0].0
145 INPUT_REG_START=1
146 INPUT_REG_LENGTH=1
147 OUTPUT_REG_START=1
148 OUTPUT_REG_LENGTH=1
149 ; If INPUT_ACK_ALL is >= 0, then the given bit is used as a global ACK value.
150 ; This is treated identically to a bin, so it might need an additional digital
151 ; input module.
152 ; Default: -1 (if missing or empty), i.e. disabled.
153 INPUT_ACK_ALL=-1
154 ; Set to non-zero to make the requests output flash at 1Hz
155 OUTPUT_FLASH=1
156
157

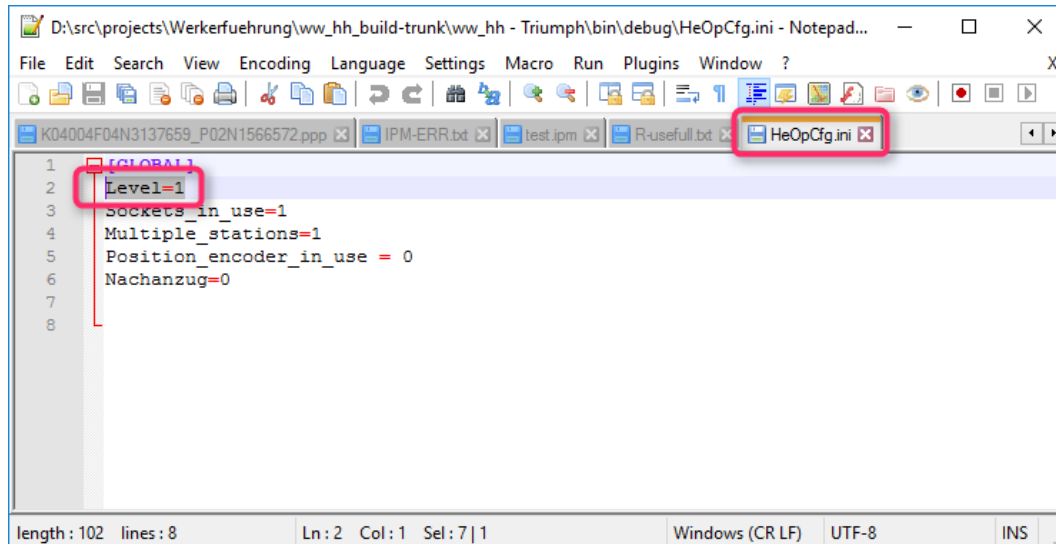
```

15.2.3 Editor Configuration

The Pick2Light driver/tool is not available in the heOpCfg editor by default. To enable it, follow the steps described below.

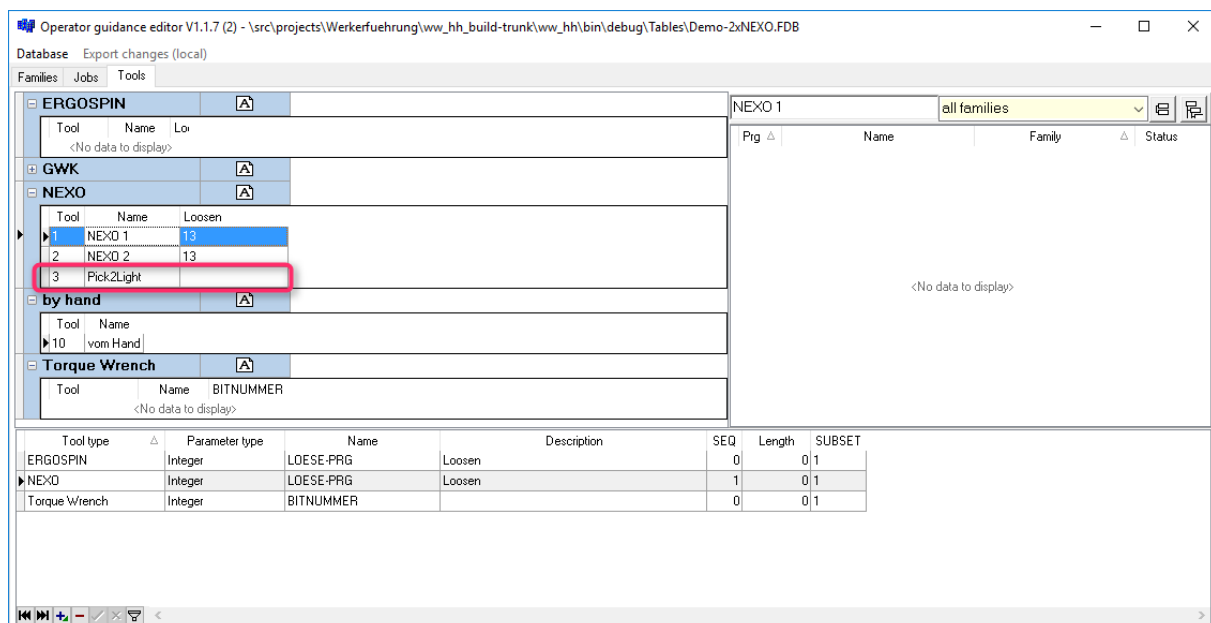
15.2.3.1 Enable Level 1 configuration

To allow Pick2Light configuration, the editor must be set to Level 1 configuration. See the following screenshot on how to enable configuration level 1 in heOpCfg.ini (located in the installation directory of heOpCfg.exe):



15.2.3.2 Add and configure the Pick2Light tool

The Pick2Light tool is simply added as a new Tool under the “NEXO” category. See the following screenshot:



15.3 Barcode scanner

To use a barcode scanner with heOGS, please uncomment the following line in the station.ini:

```
[SERIAL]
;Profibus    =    COM4 187500,e,8|halbduplex
Barcode =    COM4 115200,n,8|halbduplex
;SocketTray =    COM8 115200,e,8|halbduplex
;Barcode_Rfid =    COM41 9600,n,8
```

serial port

CRLF

Please make sure to set the serial connection parameters to the one used by your barcode scanner.

15.4 User configuration

Adding users to a station is done in the [USER] section of the station.ini. The syntax is as documented in the screenshot below:

<username>=<password>,<userlevel(optional; default = 1)>,<RFID-Code(optional, see section 15.4.1)>

```
[USER]
; Define logins
; Format:
;   <username>=<password>[,<userlevel>]
; If <userlevel> is omitted, 1 is used as default.
; Allowed userleves are 1=Operator, 2=Supervisor, 3=Administrator
; The <username> "autologon" is specially treated, and must follow
; the format:
;   autologon=<username>
; If autologon is available and a non-empty username is given, then an
; operator with the given <username> is logged on (userlevel is always
; = 1).
;
autologon=TESTUSER
TESTUSER=testpassword,1,U40203ABCDE
```

The user in the screenshot was created for demonstration purposes with the following parameters:

- 1.) Username: TESTUSER
- 2.) Password: testpassword
- 3.) Userlevel: 1
- 4.) RFID-Code: U40203ABCDE
- 5.) Autologon is activated for this user. If autologon is available and a non-empty username is given, then an operator with the given username is logged on automatically when HeOGS starts. The userlevel of the autologon user is always 1.

15.4.1 RFID reader logon/logoff

The RFID-Codes are defined per user as shown in chapter 15.4. The activation of the RFID-Reader is done in the serial section. Simply add a line with the serial connection parameters, e.g.:

Barcode_Rfid = <COM-port> <baud rate>,<parity>,<data bits>

For an example RFID-Reader configuration, please see the screenshot below:

```
[SERIAL]
;Profibus   =   COM4 187500,e,8|halbduplex
;Barcode    =   COM4 115200,n,8|halbduplex
;SocketTray =   COM8 115200,e,8|halbduplex
Barcode_Rfid =   COM41 9600,n,8
```

NOTE: Using an RFID reader for logon/logoff also requires modifying the code in barcode.lua to setup tag ⇔ user lookup.